

Defense and Repair

Immune Regulation of Inflammation



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APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Bryan Stepanenko, MD, disclosed no relevant financial relationships with any ineligible companies.

Robert Luby, MD, disclosed no relevant financial relationships with any ineligible companies.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

1. Explain key concepts and features that distinguish chronic inflammation from the acute inflammatory response.
2. Recall the clinical findings and laboratory biomarkers of the acute inflammatory response and chronic inflammation.
3. List the most common antecedents, triggers, and mediators of chronic inflammation.
4. Explain how modifiable lifestyle factors and mental, emotional, and spiritual factors may be mediators of chronic inflammation.
5. Explain the foundational mechanisms of chronic inflammation and the mechanisms by which inflammation is a mediator of dysfunction in other bodily systems.

IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Identify and document the patient's clinical findings, antecedents, triggers, mediators, and effects of chronic inflammation on the IFM Timeline and Matrix.
2. Explain to a patient how their antecedents, triggers, and mediators of chronic inflammation underpin their clinical findings and current concerns.
3. Co-prioritize with the patient to initiate dietary, lifestyle, and gut health interventions as foundational therapies to address the patient's antecedents, triggers, mediators, and effects of inflammation.
4. Initiate foundational nutraceuticals, botanicals, and supplements as clinically indicated.

Functional Medicine Matrix

Clinical Imbalances and Physiological Dysfunctions

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Long Description: The Functional Medicine Matrix

- The Functional Medicine Matrix, with emphasis on the defense and repair node.

Clinical Focus

- The acute inflammatory response
- “A process with a purpose”



Acute Inflammation Clinical Findings

Physical Exam Signs & Symptoms

- Heat (*calor*)
- Redness (*rubor*)
- Swelling (*tumor*)
- Pain (*dolor*)
- Loss of function (*functio laesa*)

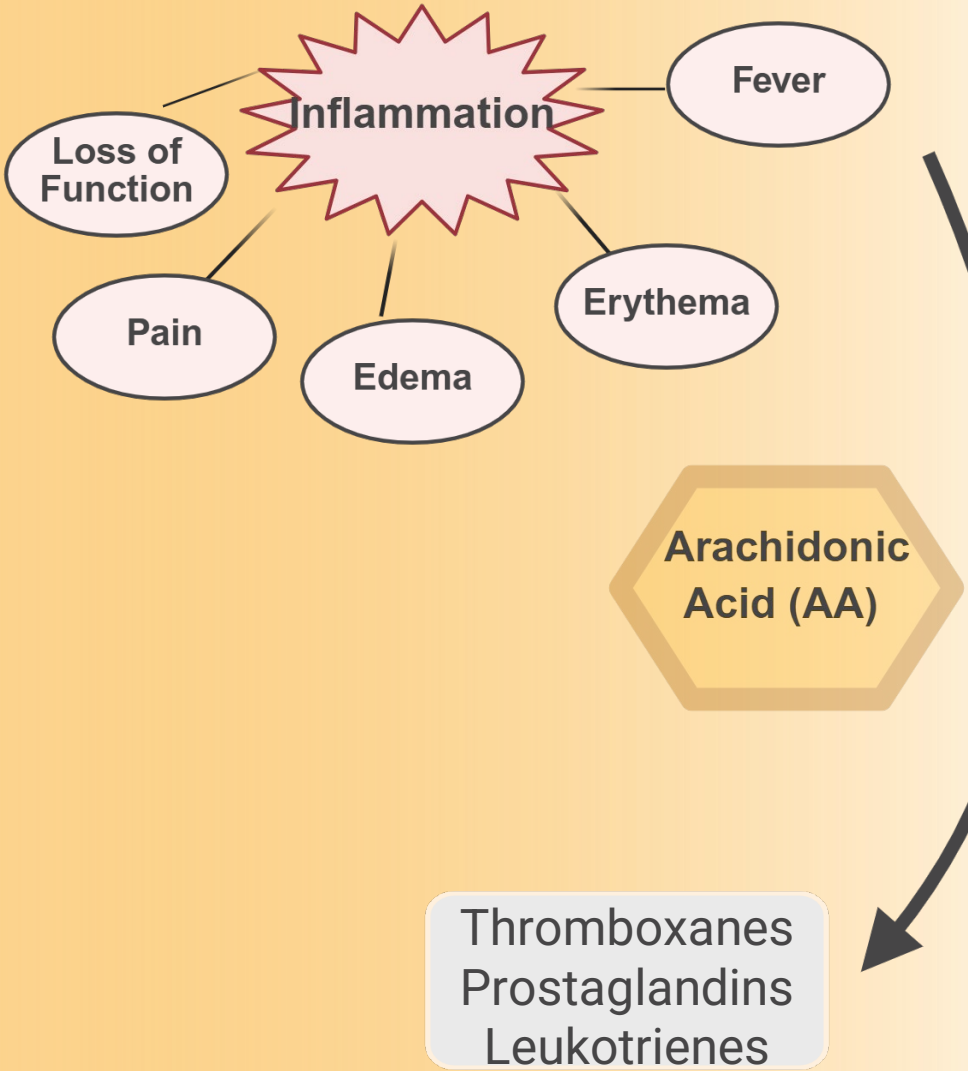


Acute Inflammation

Key Physiological Concepts

- The acute inflammatory response is the body's normal physiologic attempt to defend against microbes and repair the effects of injury/insult.
- Injury/insult most often results from diet, lifestyle, trauma, microbes, and toxins.
- Most acute inflammatory triggers are molecular agents that activate signal transduction.

Initiation of Inflammatory Response

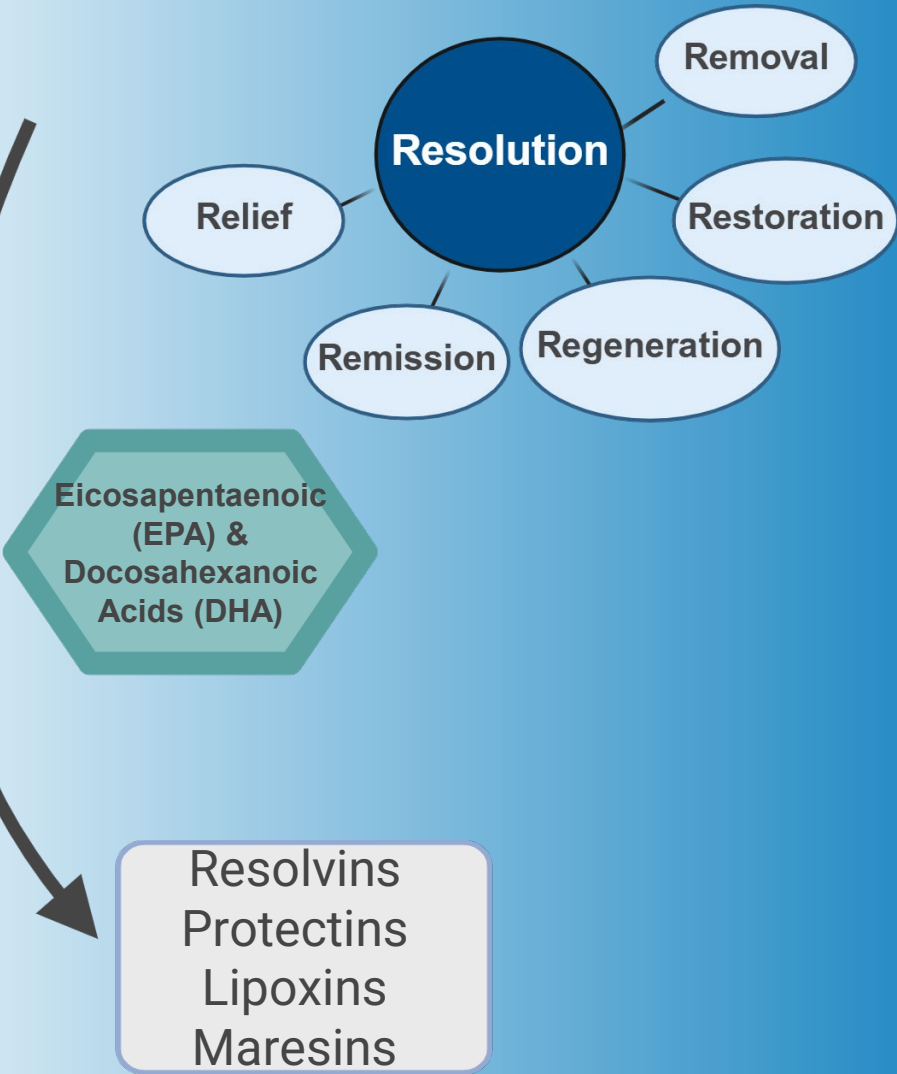


enzymes driving BOTH processes

COX

LOX

Resolution of Inflammatory Response



Long Description: The Inflammation Response

- The cardinal signs of inflammation are loss of function, pain, edema, erythema, and fever. During an inflammatory response, cell membranes release arachidonic acid, which is metabolized by cyclooxygenase and lipoxygenase enzymes, producing inflammatory mediators such as thromboxanes, prostaglandins, and leukotrienes. Resolution of the inflammatory response involves removal, restoration, regeneration, remission, and relief. In this process, omega 3 fatty acids including E P A and D H A are metabolized by lipoxygenase into resolvins, protectins, maresins, and lipoxins, otherwise known as specialized pro-resolving mediators.

The Acute Inflammatory Response

A Process With a Purpose

- **DEFEND** against invaders by
 - Chemical attack
 - Engulfing
- **REPAIR** damaged tissue
 - Sequester source of injury
 - Activate healing response
 - Eradication or sequestration of agent/stressor
- Usually localized (anaphylaxis excepted)
- Usually adaptive (allergy is exception)

The Acute Inflammatory Response

A Process With a Purpose

- Leukocyte stimulation/maturation
 - Adhesion
 - Migration
 - Activation
- Altered vascular rheology
 - Changes in blood flow
 - Coagulation—fibrinolysis cascades

Physiology of Inflammation

A Process With a Purpose

- Mucous production (mucosa)
- Smooth muscle contraction
- Nociceptor activation
- Psycho-neuroendocrine reflex

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PHYSIOLOGY OF INFLAMMATION

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Physiology of Inflammation

A Process With a Purpose

- Release (degranulation) of:
 - Inflammatory cytokines
 - Proteases
- Activation of complement cascade
- Mobilization of cell adhesion molecules
- Antimicrobial peptides (defensins)

Acute Inflammatory Response

Increased Production of:

- **Free radicals** (oxidative burst)
 - NADPH oxidase: O_2^- (superoxide)
 - Myeloperoxidase: HClO (hypochlorous acid)
- **Inflammatory cytokines & chemokines**
 - Thromboxanes, prostaglandins, and leukotrienes
 - IL-1, IL-6, TNF- α , NF-kB
- **Acute-phase reactants** (CRP)

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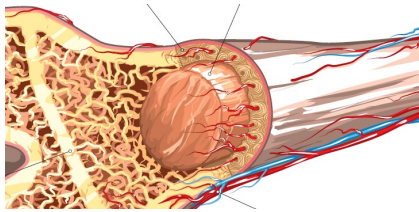
Components of the Acute Phase Inflammatory Response

Inflammatory stimulus:
trauma, infection, tissue
injury, stress, etc.

Local acute
inflammatory
response

IL-6, TNF- α

Bone marrow
($\uparrow$ CSF by
stromal cells
and
macrophages)

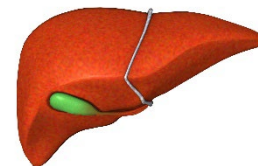


Leukocytosis
($\uparrow$ WBC)

Increases in acute-phase
proteins:

- C-reactive protein (CRP)
- Serum amyloid A (SAA)
- Fibrinogen
- Mannose-binding protein
- Complement components

IL-1, TNF α , IL-6, LIF, OSM

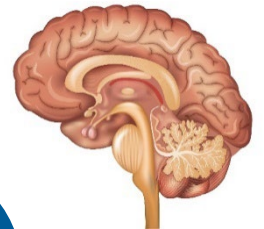


**Hepatic Acute-
Phase Response**

IL-1, TNF- α , IL-6

Fever

Prostaglandins $\uparrow$



The hypothalamus
responds to level
of cortisol

ACTH released
(via pituitary gland)

Corticosteroids

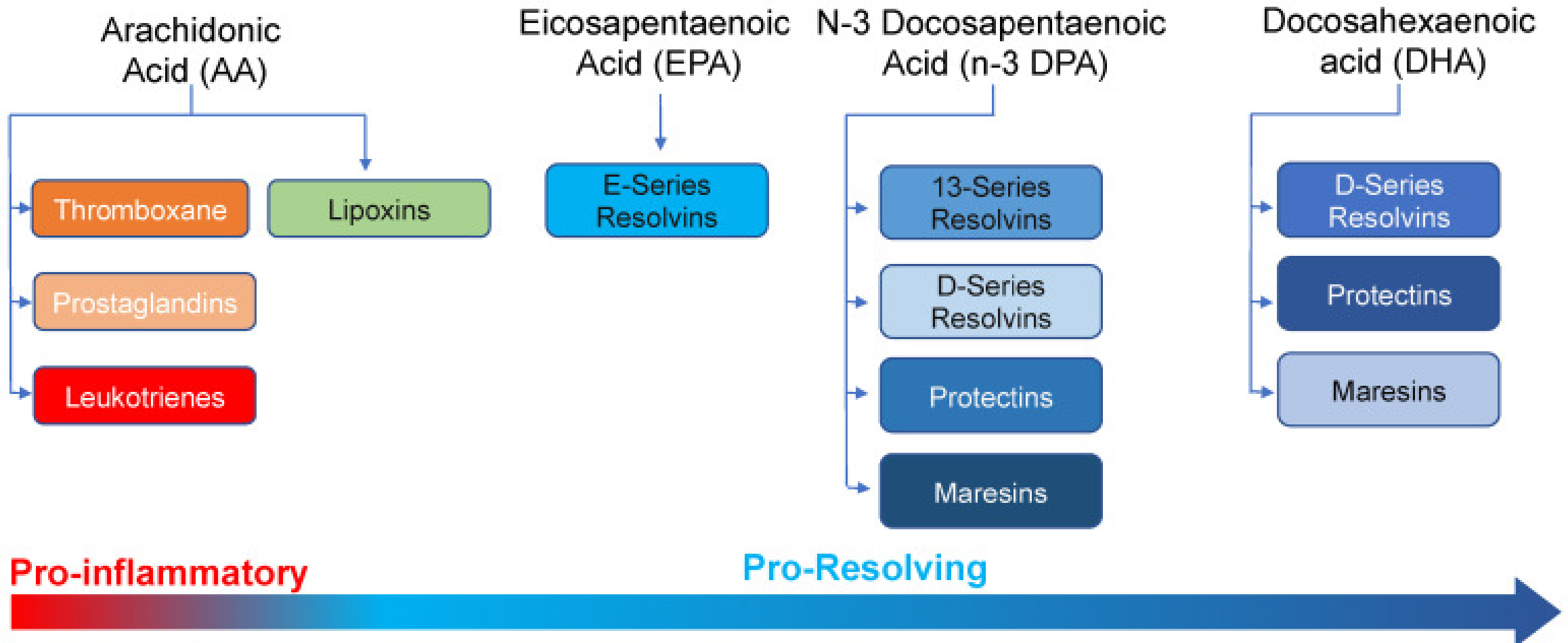
Adrenal
gland



Long Description: Components of the Acute Phase Inflammatory Response

- A soft tissue injury is a stimulus that can initiate a local acute inflammatory response through the release of acute-phase proteins such as C R P. A soft tissue injury such as a sprained ankle causes the release of I L 1, T N F alpha, and I L 6, which stimulate the release of prostaglandins and A C T H. A C T H stimulates the release of corticosteroids from the adrenal glands. Corticosteroids and other inflammatory cytokines including I L 1, T N F alpha, I L 6, L I F, and O S M directly stimulate the hepatic acute-phase response, causing the release of acute-phase proteins like C R P, serum amyloid A, fibrinogen, mannose-binding protein, and complement components from the liver.

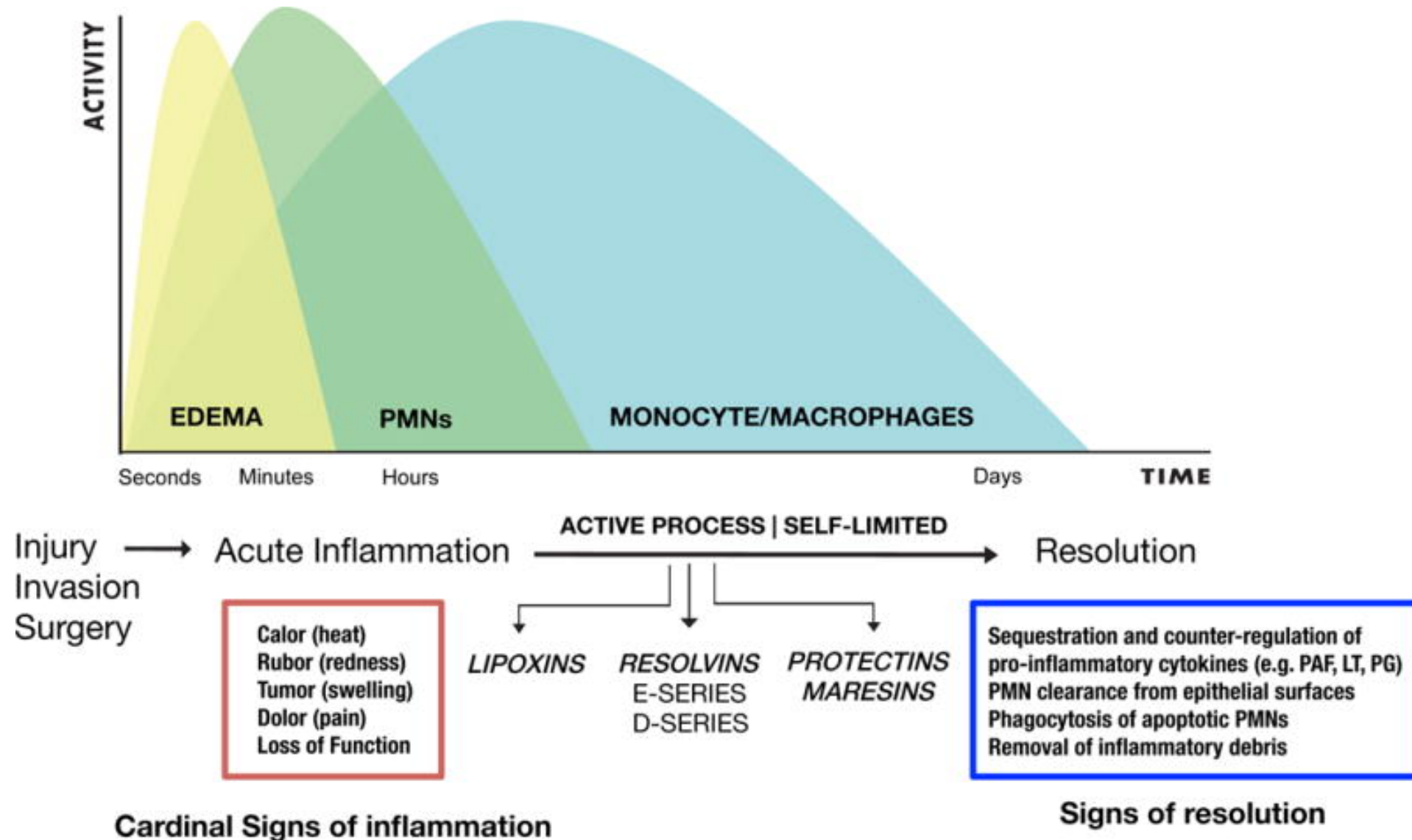
Lipid-Mediated Resolution



Long Description: Lipid-Mediated Resolution

- Thromboxane, prostaglandins, and leukotrienes are pro-inflammatory lipid mediators derived from arachidonic acid. Lipoxins are also derived from arachidonic acid but are pro-resolving. Pro-resolving lipid mediators also include E series resolvins, derived from E P A, 13-series resolvins, D series resolvins, protectins, and maresins, derived from D P A, and D series resolvins, protectins, and maresins derived from D H A.

Resolution of Acute Inflammation



Long Description: Resolution of Acute Inflammation

- Acute inflammation is characterized by calor (heat), rubor (redness), tumor (swelling), dolor (pain), and loss of function. In an acute inflammatory response, edema formation starts within seconds of the initiating event, followed by polymorphonuclear neutrophil (P M N) infiltration and subsequently, their clearance by monocytes and macrophages. Specialized pro-resolving mediators (S P M) including lipoxins, resolvins, protectins, and maresins, promote the resolution of inflammation. Signs of resolution include the sequestration and counter-regulation of pro-inflammatory cytokines, P M N clearance from epithelial surfaces, phagocytosis of apoptotic P M N's, and removal of inflammatory debris.

Clinical Focus

- Transition from acute to chronic inflammation.
- Chronic Inflammation: Molecular Mechanisms and Mediators



Chronic vs Acute Inflammation

Key Physiological Concepts

- Chronic inflammation happens when the acute inflammatory response persists.
- Chronic inflammation is often an effect of recurrent or **persistent triggers** (dietary, pathogen, toxin, injury) AND **ineffective resolution** (potentiated by adverse dietary intake and/or lifestyle behaviors).
- The chronic inflammatory response becomes a mediator of broader physiological dysfunction.

The Transition

From Local and Acute to Systemic and Chronic Inflammation

The Starting Point:
Acute Inflammation
Often local insults

Key Players:
PAMPs & DAMPs
(bugs, drugs, foods,
trauma, toxins)

Activate

Pattern
Recognition
Receptors on
Immune Cells

Mechanistic Mediators (see next slide*):

- Continuous PAMP/DAMP exposure
- Loss of Tolerance
- Lack of Resolution
- Immune Dysregulation
- Bioaccumulation of Toxins

Mediated by: Histamine,
Prostaglandins, and Leukotrienes,
Cytokines, NLRP-3 Inflammasome,
and NFκB Activation

The Outcome:
Chronic Inflammation
Often systemic effects

Long Description:

The Transition from Local and Acute to Systemic and Chronic Inflammation

- In acute inflammation, pathogen and damage associated molecular patterns including bugs, drugs, foods, trauma, and toxins activate pattern recognition receptors on immune cells. Acute inflammation is mediated by histamine, prostaglandins, leukotrienes, cytokines, N L R P 3 inflammasome, and N F kappa B activation and often involves local insults.
- In chronic inflammation, continuous exposure to pathogen- and damage-associated molecular patterns, loss of tolerance, lack of resolution, immune dysregulation, and bioaccumulation of toxins lead to chronic inflammation, often with systemic effects.

Chronic Inflammation

Molecular Mediators and Mechanisms

Continuous PAMP/DAMP Exposure

- Leading to tissue damage and immune system exhaustion

Loss of Tolerance

- Autoimmune conditions

Lack of Resolution

- Leads to endothelial dysfunction, oxidative stress, and chronic tissue damage

Immune Dysregulation

- Overactive or underactive immune function

Bioaccumulation of Toxins

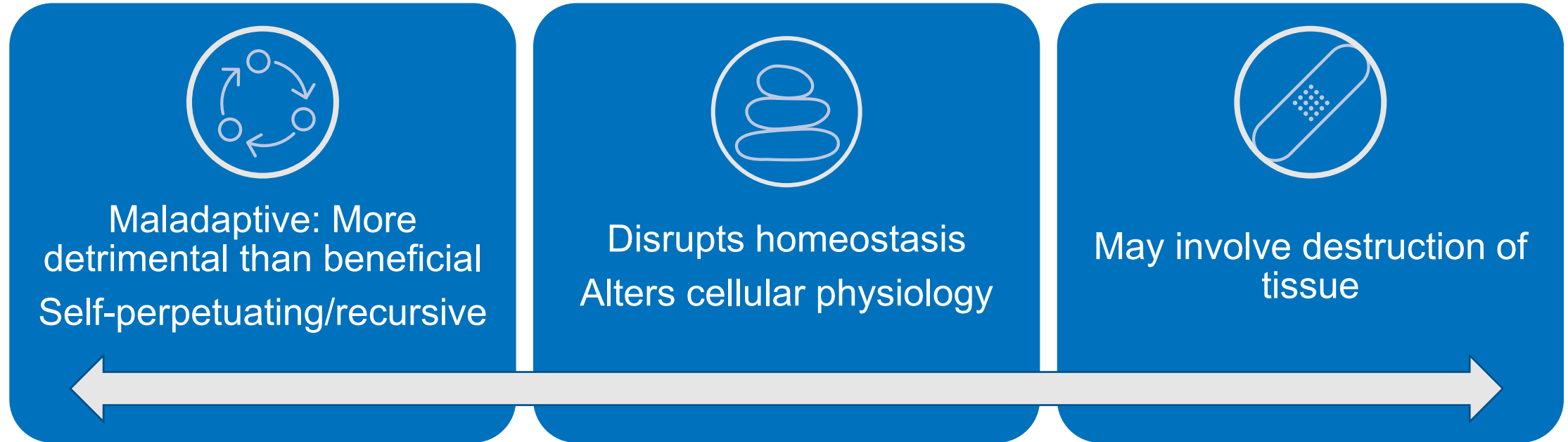
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FROM LOCAL AND ACUTE TO SYSTEMIC AND CHRONIC INFLAMMATION

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Chronic Inflammation

Persistent Acute-Phase Response



Chronic inflammation

- Usually secondary to an impairment in the resolution phase of inflammation.
- Often associated with a loss of tolerance due to dietary patterns, pathogens, or environmental triggers.

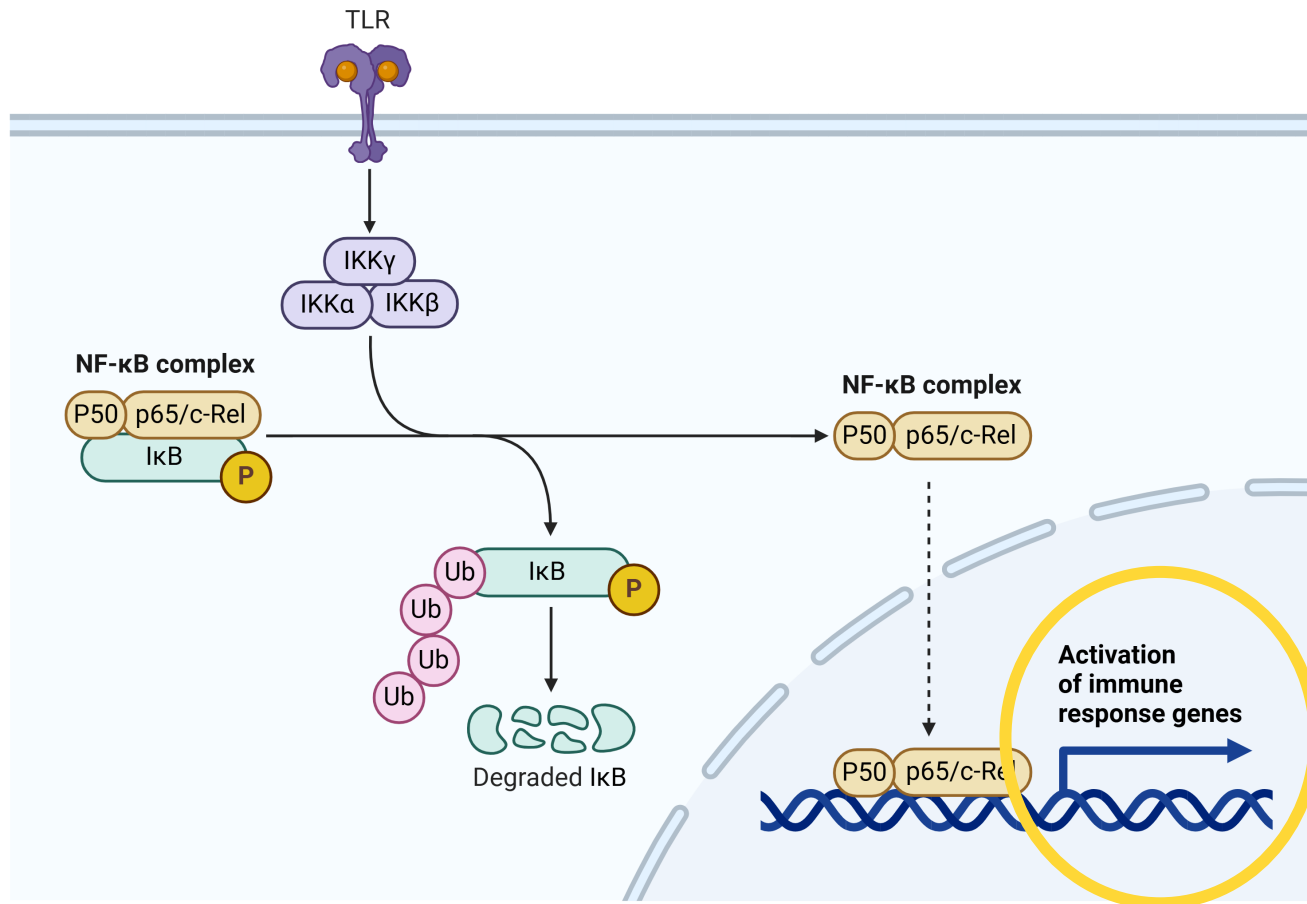
Chronic Inflammation

Molecular and Cellular Mechanisms of Action

- Overabundance of inflammatory precursors (ex: high AA or high NLRP3 activity, baseline inflammation from exposures/infections)
- Lack of antioxidant phytochemicals (ex: ultra-processed foods, Low Fruit and Vegetable intake)
- Imbalanced dietary intake of fatty acids (ex: high Omega-6, low Omega-3)
- Imbalanced immune system that is primed for inflammation (low tolerance)
 - Low T regulatory cells (Treg)
 - High T effector cells (ex: Th1, Th2, Th17)
 - M1 macrophage phenotype

Nuclear Factor Kappa B (NF-κB)

Transcriptional/Epigenetic Upregulation in Chronic Inflammation



- Discovered in 1986 by Ranjan Singh (working in lab of Dr. David Baltimore)**
- Bound in cytoplasm to inhibitor protein complex: IκB
- Inflammatory stimuli lead to phosphorylation of IκB by IκB-kinase, which causes it to dissociate from NFκB
- IκB is then degraded by enzymes
- **TRANSCRIPTIONAL EFFECT:** Liberated NFκB then moves to cell nucleus, where it binds to specific regions of DNA & *upregulates the expression of inflammatory genes*

Adapted from "NF-κB Signaling Pathway" by BioRender.com (2025). Retrieved from: <https://app.biorender.com/biorender-templates>

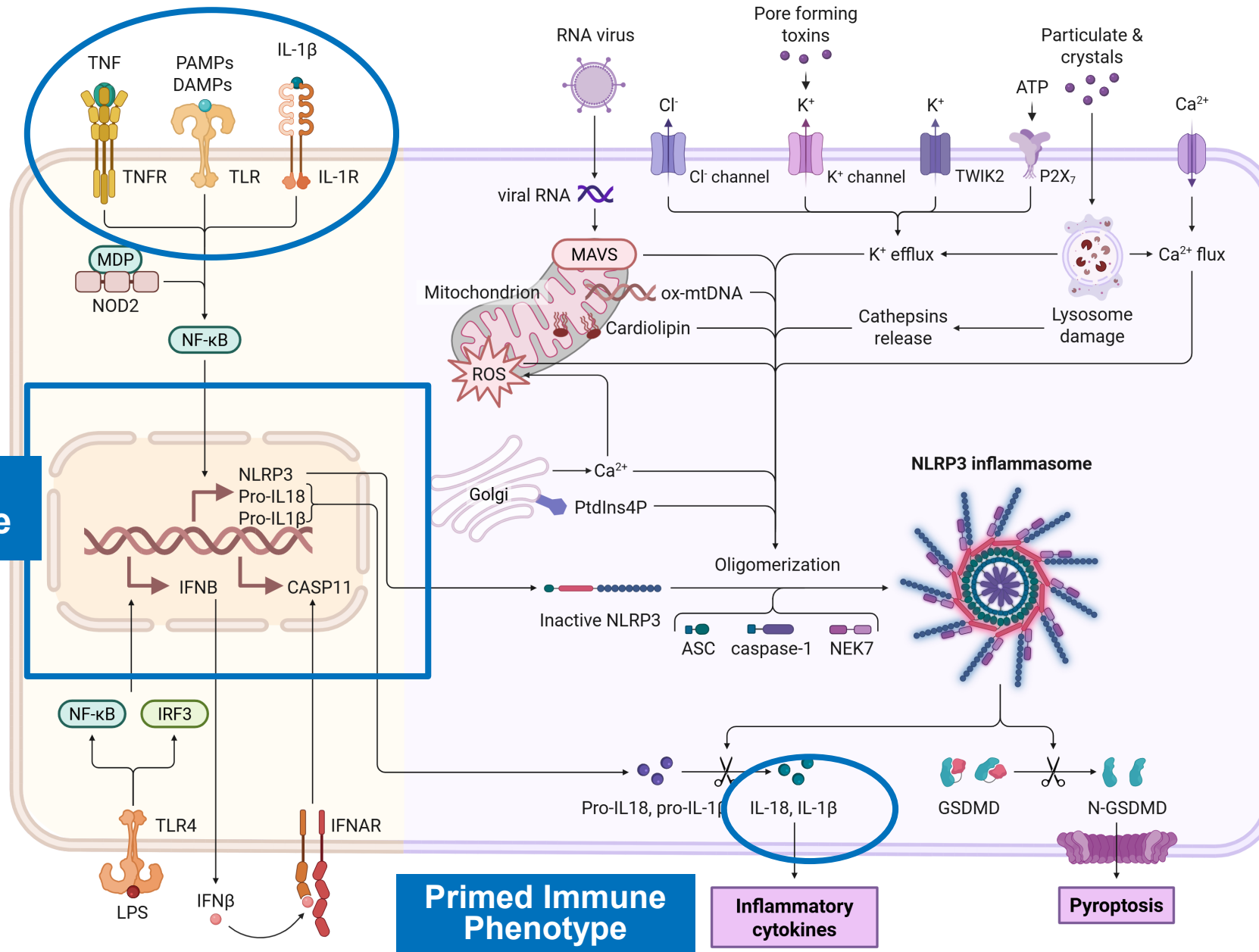
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Step 1: priming

Step 2: activation

NLRP3 Inflammasome

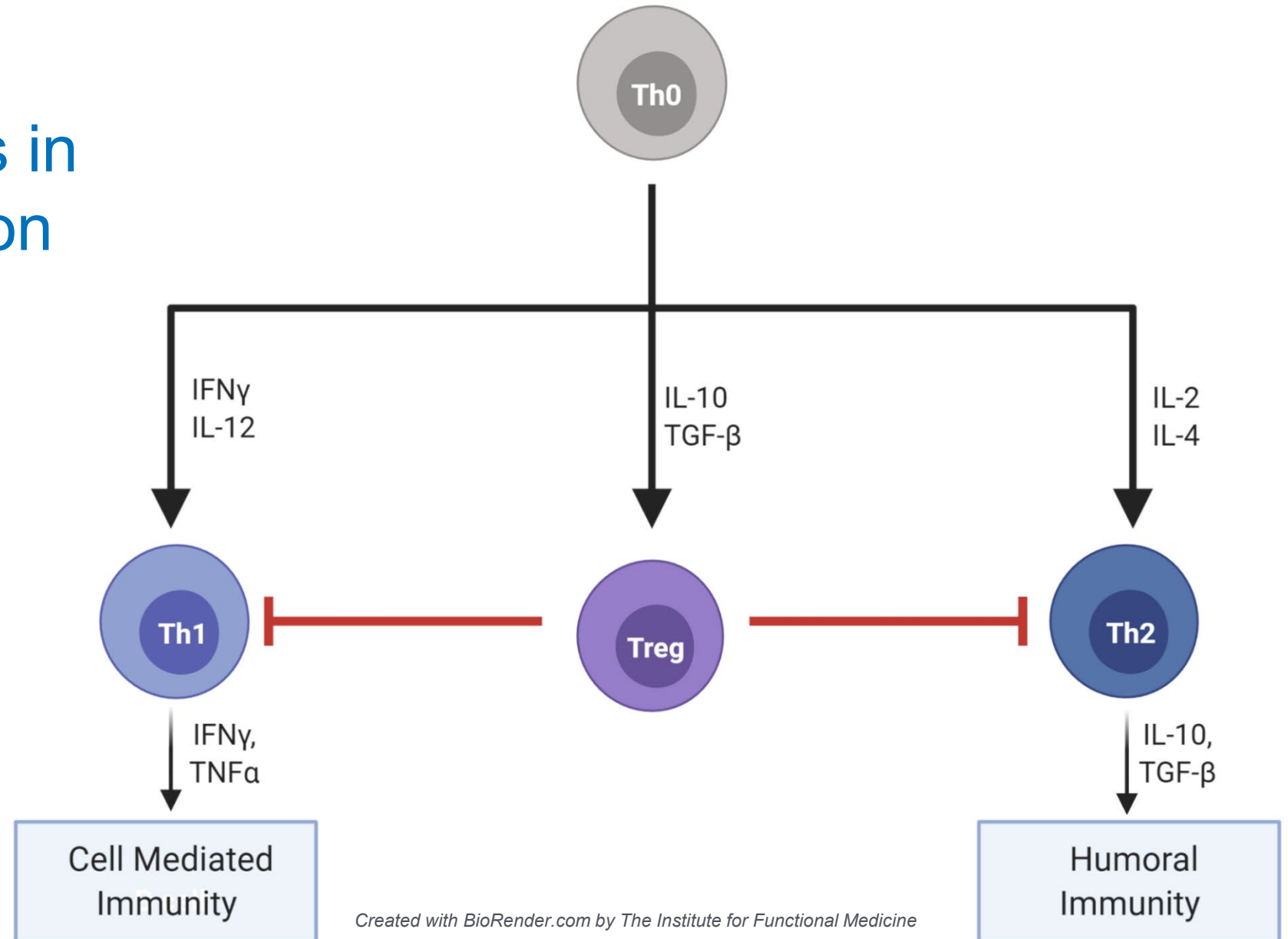


Adapted from "NLRP3 Inflammasome Activation" by BioRender.com (2025)
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Long Description: Nuclear Factor Kappa B

- NLRP3 inflammasome activation requires two steps: priming and activation. In priming, a local insult, such as a pathogen or damage-associated molecular pattern, activates NF-κB. In activation, a trigger such as RNA virus, pore forming toxins, ATP, or crystals leads to NLRP3 inflammasome assembly, triggering inflammation and inflammatory cell death, called pyroptosis.

The Role of Regulatory T-Cells in Immunomodulation

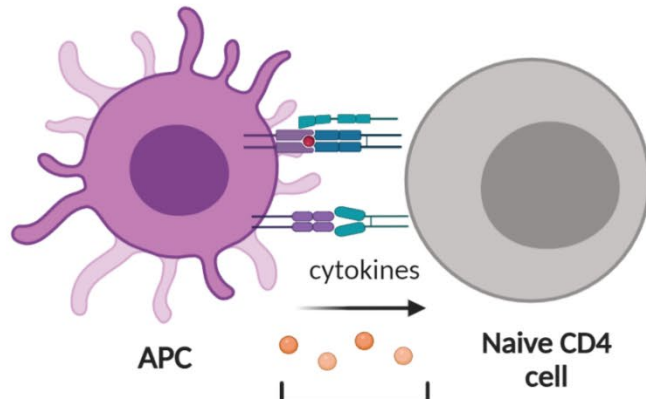


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Long Description: The Role of Regulatory T-Cells in Immunomodulation

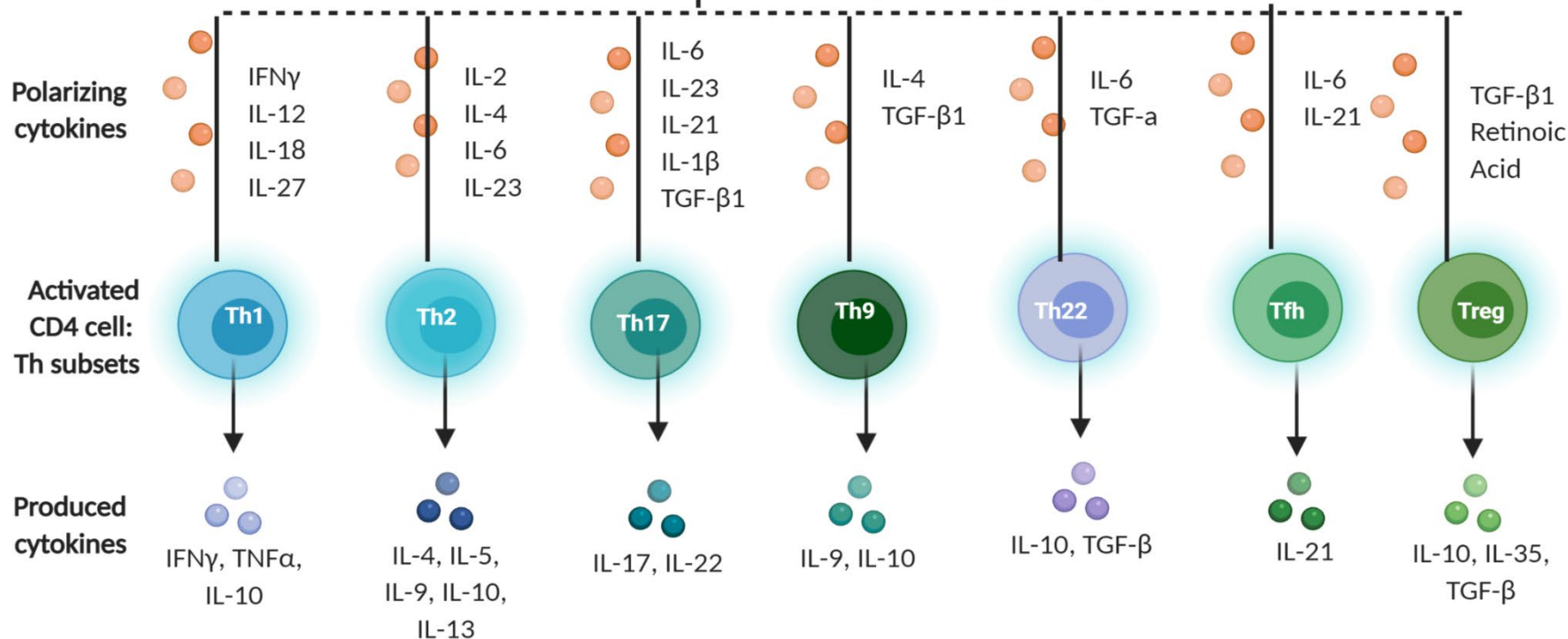
- Naïve T cells differentiate into T_H 1, T_H 2, or T regulatory cells based on cytokine involvement. I F N gamma and I L 12 drive T_H 1 differentiation, promoting cell mediated immunity, whereas I L 2 and I L 4 drive T_H 2 differentiation, promoting humoral immunity. I L 10 and T G F beta drive T regulatory differentiation, which regulate both the T_H 1 and T_H 2 responses.

T-Cell Differentiation



Different combinations of cytokines influence T cell differentiation into distinct effector T cell subtypes that produce signature cytokines.

- After a trigger, the GOALS are:
- Remove the trigger (if possible)
 - Resolve inflammation
 - Make more T reg cells



Adapted from "T cell activation and differentiation" by BioRender.com (2020). Retrieved from <https://app.biorender.com/biorender-templates>

Long Description: T-Cell Differentiation

- When a naïve C D 4 T cell recognizes its specific antigen on an antigen-presenting cell, the cytokine environment influences T cell differentiation into distinct effector T cell subtypes that produce their own signature cytokines.

Acute vs Chronic Inflammation

Distinguishing Features



Acute Inflammation

- Triggered by infection from pathogen-associated molecular patterns (PAMPs) or cellular injury from damage-associated molecular patterns (DAMPs)
- Generally short term, resolving
- Not age-related



Systematic Chronic Inflammation

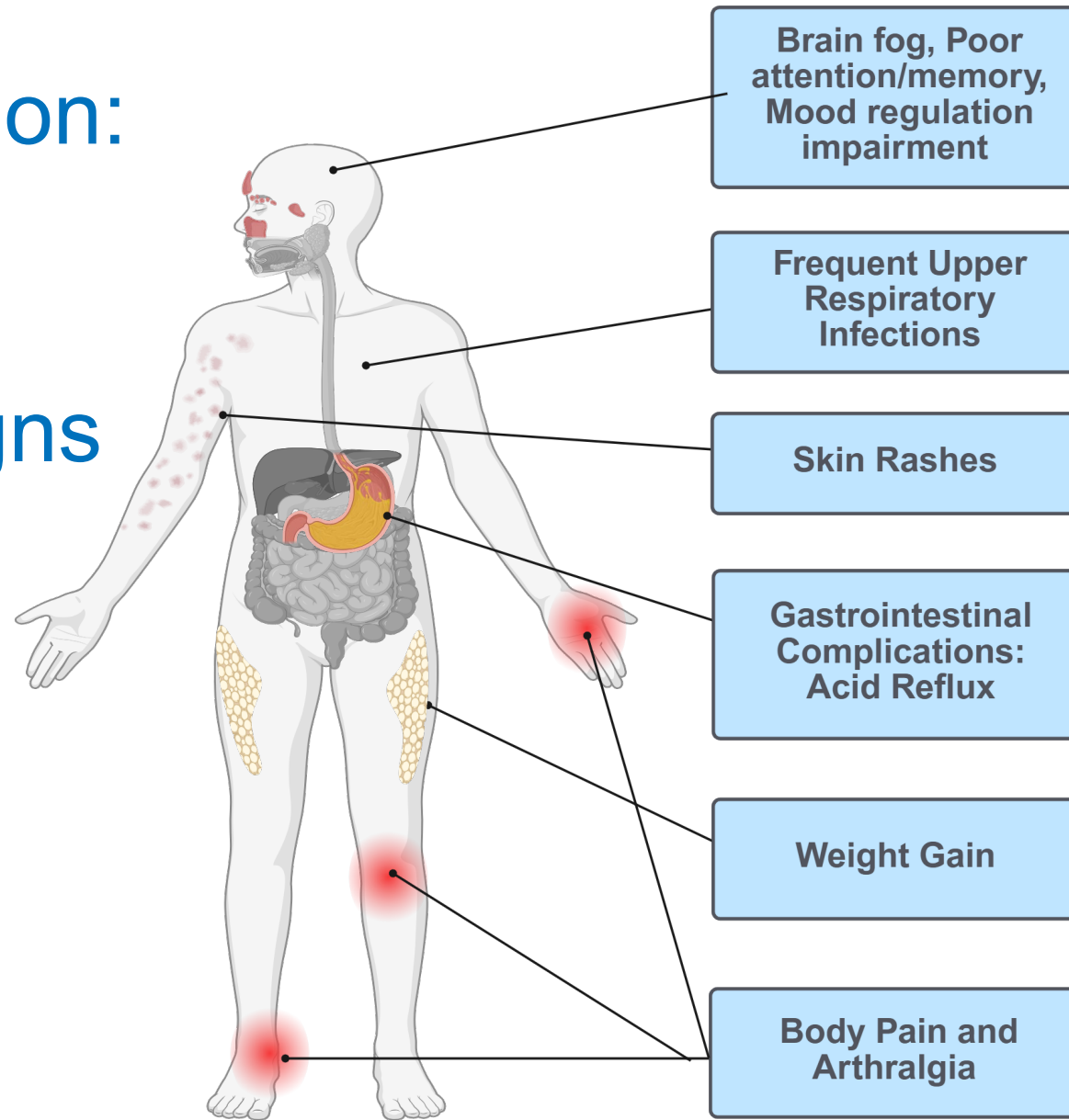
- Triggered by damage-associated molecular patterns (DAMPs) from the exposome, metabolic dysfunction, and tissue damage
- Persistent, recurrent exposure to PAMPs, DAMPs, or other antigens
- Can be age-related, potentiated by genetic, dietary, lifestyle, and exposure factors

Clinical Focus

- Clinical findings and presentation of chronic inflammation.



Chronic Inflammation: Clinical Findings Examples of Symptoms and Signs



Long Description: Chronic Inflammation

- Signs and symptoms of chronic inflammation include:
- Brain fog, poor attention, poor memory, mood regulation impairment
- Frequent upper respiratory infections
- Gastrointestinal complications: acid reflux
- Weight gain
- Body pain and arthralgia

Chronic Inflammation

Disease Associations

- Atopic syndrome/eczema
- Rheumatologic: autoimmune disorders
- Gastroenterological: IBD, gluten intolerance, IBS, SIBO, candida
- Cardiovascular disease: atherosclerosis
- Neurodegenerative disease: PD, AD, MS
- Cancer
- Endocrinologic: diabetes
- Psychiatric disorders: schizophrenia, depression, anxiety, mood disorders

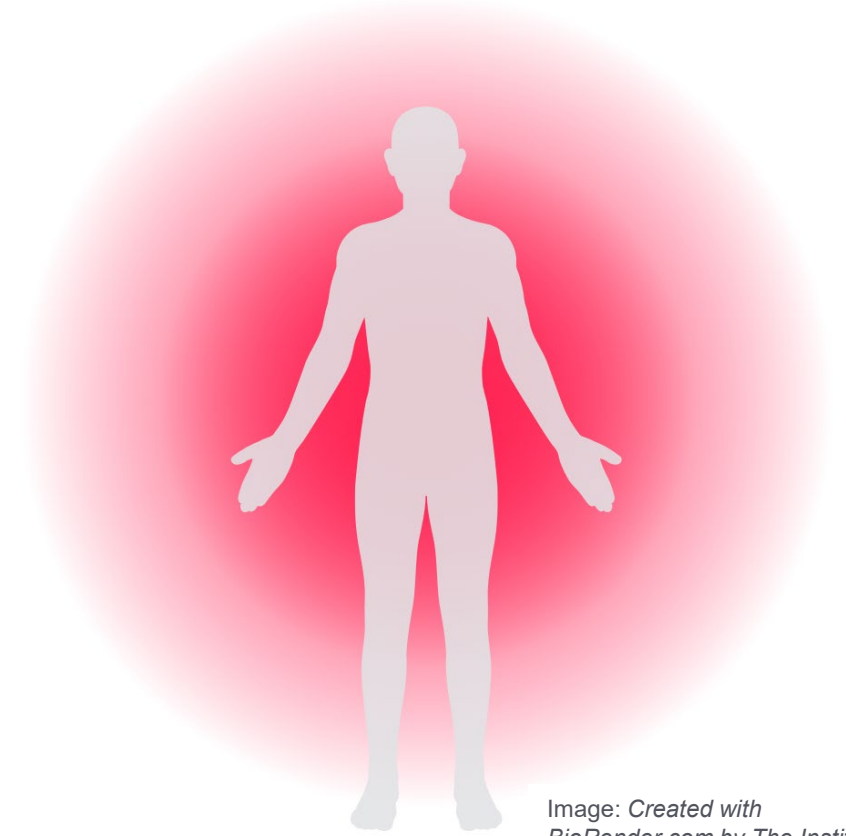


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Functional Medicine

Chronic Inflammation and Immune Dysregulation

Clinical Findings

Immune Response

Antigens	<i>Excess</i>	<i>Deficiency</i>
Internal (self)	Autoimmunity Cardiovascular Neurodegeneration	Cancer
External (non-self)	Allergy, Atopy, Hypersensitivity	Infection

Functional Medicine Matrix

Clinical Findings of Chronic Inflammation

Clinical Imbalances and Physiological Dysfunctions

The Patient's Story

Antecedents (Inherited & Acquired)

- Prenatal history/exposure
- Adverse childhood events (ACEs)
- Strong family history (autoimmune or CVD)

Triggers & Precipitating Events

- Infection
- Injury
- Toxin exposure
- Traumatic life event

Mediators/Perpetuators

- Chronic stress
- Gut dysfunction
- Environmental exposures
- Obesity

Assimilation

- Constipation, diarrhea, and acid reflux
- Gastroenterological: IBD, gluten intolerance
- Malabsorption

Structural Integrity

- Weight gain or loss (unintentional)
- Sarcopenia
- Obesity

Communication

- Insulin resistance
- Hormonal dysfunction
- Thyroid dysfunction

Mental

- Depressed moods
- Feeling anxious
- Scattered thinking

Emotional

Spiritual

Defense & Repair

- Frequent infections
- Allergy, atopy, sensitivities
- Cancer
- Autoimmune

Energy

- Body pain, arthralgia, myalgia
- Chronic fatigue and insomnia
- Mental fatigue
- Chronic sleep dysfunction

Mitigation & Clearance

- Toxin exposure (mold, metal, chemical)
- Multiple chemical sensitivity history
- Constipation
- Lack of sweating

Transport

- Cardiovascular disease: atherosclerosis, hypertension
- Edema, lymphedema

Modifiable Lifestyle Factors

Sleep

- Sleep dysfunction/disorders
- Chronic sleep deprivation
- Sleep apnea

Physical Activity

- Sedentary lifestyle
- Exercise intolerance
- Excessive exercise

Nutrition

- Ultra-processed foods
- Low protein
- Nutrient devoid
- Calorie excess

Stressors

- Chronic stress
- Post-traumatic stress
- Moral injury
- Existential crisis

Relationships

- Social isolation
- Low power/autonomy
- Marital strain
- Social rejection

Pahwa R, Goyal A, Jialal I. *Chronic Inflammation*. StatPearls Publishing; 2023. Updated August 7, 2023. Accessed May 29, 2025.
<https://www.ncbi.nlm.nih.gov/books/NBK493173/>

Accessible version in your toolkit.

Long Description: Functional Medicine Matrix: Clinical Findings of Chronic Inflammation

- The patient's story
 - Antecedents -Prenatal history and exposure, Adverse childhood events, Strong family history
 - Triggers and precipitating factors
 - Infection, Injury, Toxin exposure, Traumatic life event
 - Mediators and perpetuators
 - Chronic stress, Gut dysfunction, Environmental exposures, Obesity
- Modifiable lifestyle factors
 - Sleep: Sleep dysfunction and disorders, Chronic sleep deprivation, Sleep apnea
 - Physical Activity: Sedentary lifestyle, Exercise intolerance, Excessive exercise
 - Nutrition: Ultra-processed foods, Low protein, Nutrient devoid, Calorie excess
 - Stressors: Chronic stress, P T S D, Moral injury, Existential crisis
- Relationships: Social isolation, Low power or autonomy, Marital strain, Social rejection
- Clinical imbalances and physiological dysfunctions
 - Assimilation
 - Constipation, diarrhea, and acid reflux
 - Gastroenterological: I B D, gluten intolerance, malabsorption
- Defense and repair
 - Frequent infections
 - Allergy, atopy, sensitivities
 - Cancer, autoimmune
- Energy
 - Body pain, arthralgia, myalgia
 - Chronic fatigue and insomnia
 - Mental fatigue, chronic sleep dysfunction
- Mitigation and clearance
 - Toxin exposure: mold, metal, chemical
 - Multiple chemical sensitivity history
 - Constipation, lack of sweating
- Transport
 - Cardiovascular disease: atherosclerosis, hypertension
 - Edema, lymphedema
- Communication
 - Insulin resistance, Hormonal dysfunction, Thyroid dysfunction
- Structural integrity
 - Unintentional weight gain or loss, sarcopenia, obesity
- Mental, emotional, spiritual
 - Depressed moods, feeling anxious, scattered thinking

Clinical Focus

- Laboratory biomarkers of inflammation.



Laboratory Biomarker of Inflammation

Neutrophilia

Antecedents

- Familial

Triggers

- Drugs, infection, pregnancy
- Post-splenectomy

Mediators

- Smoking
- Obesity
- Smoking and obesity
- Idiopathic, idiopathic-inflammatory
- Myeloproliferative disorders
- Lymphoproliferative disease
- Multiple causes
- Inflammatory conditions
- Tumors

Laboratory Biomarker of Inflammation

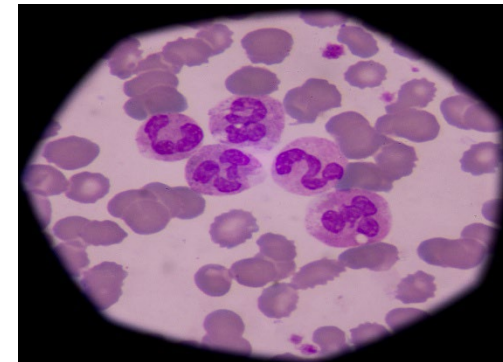
Leukocytosis

Triggers:

- Acute phase reactant
- Infection
- Allergic reactions
- Etiology by subclass (obtain differential)
 - Neutrophilia: acute bacterial infection
 - Lymphocytosis: acute viral infection
 - Monocytosis: chronic infection, inflammation
 - Eosinophilia: allergic, parasitic

Mediators:

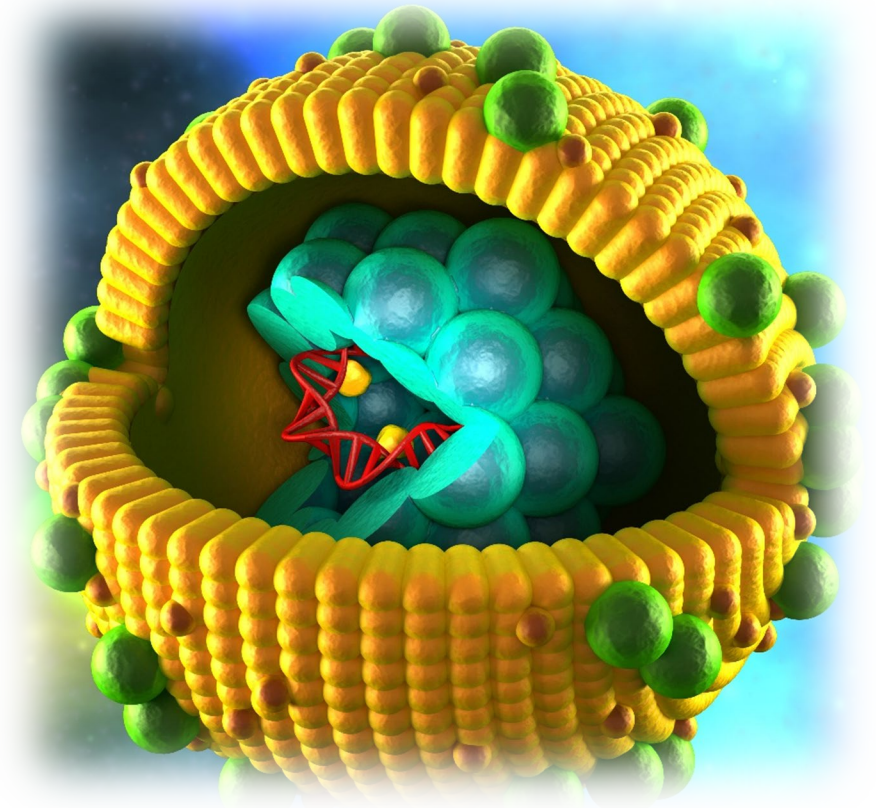
- Malignancy: leukemia, myeloproliferative disorders
- Obesity
- Stressors: trauma, overexertion
- Seizures, emotional stress



Laboratory Biomarker of Inflammation

Leukopenia

- WBC $<4,300$ cells/mm³
 - 3,000-4,000 cells/mm³ often benign
 - **Downward trend more significant** than absolute number – rule out myelodysplastic syndromes
- Clinical Findings and Mediators:
 - **Chronic viral infections** (most common cause); HIV, herpes, hepatitis, etc.
 - **Vitamin C, folate, B-12, or copper deficiency**
 - **Protein insufficiency or maldigestion**



Laboratory Biomarker of Inflammation

ESR (Erythrocyte Sedimentation Rate)

ESR Considerations:

- Non-specific marker of inflammation
- Blood test that measures the rate at which blood cells settle at the bottom of a test tube
- Elevated ESR is an indicator of infection, autoimmune disorders, and chronic inflammatory conditions

ESR Clinical Utility:

- Distinguishing **inflammatory** vs. **allergic conditions**
- **Chronic or persistent infections**
 - Osteomyelitis
 - Pelvic inflammatory disease
 - Tuberculosis

Diagnosis and/or Monitoring:

- Systemic lupus erythematosus
- Polymyalgia rheumatica
- Temporal arteritis
- Rheumatoid arthritis
- Inflammatory bowel disease
- Multiple myeloma
- Hodgkin's lymphoma
- Chronic kidney disease

References

ESR DIAGNOSIS AND MONITORING

SLE, PMR, Temporal Arteritis, RA, HL:

- Bray C, Bell LN, Liang H, et al. Erythrocyte sedimentation rate and C-reactive protein measurements and their relevance in clinical medicine. *WMJ*. 2016;115(6):317-321.

Inflammatory Bowel Disease, Osteomyelitis, Tuberculosis:

- Litao MK, Kamat D. Erythrocyte sedimentation rate and C-reactive protein: how best to use them in clinical practice. *Pediatr Ann*. 2014;43(10):417-420. doi:10.3928/00904481-20140924-10

Multiple Myeloma:

- Koshiaris C, Van den Bruel A, Oke JL, et al. Early detection of multiple myeloma in primary care using blood tests: a case-control study in primary care. *Br J Gen Pract*. 2018;68(674):e586-e593. doi:10.3399/bjgp18X698357

Chronic Kidney Disease:

- Guo S, Wang M, Yu Y, et al. The association of erythrocyte sedimentation rate, high-sensitivity C-reactive protein and diabetic kidney disease in patients with type 2 diabetes. *BMC Endocr Disord*. 2020;20(1):103. doi:10.1186/s12902-020-00584-7

Pelvic Inflammatory Disease:

- Jaiyeoba O, Soper DE. A practical approach to the diagnosis of pelvic inflammatory disease. *Infect Dis Obstet Gynecol*. 2011;2011:753037. doi:10.1155/2011/753037

Laboratory Biomarker of Inflammation

CRP (C-reactive Protein)

CRP Considerations:

- Does not increase with all types of inflammation
- May have normal levels with lupus, cancer
- May be blunted with liver disease and protein deficiency

CRP Clinical Utility:

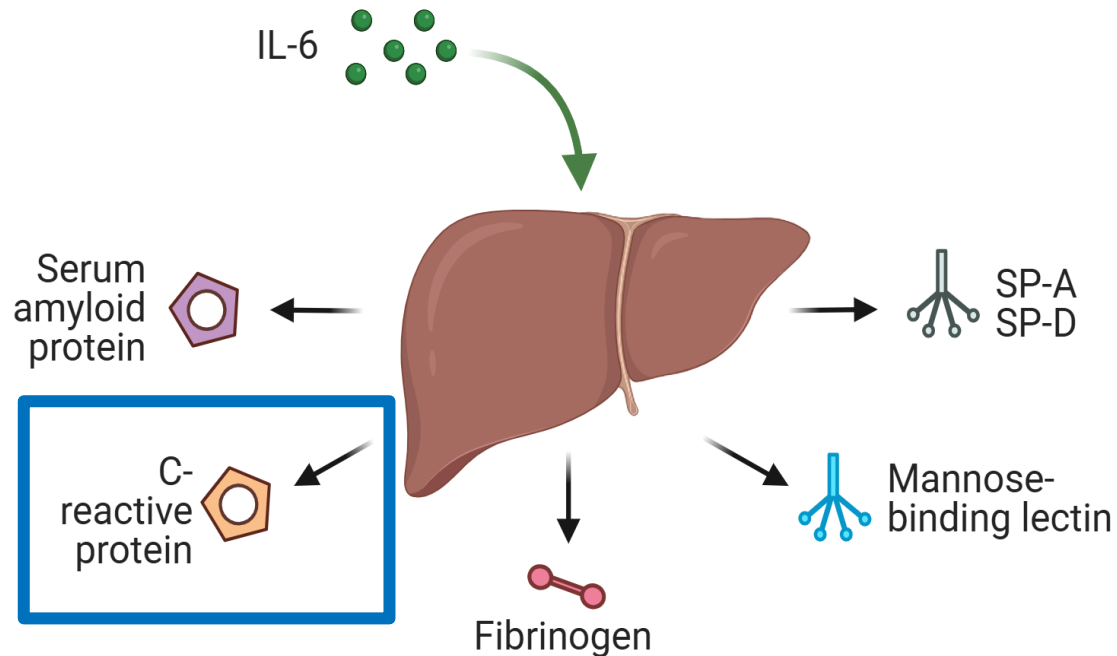
- Direct measure of acute phase response
- Can correspond with disease activity or severity
- Can correspond with risk of disease progression

Diagnosis and/or monitoring:

- Infectious conditions
- Post-acute infection
- Autoimmune conditions
- Atopy conditions
- Cancer conditions
- Cardiovascular conditions
- Intestinal inflammatory conditions
- Inflammatory arthritis
- Metabolic dysfunction

Acute Phase Reactants: C-Reactive Protein

Bacteria induce macrophage to produce IL-6, which acts on hepatocytes to produce acute-phase proteins



Limitations

- Does not increase with all types of inflammation
- May have normal levels with:
 - Active inflammatory disease, cancer, SLE
- May be blunted with liver disease and protein deficiency

Long Description: Acute Phase Reactant: C-Reactive Protein

- Bacteria induce macrophages to produce I L 6, which acts on hepatocytes to produce acute-phase proteins including serum amyloid protein, C reactive protein, fibrinogen, mannose-binding lectin, S P A, and S P D.

References

CRP

- Lapić I, Padoan A, Bozzato D, Plebani M. Erythrocyte sedimentation rate and c-reactive protein in acute inflammation. *Am J Clin Pathol*. 2020;153(1):14-29. doi:10.1093/ajcp/aqz142
- Litao MK, Kamat D. Erythrocyte sedimentation rate and C-reactive protein: how best to use them in clinical practice. *Pediatr Ann*. 2014;43(10):417-420. doi:10.3928/00904481-20140924-10
- **Cancer-Specific:** Shrotriya S, Walsh D, Nowacki AS, et al. Serum C-reactive protein is an important and powerful prognostic biomarker in most adult solid tumors. *PLoS One*. 2018;13(8):e0202555. doi:10.1371/journal.pone.0202555
- Milazzo D, Biasucci LM, Luciani N, et al. Elevated levels of C-reactive protein before coronary artery bypass grafting predict recurrence of ischemic events. *Am J Cardiol*. 1999;84(4):459-A9. doi:10.1016/s0002-9149(99)00333-1
- Muhammad IF, Borné Y, Hedblad B, Nilsson PM, Persson M, Engström G. Acute-phase proteins and incidence of diabetes: a population-based cohort study. *Acta Diabetol*. 2016;53(6):981-989. doi:10.1007/s00592-016-0903-8
- Sabatine MS, Morrow DA, Jablonski KA, et al. Prognostic significance of the Centers for Disease Control/American Heart Association high-sensitivity C-reactive protein cut points for cardiovascular and other outcomes in patients with stable coronary artery disease. *Circulation*. 2007;27;115(12):1528-1536. doi:10.1161/circulationaha.106.649939
- Zebrack JS, Muhlestein JB, Horne BD, Anderson JL; Intermountain Heart Collaboration Study Group. C-reactive protein and angiographic coronary artery disease: independent and additive predictors of risk in subjects with angina. *J Am Coll Cardiol*. 2002;39(4):632-637. doi:10.1016/s0735-1097(01)01804-6

Examples of Associated Inflammatory Disease States

C-Reactive Protein

- Cardiovascular diseases
 - Atherosclerotic disease
 - Congestive heart failure
 - Atrial fibrillation
 - Myocarditis
 - Aortic value disease
- Type 2 diabetes mellitus
- Age-related macular degeneration
- Hemorrhagic stroke
- Alzheimer's disease
- Parkinson's disease
- Systemic lupus erythematosus
- Patients receiving hormone replacement therapy
- Rheumatoid arthritis
- Elevated BMI
- Post-acute infection sequelae
- Cholecystitis

Clinical Pearl: hs-CRP

- Assessment of **risk in cardiovascular or metabolic disease** with other compounding risk factors present
- **Assess low-grade inflammation in person without overt symptoms such as inflammatory arthritis**
- If result comes back above **>3-10 mg/L**, consider doing regular CRP at that time (depending on type of analyzer used – indeterminate at higher levels)

Who Should Be Tested?

- Patients with stable coronary artery disease
- Stable and unstable angina
- Post-CABG surgery
- AFib
- Persons with T2DM

What Decisions Does This Biomarker Help With?

- Statin therapy
- Dietary recommendations
- Hormone replacement therapy (BHRT)
- Continued investigation into ATMs

References

CLINICAL PEARL: HS-CRP

- Kott KA, Bishop M, Yang CHJ, et al. Biomarker development in cardiology: reviewing the past to inform the future. *Cells*. 2022;11(3):588. doi:10.3390/cells11030588

AFib:

- Machado AM, Ferraz AS, Pereira MG, Leite F. Beyond the beats: a systematic review of the underlying inflammatory pathways between atrial fibrillation and cognitive decline. *Neurol Sci*. Published online February 20, 2025. doi:10.1007/s10072-025-08040-x

Steatotic Liver Disease:

- Liang B, Qiu X, Huang J, et al. Nonlinear associations of the hs-CRP/HDL-C index with metabolic dysfunction-associated steatotic liver disease and advanced liver fibrosis in US adults: insights from NHANES 2017-2018. *Sci Rep*. 2025;15(1):4029. doi:10.1038/s41598-025-88685-y

T2DM/Diet Recommendations:

- Ghule A, Kamble TK, Talwar D, et al. Association of serum high sensitivity C-reactive protein with pre-diabetes in rural population: a two-year cross-sectional study. *Cureus*. 2021;13(10):e19088. doi:10.7759/cureus.19088

Hormone Therapy:

- Sproston NR, Ashworth JJ. Role of C-reactive protein at sites of inflammation and infection. *Front Immunol*. 2018;9:754. doi:10.3389/fimmu.2018.00754

Clinical Pearl: CRP vs ESR

	CRP	ESR
Specimen Requirement & Storage	Serum or plasma, can be stored for ≤7 days	Requires fresh, whole blood and must be tested day of collection
Measurement Methods	Direct measurement of the acute phase response	Indirect measurement of the rise of fibrinogen
Rate of Change	Acute	Prolonged/delayed
Physiological Variables	• Does not increase with age or gender • Slight increase with pregnancy	• May increase with age • Decrease with RBC disorders • Impacted by environmental factors • For example: • Salicylates may decrease ESR • Smoking may increase ESR

References

C-REACTIVE PROTEIN/ESR CLINICAL PEARLS

Value Range, Age-Related Changes:

- Litao MK, Kamat D. Erythrocyte sedimentation rate and C-reactive protein: how best to use them in clinical practice. *Pediatr Ann.* 2014;43(10):417-420. doi:10.3928/00904481-20140924-10
- Farrell SF, Armfield NR, Cabot PJ, et al. C-reactive protein (CRP) is associated with chronic pain independently of biopsychosocial factors. *J Pain.* 2024;25(2):476-496. doi:10.1016/j.jpain.2023.09.008
- Tishkowski K, Zubair M. *Erythrocyte Sedimentation Rate*. StatPearls Publishing; 2023. Updated April 23, 2023. Accessed May 29, 2025. <https://www.ncbi.nlm.nih.gov/books/NBK557485/>
- Singh B, Goyal A, Patel BC. *C-Reactive Protein: Clinical Relevance and Interpretation*. StatPearls Publishing; 2025. Updated May 3, 2025. Accessed May 29, 2025. <https://www.ncbi.nlm.nih.gov/books/NBK441843/>
- Lapić I, Padoan A, Bozzato D, Plebani M. Erythrocyte sedimentation rate and C-reactive protein in acute inflammation. *Am J Clin Pathol.* 2020;153(1):14-29. doi:10.1093/ajcp/aqz142

Sepsis:

- Bray C, Bell LN, Liang H, et al. Erythrocyte sedimentation rate and C-reactive protein measurements and their relevance in clinical medicine. *WMJ.* 2016;115(6):317-321.

Systemic Inflammation & Prognostic Value:

- Shrotriya S, Walsh D, Nowacki AS, et al. Serum C-reactive protein is an important and powerful prognostic biomarker in most adult solid tumors. *PLoS One.* 2018;13(8):e0202555. doi:10.1371/journal.pone.0202555

Correlations With Loneliness and Immune Function

C-Reactive Protein

Higher CRP levels were observed in study participants who reported that they were lonely on UCLA Loneliness scales.

- A 2018 observational study (n=927) indicated that those who reported that they were lonely had **significantly increased inflammatory markers**, including IL-6, fibrinogen, and CRP.
- In a 2021 observational study reviewing the effects of the COVID-19 pandemic in Japan (n=624), it was indicated that there were correlations with immune function and increased loneliness. Those who reported being **more isolated and lonelier** had higher CRP levels. This was especially true for men.



Laboratory Clinical Findings of Chronic Inflammation

Lab	Associated Clinical Indicators/ATMs	Clinical Pearls
Leukopenia WBC <4,300 cells/mm ³	- Chronic viral infections are the most common cause; HIV, herpes, hepatitis, etc. - Folate, B-12, or copper deficiency	3,000-4,000 cells/mm³, often benign - Downward trend more significant than absolute number – rule out myelodysplastic syndromes
Leukocytosis WBC >11K	- Infection - Inflammation - Allergic reactions - Malignancy: leukemia, myeloproliferative disorders - Obesity - Stressors: trauma, overexertion, seizures, emotional stress Etiology by subclass (obtain differential) - Neutrophilia: acute bacterial infection - Lymphocytosis: acute viral infection - Monocytosis: chronic infection, inflammation - Eosinophilia: allergic, parasitic	Acute phase reactant

Laboratory Clinical Findings of Chronic Inflammation

Lab	Associated Clinical Indicators/ATMs	Clinical Pearls
Neutrophilia	Antecedent: Familial Triggers: Drugs, infection, pregnancy, post-splenectomy Mediators: • Smoking • Obesity • Smoking and obesity • Idiopathic, idiopathic-inflammatory • Myeloproliferative disorders • Lymphoproliferative disease • Multiple causes • Inflammatory conditions • Tumors	Drug-Induced Neutropenia (often administered for chronic inflammatory conditions) • Antibiotics • Anticonvulsants (carbamazepine, valproate) • Anti-thyroid (PTU, methimazole) • Anti-depressants & antipsychotics (clozapine) • Anti-inflammatories • Cardiovascular meds • Cancer chemotherapy: can be long term • Environmental toxins • Heavy metals: arsenic, mercury, gold
Elevated ESR	Diagnosis and/or monitoring • Chronic or persistent infections • Osteomyelitis • Pelvic inflammatory disease • Tuberculosis • Systemic lupus erythematosus • Polymyalgia rheumatica • Temporal arteritis • Rheumatoid arthritis • Inflammatory bowel disease • Multiple myeloma • Hodgkin's lymphoma • Chronic kidney disease	Used to help distinguish between inflammatory vs. allergic conditions

Laboratory Clinical Findings of Chronic Inflammation

Lab	Associated Clinical Indicators/ATMs	Clinical Pearls
Elevated CRP	• Inflammatory bowel disease • Coronary artery disease risk • Metabolic syndrome/T2DM • Myocardial damage after ischemia • Periodontal disease • Obesity • Arthritis (some forms) • Pelvic inflammatory disease • Systemic lupus erythematosus • Infection after surgery • Thyroid disease (hs-CRP)	Limitations • Does not increase with all types of inflammation • May have normal levels with active inflammatory disease, cancer, SLE • May be blunted with liver disease and protein deficiency Correlations With Loneliness and Immune Function • Higher CRP levels have shown to correlate with loneliness and immune function.
Elevated hs-CRP	• Patients with stable coronary artery disease • Stable and unstable angina • Post-CABG surgery • AFib • Persons with T2DM	• Assessment of risk in cardiovascular or metabolic disease with other compounding risk factors present • Assess low-grade inflammation in person without overt symptoms such as inflammatory arthritis • If result comes back above >3-10 mg/L, consider doing regular CRP at that time (depending on type of analyzer used – indeterminate at higher levels) Helps make decisions with: • Statin therapy • Dietary recommendations • Hormone replacement therapy (E2)

References

LABORATORY CLINICAL FINDINGS OF CHRONIC INFLAMMATION

- Ferri FF. *Ferri's Best Test: A Practical Guide to Clinical Laboratory Medicine and Diagnostic Imaging*. 1st ed. Mosby; 2004: Section II:103-190.
- Anitha V, Nair S, Shivakumar V, Shanmugam M, Priya BM, Rajesh P. Estimation of high sensitivity C-reactive protein in patients with periodontal disease and without coronary artery disease. *Indian J Dent Res*. 2015;26(5):500-503. doi:10.4103/0970-9290.172048
- Emerging Risk Factors Collaboration, Kaptoge S, Di Angelantonio E, et al. C-reactive protein concentration and risk of coronary heart disease, stroke, and mortality: an individual participant meta-analysis. *Lancet*. 2010;375(9709):132-140. doi:10.1016/S0140-6736(09)61717-7
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- Ridker PM. A test in context: high-sensitivity C-reactive protein. *J Am Coll Cardiol*. 2016;67(6):712-723. doi:10.1016/j.jacc.2015.11.037
- Roy S, Banerjee U, Dasgupta A. Protective role of bilirubin against increase in hsCRP in different stages of hypothyroidism. *Indian J Clin Biochem*. 2016;31(1):43-49. doi:10.1007/s12291-015-0495-z

Defense and Repair: Part 2

Immune Regulation of Inflammation

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

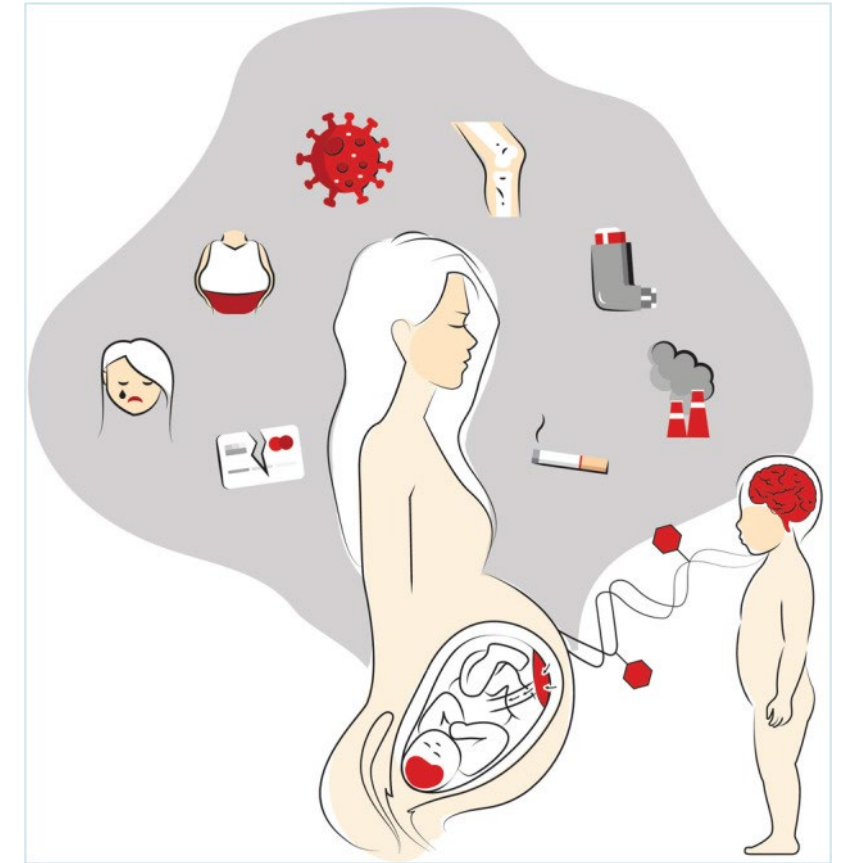
- Antecedents, triggers, and mediators of chronic inflammation.



Maternal Exposome

Antecedents of Chronic Inflammation

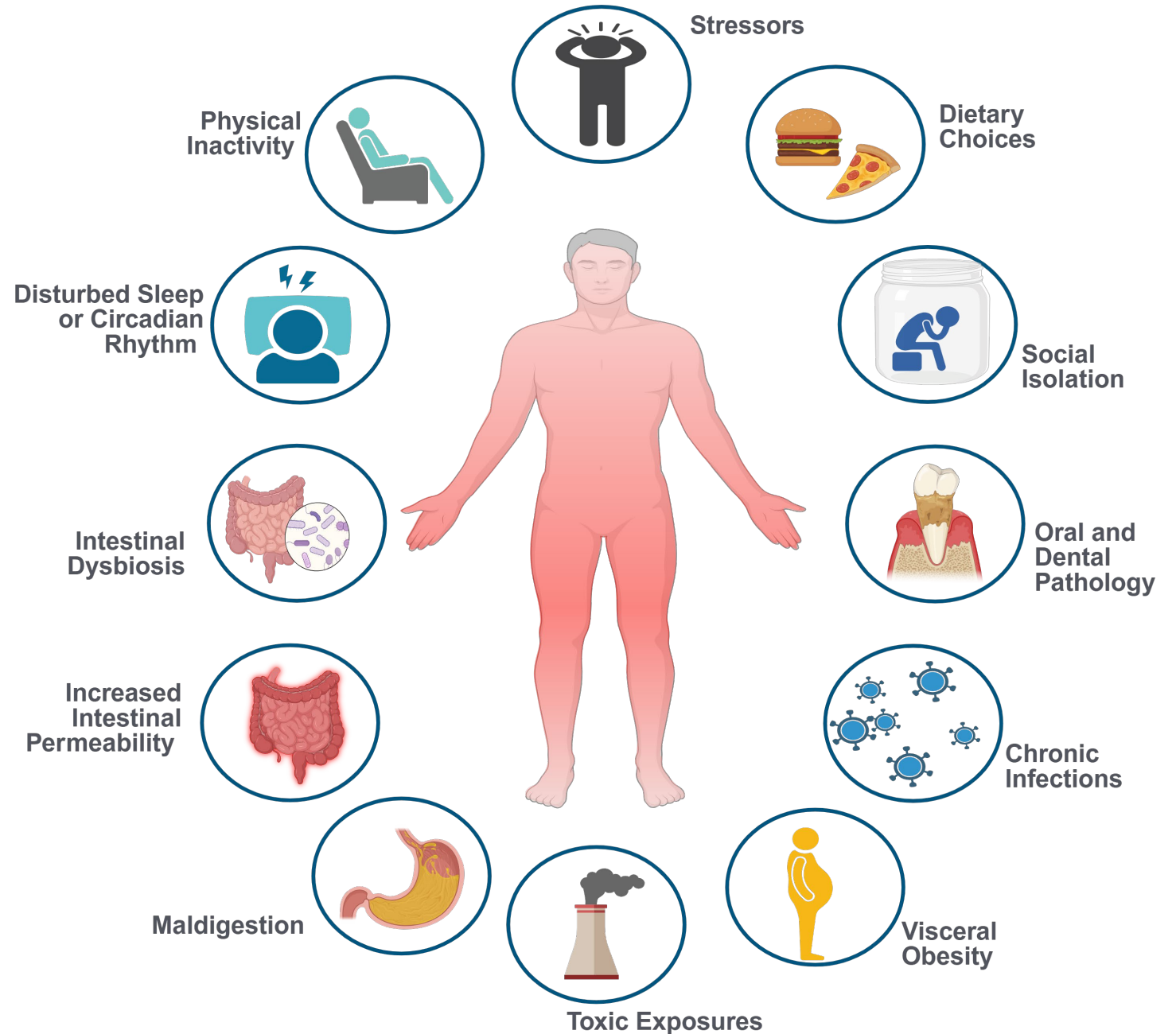
- Diet
- Physical activity
- Psychological stress
- Xenobiotic exposure (e.g., pollutants, smoking)
- SDOH (e.g., food accessibility, housing and transportation security, psychological stress, pollution)



The immune system is programmed during the prenatal period. Epigenetic changes due to antecedents (i.e., maternal exposome) create the potential for intergenerational transmission of risk of chronic inflammation systemically.

- Furman D, Campisi J, Verdin E, et al. Chronic inflammation in the etiology of disease across the life span. *Nat Med*. 2019;25(12):1822-1832. doi:10.1038/s41591-019-0675-0
- Han VX, Patel S, Jones HF, et al. Maternal acute and chronic inflammation in pregnancy is associated with common neurodevelopmental disorders: a systematic review. *Transl Psychiatry*. 2021;11(1):71. doi:10.1038/s41398-021-01198-w
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Most Common Triggers & Mediators of Chronic Inflammation

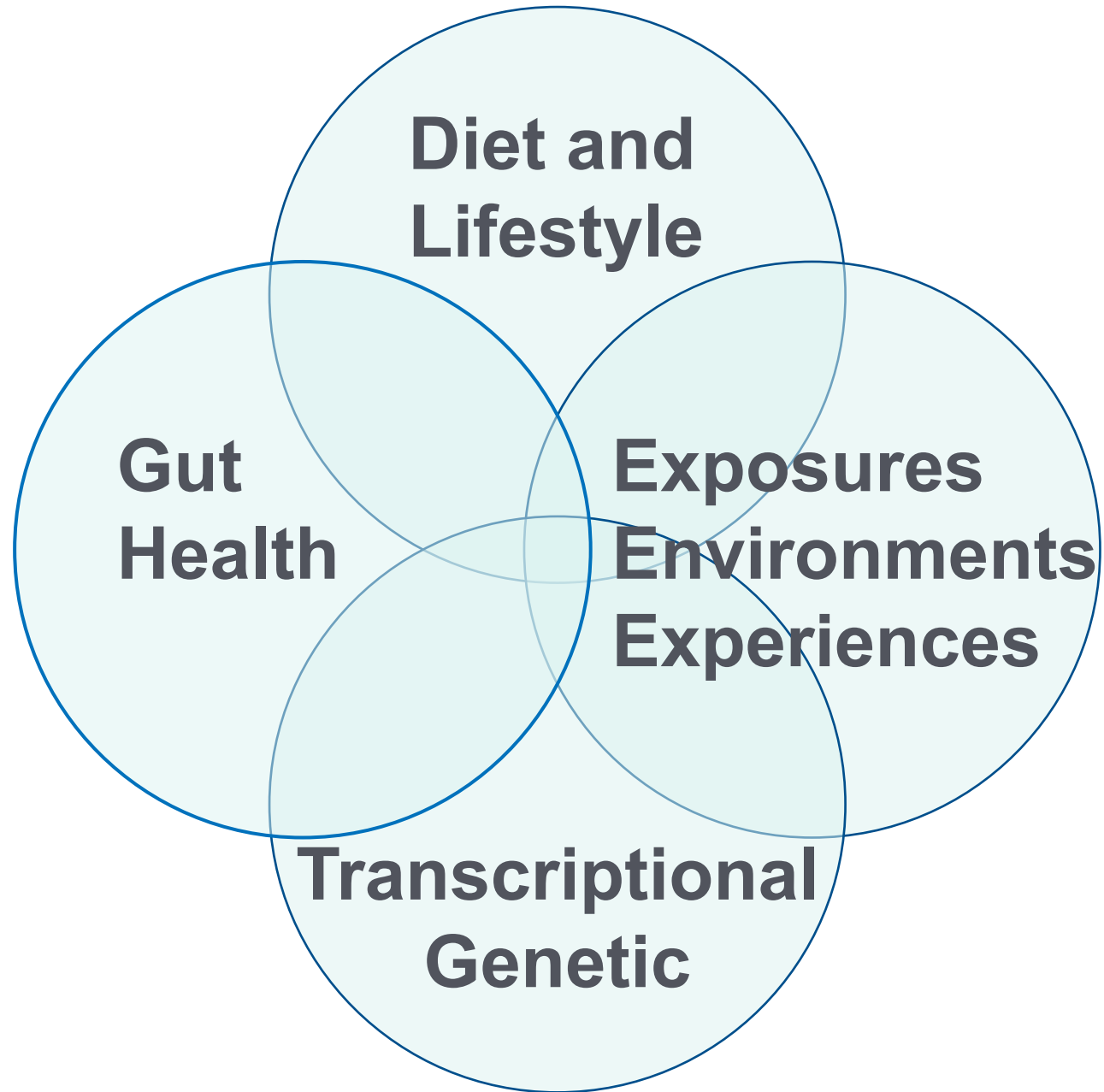


Long Description:

Most Common Triggers and Mediators of Chronic Inflammation

- Chronic infections
- Physical inactivity
- Visceral obesity
- Intestinal dysbiosis, increased intestinal permeability, maldigestion
- Dietary choices
- Social isolation
- Psychological stress
- Disturbed sleep or circadian rhythm
- Oral and dental pathology
- Exposure to xenobiotics, toxins, recreational substances

Most Common Antecedents, Triggers, and Mediators of Chronic Inflammation



Long Description: The Most Common A T Ms of Chronic Inflammation

- Overlapping circles representing the A T M's of chronic inflammation including diet and lifestyle, exposures, environments, and experiences, transcriptional, genetic, and gut health.

ATMs of Inflammation: Diet and Lifestyle

Antecedents/Mediators

The Patient's Story

Antecedents (Inherited & Acquired)

- Prenatal history/exposure
- Adverse childhood events (ACEs)
- Strong family history (autoimmune or CVD)

Triggers & Precipitating Events

- Infection
- Injury
- Toxin exposure
- Traumatic life event

Mediators/Perpetuators

- Chronic stress
- Gut dysfunction
- Environmental exposures
- Obesity

Antecedents, Triggers, Mediators

Modifiable Lifestyle Factors

Modifiable Lifestyle Factors

Sleep

- Sleep dysfunction/disorders
- Chronic sleep deprivation
- Sleep apnea

Physical Activity

- Sedentary lifestyle
- Exercise intolerance
- Excessive exercise

Nutrition

- Ultra-processed foods
- Low protein
- Nutrient devoid
- Calorie excess

Stressors

- Chronic stress
- Post-traumatic stress
- Moral injury
- Existential crisis

Relationships

- Social isolation
- Low power/autonomy
- Marital strain
- Social rejection

Long Description: A T Ms of Inflammation

- The functional medicine matrix highlighting the antecedents, triggers, mediators, and modifiable lifestyle factors involved in inflammation.

References

ATMS OF INFLAMMATION: DIET AND LIFESTYLE

Sleep:

- Irwin MR, Opp MR. Sleep health: reciprocal regulation of sleep and innate immunity. *Neuropsychopharmacology*. 2017;42(1):129-155. doi:10.1038/npp.2016.148
- Irwin MR, Olmstead R, Carroll JE. Sleep disturbance, sleep duration, and inflammation: a systematic review and meta-analysis of cohort studies and experimental sleep deprivation. *Biol Psychiatry*. 2016;80(1):40-52. doi:10.1016/j.biopsych.2015.05.014

Physical Activity:

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- Jin CH, Rhyu HS, Kim JY. The effects of combined aerobic and resistance training on inflammatory markers in obese men. *J Exerc Rehabil*. 2018;14(4):660-665. doi:10.12965/jer.1836294.147

Nutrition:

- Narula N, Wong ECL, Dehghan M, et al. Association of ultra-processed food intake with risk of inflammatory bowel disease: prospective cohort study. *BMJ*. 2021;374:n1554. doi:10.1136/bmj.n1554
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Stressors:

- Johnson TV, Abbasi A, Master VA. Systematic review of the evidence of a relationship between chronic psychosocial stress and C-reactive protein. *Mol Diagn Ther*. 2013;17(3):147-164. doi:10.1007/s40291-013-0026-7

References

ATMS OF INFLAMMATION: DIET AND LIFESTYLE

Relationships:

- Wilson SJ, Andridge R, Peng J, Bailey BE, Malarkey WB, Kiecolt-Glaser JK. Thoughts after marital conflict and punch biopsy wounds: age-graded pathways to healing. *Psychoneuroendocrinology*. 2017;85:6-13. doi:10.1016/j.psyneuen.2017.07.489
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- Cole SW, Levine ME, Arevalo JM, Ma J, Weir DR, Crimmins EM. Loneliness, eudaimonia, and the human conserved transcriptional response to adversity. *Psychoneuroendocrinology*. 2015;62:11-17. doi:10.1016/j.psyneuen.2015.07.001

ATMs of Inflammation: Gut Health

Antecedents/Mediators

D

Maldigestion and Malabsorption

- Deficiencies of phytonutrients, minerals, antioxidants, etc.

I

Intestinal Permeability

- LPS and other toxin/toxicant seepage/exposure

G

Gut Microbiome: Dysbiosis and Imbalances

- Immunomodulation and gut/brain effects

I

Intestinal Inflammation (& Oral/Dental Sources; Periodontitis)

- Root cause of systemic inflammation

N

Enteric Nervous System Dysfunction

- Correlated with stress, dysbiosis, and gut/brain dysfunction

References

ATMS OF INFLAMMATION: GUT HEALTH

Maldigestion and Malabsorption

- Kurjatko A, Galet C, Weigel I, Suarez J, Wibbenmeyer L. Trace element supplementation in burn patients: exploring the relationship between burn size and mineral needs. *J Burn Care Res.* 2025;46(2):411-418. doi:10.1093/jbcr/irae175
- Razzaque MS, Wimalawansa SJ. Minerals and human health: from deficiency to toxicity. *Nutrients.* 2025;17(3):454. doi:10.3390/nu17030454

Intestinal Hyperpermeability

- López-Moreno J, García-Carpintero S, Jimenez-Lucena R, et al. Effect of dietary lipids on endotoxemia influences postprandial inflammatory response. *J Agric Food Chem.* 2017;65(35):7756-7763. doi:10.1021/acs.jafc.7b01909
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Gut Microbiome

- Afzaal M, Saeed F, Shah YA, et al. Human gut microbiota in health and disease: unveiling the relationship. *Front Microbiol.* 2022;13:999001. doi:10.3389/fmicb.2022.999001
- Wei P, Keller C, Li L. Neuropeptides in gut-brain axis and their influence on host immunity and stress. *Comput Struct Biotechnol J.* 2020;18:843-851. doi:10.1016/j.csbj.2020.02.018

Intestinal Inflammation

- Majster M, Almer S, Malmqvist S, Johannsen A, Lira-Junior R, Boström EA. Salivary calprotectin and neutrophils in inflammatory bowel disease in relation to oral diseases. *Oral Dis.* 2025;31(1):286-297. doi:10.1111/odi.15036

Enteric Nervous System

- Sharkey KA, Mawe GM. The enteric nervous system. *Physiol Rev.* 2023;103(2):1487-1564. doi:10.1152/physrev.00018.2022
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- Ge L, Liu S, Li S, et al. Psychological stress in inflammatory bowel disease: psychoneuroimmunological insights into bidirectional gut-brain communications. *Front Immunol.* 2022;13:1016578. doi:10.3389/fimmu.2022.1016578

ATMs of Inflammation: Exposures & Experiences

Antecedents/Triggers/Mediators



Adverse Childhood Events

- Abuse, neglect, bullying, traumas, etc.

Social Determinants of Health

- Community, financial, educational, social, & healthcare quality

Social Needs of Health

- Housing, transportation, nutrition, and healthcare insecurity

Precipitating Life Events

- “My health was never the same since...”

Toxins & Trauma; Infections & Infestations; Antigens

- Mold, Lyme, dental, occupational, environmental, dietary, etc.

References

ATMS OF INFLAMMATION: EXPOSURES & EXPERIENCES

ACEs:

- John-Henderson NA, Kampf TD, Alvarado G, Counts CJ, Larsen JM, Palmer CA. Childhood adversity is associated with heightened inflammation after sleep loss. *Sleep Health*. 2022;8(3):283-287. doi:10.1016/j.sleh.2022.03.006
- Zagaria A, Fiori V, Vacca M, Lombardo C, Pariante CM, Ballesio A. Inflammation as a mediator between adverse childhood experiences and adult depression: a meta-analytic structural equation model. *J Affect Disord*. 2024;357:85-96. doi:10.1016/j.jad.2024.04.072

SDOH:

- Powell-Wiley TM, Baumer Y, Baah FO, et al. Social determinants of cardiovascular disease. *Circ Res*. 2022;130(5):782-799. doi:10.1161/CIRCRESAHA.121.319811

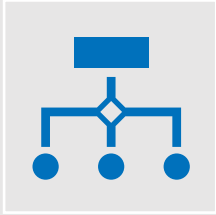
Social Needs (Food Insecurity Specifically):

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Toxins and Trauma:

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- Vineis P, Barouki R. The exposome as the science of social-to-biological transitions. *Environ Int*. 2022;165:107312. doi:10.1016/j.envint.2022.107312

ATMs of Inflammation: Genetic/Transcriptional Antecedents/Mediators



Family History of Inflammatory Conditions

Cardiovascular/metabolic, autoimmunity, diabetes, cancer, etc.



Single Nucleotide Polymorphisms and Polygenic Effects

Multiple examples



Epigenetic/Transcriptional

Diet/nutrients, esp. phytochemicals; stress reduction, exercise

- David T, Ling SF, Barton A. Genetics of immune-mediated inflammatory diseases. *Clin Exp Immunol*. 2018;193(1):3-12. doi:10.1111/cei.13101
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Other Common ATMs of Chronic Inflammation



Medications



Trauma



Toxins



Infections

References

OTHER COMMON ATMS OF CHRONIC INFLAMMATION

Trauma/Stress:

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Medications:

- Muñoz-Cano R, Pascal M, Bartra J, et al. Distinct transcriptome profiles differentiate nonsteroidal anti-inflammatory drug-dependent from nonsteroidal anti-inflammatory drug-independent food-induced anaphylaxis. *J Allergy Clin Immunol*. 2016;137(1):137-146. doi:10.1016/j.jaci.2015.05.042
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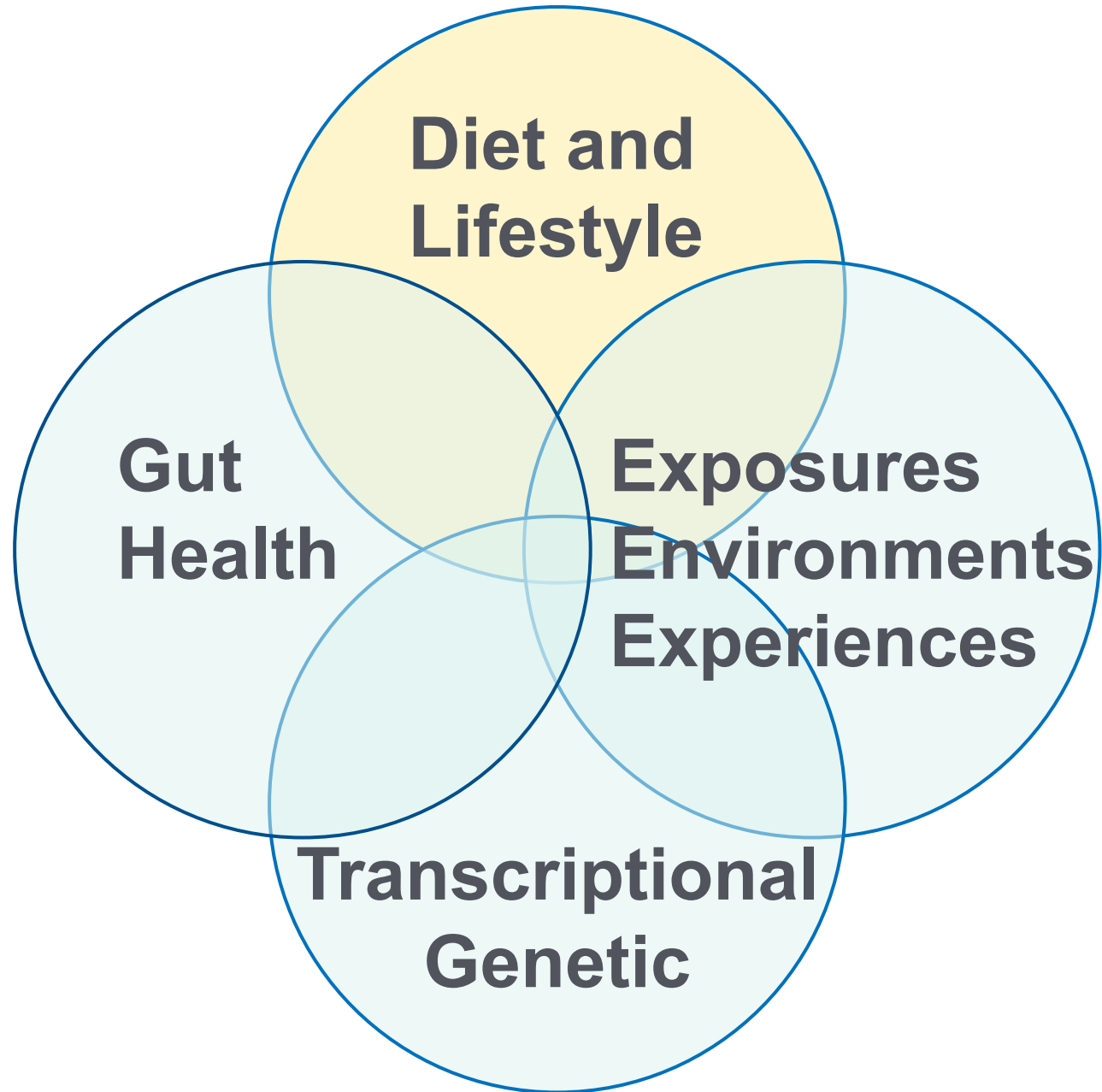
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Clinical Focus

- Pathophysiological mechanisms of dietary triggers and mediators of chronic inflammation.



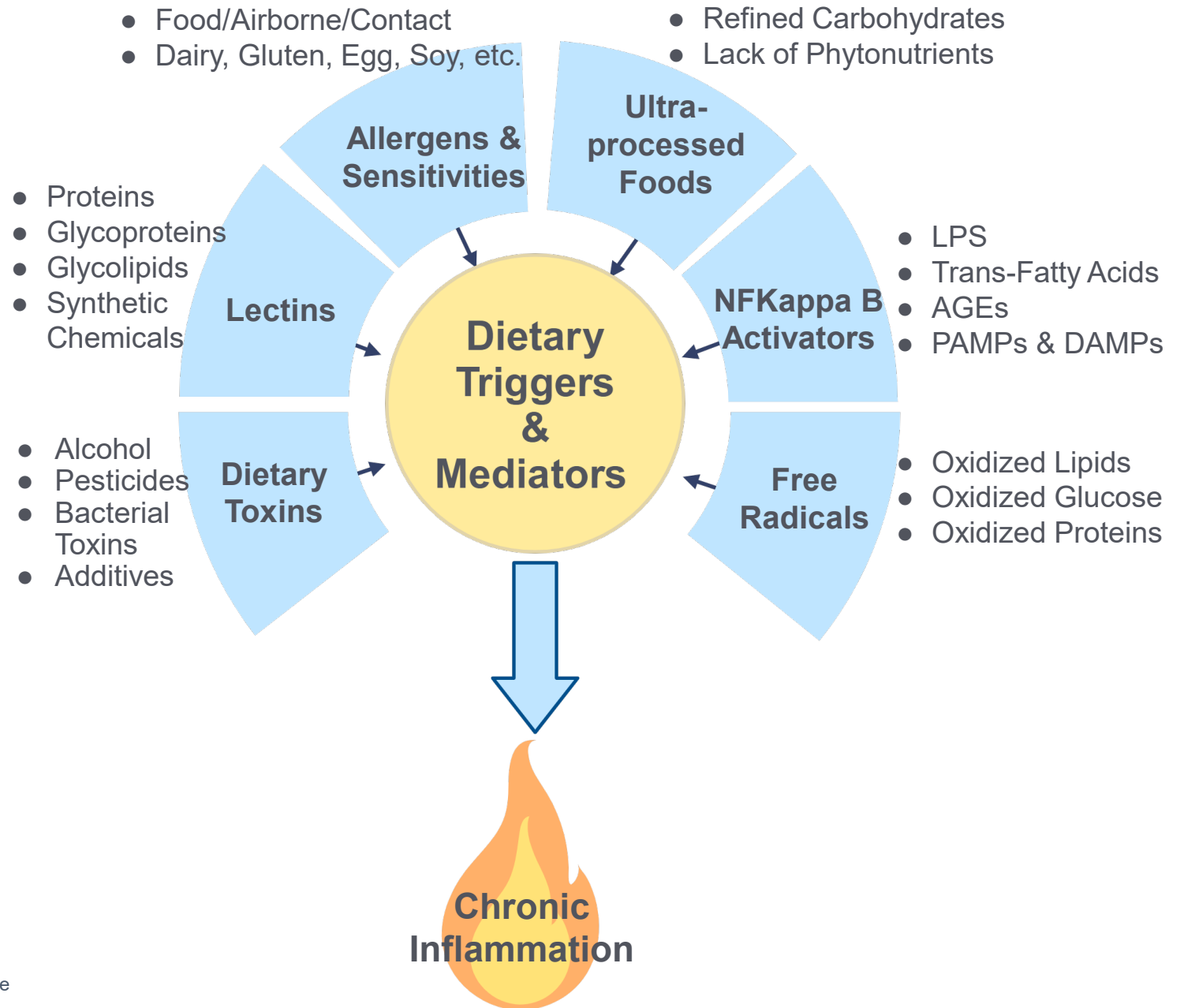
Most Common Antecedents, Triggers, and Mediators of Chronic Inflammation



Long Description: Most Common A T Ms of Chronic Inflammation

- Overlapping circles representing the A T M's of chronic inflammation with a focus on diet and lifestyle.

Chronic Inflammation: Dietary Triggers and Mediators



Long Description: Chronic Inflammation: Dietary Triggers and Mediators

- Dietary toxins
 - Alcohol
 - Pesticides
 - Bacterial toxins
 - Additives
- Lectins
 - Proteins
 - Glycoproteins
 - Glycolipids
 - Synthetic chemicals
- Allergens and sensitivities
 - Food, airborne, contact
 - Dairy, gluten, egg, soy, etc.
- Ultra-processed foods
 - Refined carbohydrates
 - Lack of phytonutrients
- N F kappa B activators
 - L P S
 - Trans-fatty acids
 - A G E's
 - PAMPs and DAMPs
- Free radicals
 - Oxidized lipids
 - Oxidized glucose
 - Oxidized proteins

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DIETARY TRIGGERS AND MEDIATORS OF CHRONIC INFLAMMATION

Trauma:

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Considering Meal Order

- Isocaloric meal after a 12-hour fast in adults with type 2 DM (metformin controlled)
 - Week 1: Carbohydrate 15 minutes prior to protein and vegetable salad
 - Week 2: Reversed order



- Conclusions: Post-meal glucose levels significantly decreased by 28.6%, 36.7%, and 16.8% at 30, 60, 120 minutes pp, respectively.
- Additionally, when the protein and vegetables were consumed first, the incremental area under the curve was reduced by 73% compared to when the carbohydrates were consumed first.

Proinflammatory Effect of Mixed Meal

Representative Example From the Medical Literature

- A fasting blood sample was obtained, and the subjects were asked to eat a mixed meal containing 910 kcal (egg-muffin and sausage-muffin sandwiches and two hash browns).
 - 81g carbohydrate, 51g fat, and 32g protein) over 15 minutes.
 - “...a moderate-sized mixed meal results in a significant increase in the proinflammatory transcription factor NF- κ B in the nucleus, a decrease in I κ B α and a parallel increase in the cellular p47phox subunit, which is the key protein component of NADPH oxidase and in ROS generation.”

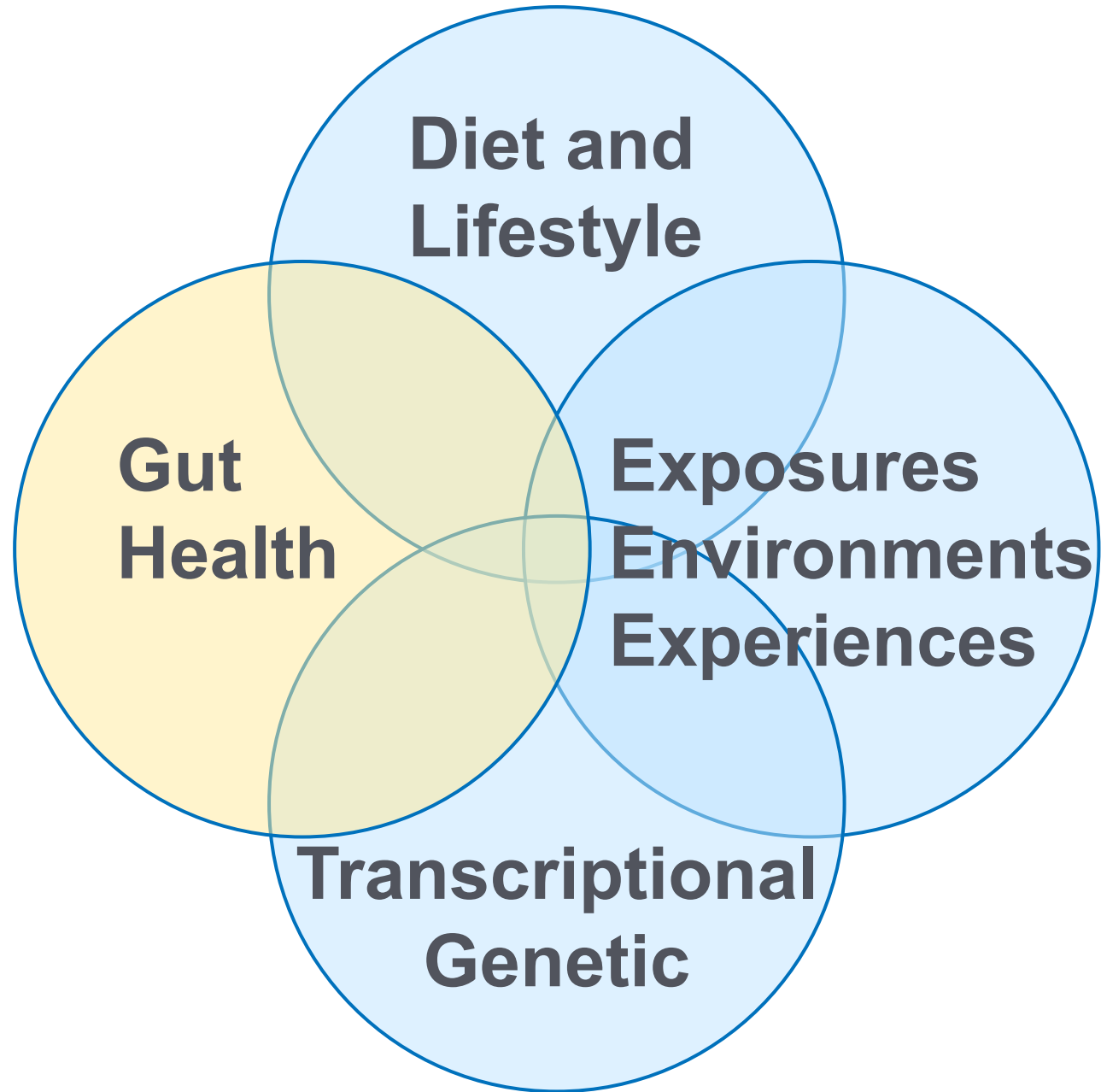


Clinical Focus

- Pathophysiological mechanisms of gut health triggers and mediators of chronic inflammation.



Most Common Antecedents, Triggers, and Mediators of Chronic Inflammation

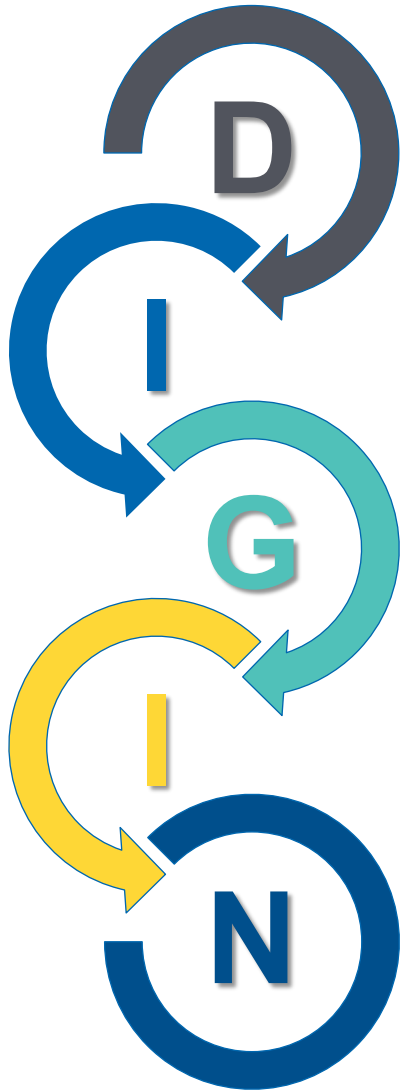


Long Description: The Most Common A T Ms of Chronic Inflammation

- Overlapping circles representing the A T M's of chronic inflammation with a focus on gut health.

The DIGIN Gut Dysfunctions

Mediators of Chronic Inflammation



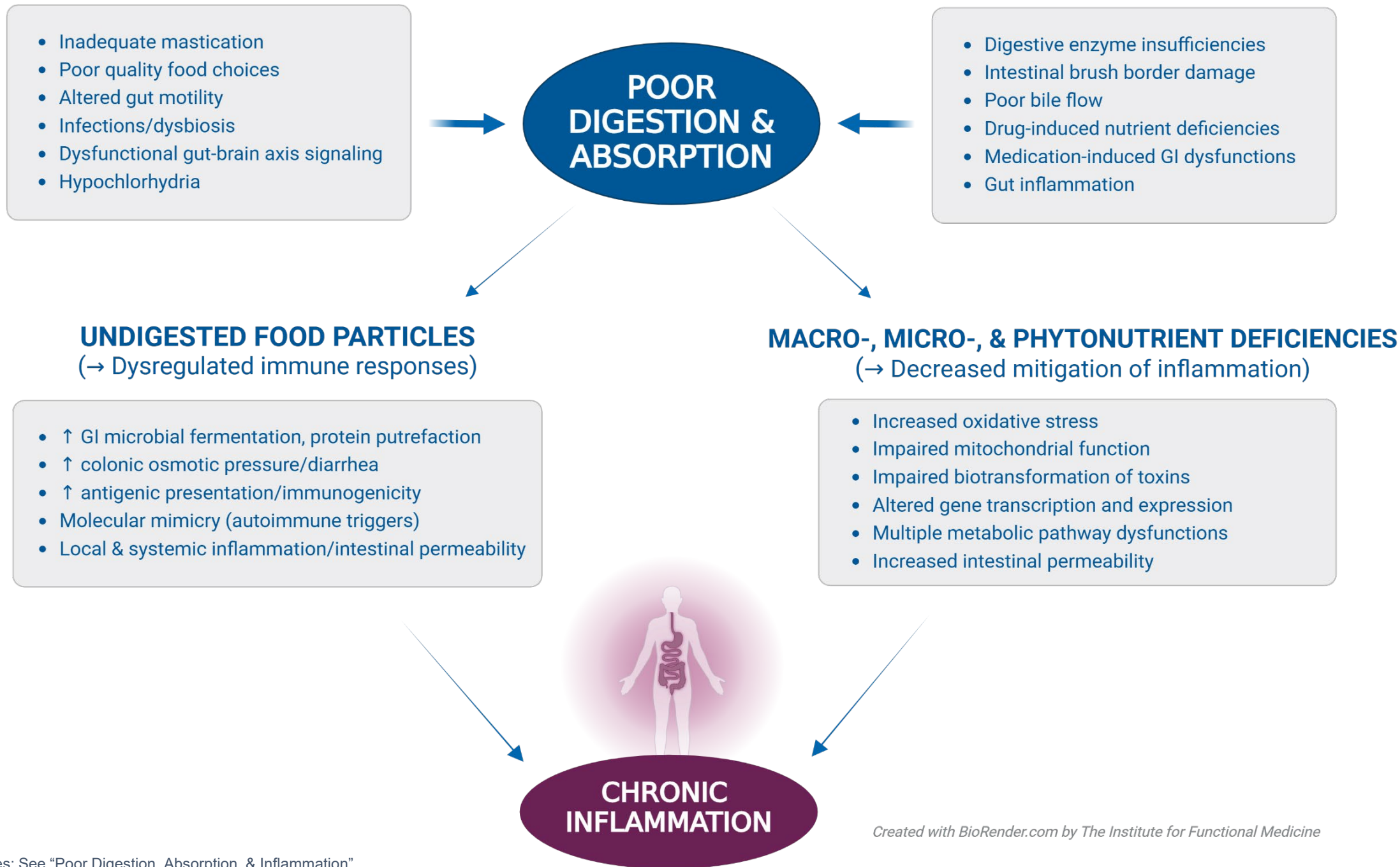
Digestion/Absorption

Intestinal Permeability

Gut Microbiota/Dysbiosis

Inflammation/Immune

Nervous System



Long Description: Chronic Inflammation and Poor Digestion and Absorption

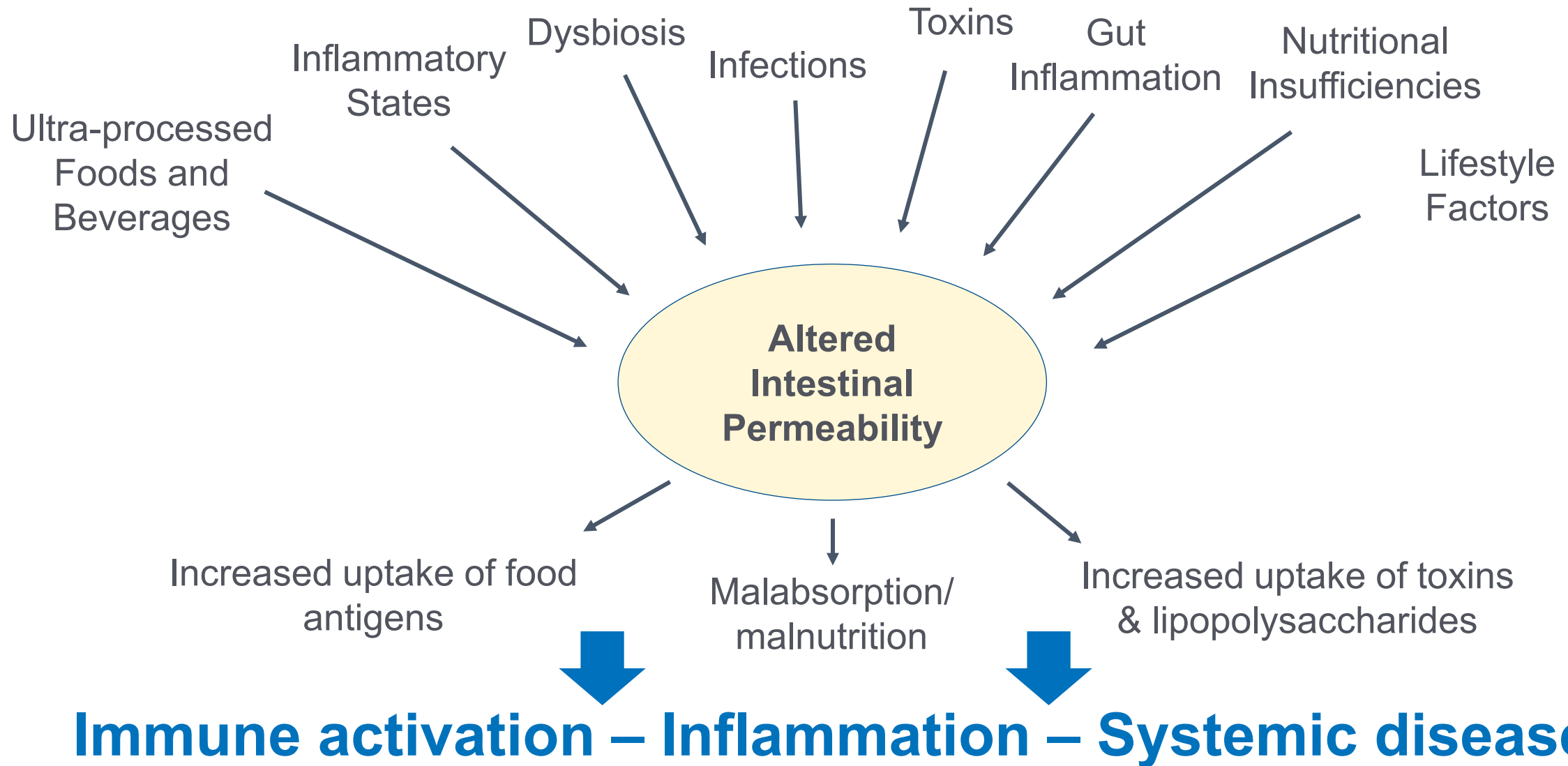
- Flow path demonstrating how poor digestion and absorption lead to chronic inflammation. Factors leading to poor digestion and absorption include inadequate mastication, poor quality food choices, altered gut motility, infections, dysbiosis, dysfunctional gut-brain axis signaling, hypochlorhydria, digestive enzyme insufficiencies, intestinal brush border damage, poor bile flow, drug-induced nutrient deficiencies, medication-induced G I dysfunctions, and gut inflammation. The resulting undigested food particles lead to dysregulated immune responses through increased G I microbial fermentation, protein putrefaction, increased colonic osmotic pressure, diarrhea, increased antigenic presentation/immunogenicity, molecular mimicry, and local and systemic inflammation and intestinal permeability. Additionally, deficiencies in macro, micro, and phytonutrients result in decreased mitigation of inflammation through increased oxidative stress, impaired mitochondrial function, impaired biotransformation of toxins, altered gene transcription and expression, multiple metabolic pathway dysfunctions, and increased intestinal permeability.

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POOR DIGESTION, ABSORPTION, & INFLAMMATION

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ATMs of Increased Intestinal Permeability



Long Description: A T Ms of Increased Intestinal Permeability

- Factors leading to altered intestinal permeability
 - Ultra-processed foods and beverages
 - Inflammatory states
 - Dysbiosis
 - Infections
 - Toxins
 - Gut inflammation
 - Nutritional insufficiencies
 - Lifestyle factors
- Consequences of altered intestinal permeability
 - Increased uptake of food antigens
 - Malabsorption and malnutrition
 - Increased uptake of toxins and L P S
 - Immune activation, inflammation, and systemic disease

References

ATMS OF INCREASED INTESTINAL PERMEABILITY

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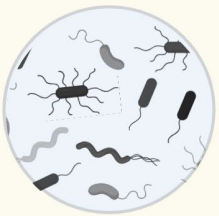
References

ATMS OF INCREASED INTESTINAL PERMEABILITY

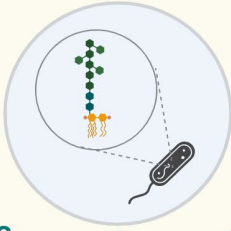
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NSAIDs, dysbiosis, enterotoxins, alcohol, gliadin, chronic systemic inflammation, Western diet (high fat, low fiber), stress, enteric pathogens.

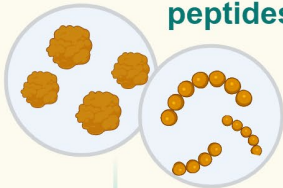
Disruption of epithelial barrier, increased permeability



Microbes/
pathogens



Bacterial
endotoxins
(LPS)



Antigenic
proteins &
peptides



Other antigens:
polysaccharides,
lipids, nucleic
acids

Gut lumen

Tight junctions

Intestinal
epithelium

Paracellular
transport

Transcellular
transport

Some ions, small
compounds, nutrients,
microbes, & other
macromolecules enter
via transcellular
mechanisms

Capillary

Immune Dysregulation and
Chronic Inflammation

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Long Description: Immune Dysregulation and Chronic Inflammation

- Non-steroidal ant-inflammatory drugs, dysbiosis, enterotoxins, alcohol, gliadin, chronic systemic inflammation, Western diet characterized by high fat and low fiber, stress, and enteric pathogens lead to disruption of the epithelial barrier and increased permeability, allowing pathogens, antigenic proteins, LPS and other antigens and to enter through paracellular transport, and other compounds entering via transcellular transport, ultimately leading to immune dysregulation and chronic inflammation.

References

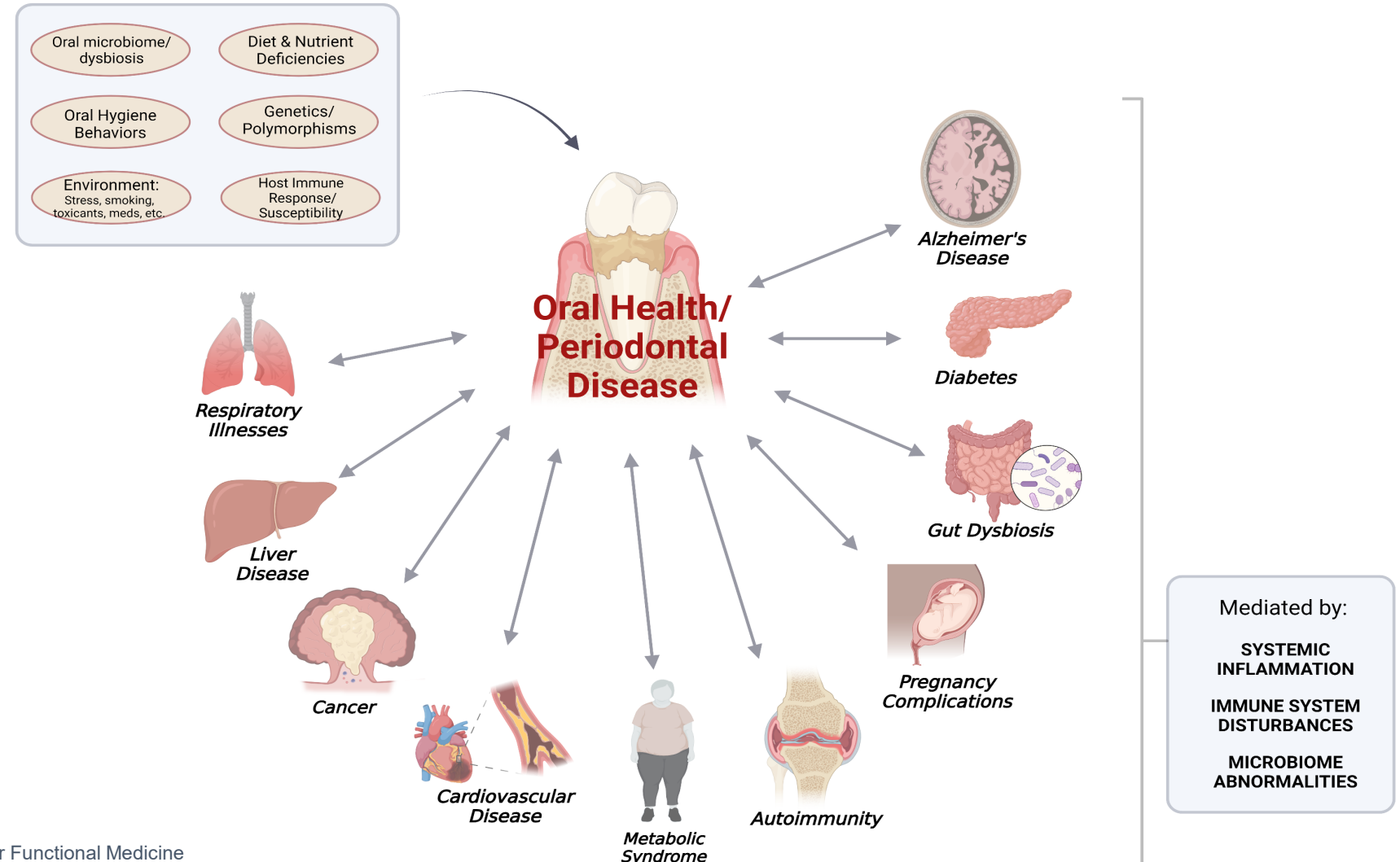
IMMUNE DYSREGULATION AND CHRONIC INFLAMMATION

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Local Inflammation to Systemic Inflammation

Example: Oral Health

- **Localized inflammation** frequently results in a **systemic inflammatory response**.
- Similarly, a **local loss of tolerance** can lead to **systemic loss of tolerance**.



Long Description: Systemic Inflammation and Oral Health

- Contributors to oral health and periodontal disease
 - Oral microbiome and dysbiosis
 - Oral hygiene behaviors
 - Environment: stress, smoking, toxicants, medications, etc.
 - Diet and nutrient deficiencies
 - Genetics and polymorphisms
 - Host immune response and susceptibility
- Conditions and disease states in bidirectional relationship with oral health mediated by systemic inflammation, immune system disturbances, and microbiome abnormalities
 - Respiratory illnesses
 - Liver disease
 - Cancer
 - Cardiovascular disease
 - Metabolic syndrome
 - Autoimmunity
 - Pregnancy complications
 - Gut dysbiosis
 - Diabetes
 - Alzheimer's disease

H. Pylori and Autoimmune Conditions

A Trigger and a Mediator

Research has linked ***Helicobacter pylori*** with autoimmune diseases across multiple systems, highlighting its broader systematic impact.

Mechanisms of Action:

- Molecular Mimicry
- Chronic Immune Activation and Inflammation
- T-cell Differentiation
- Disruption of Epithelial Barriers

Autoimmune Conditions That Have Been Linked to *H. pylori*:

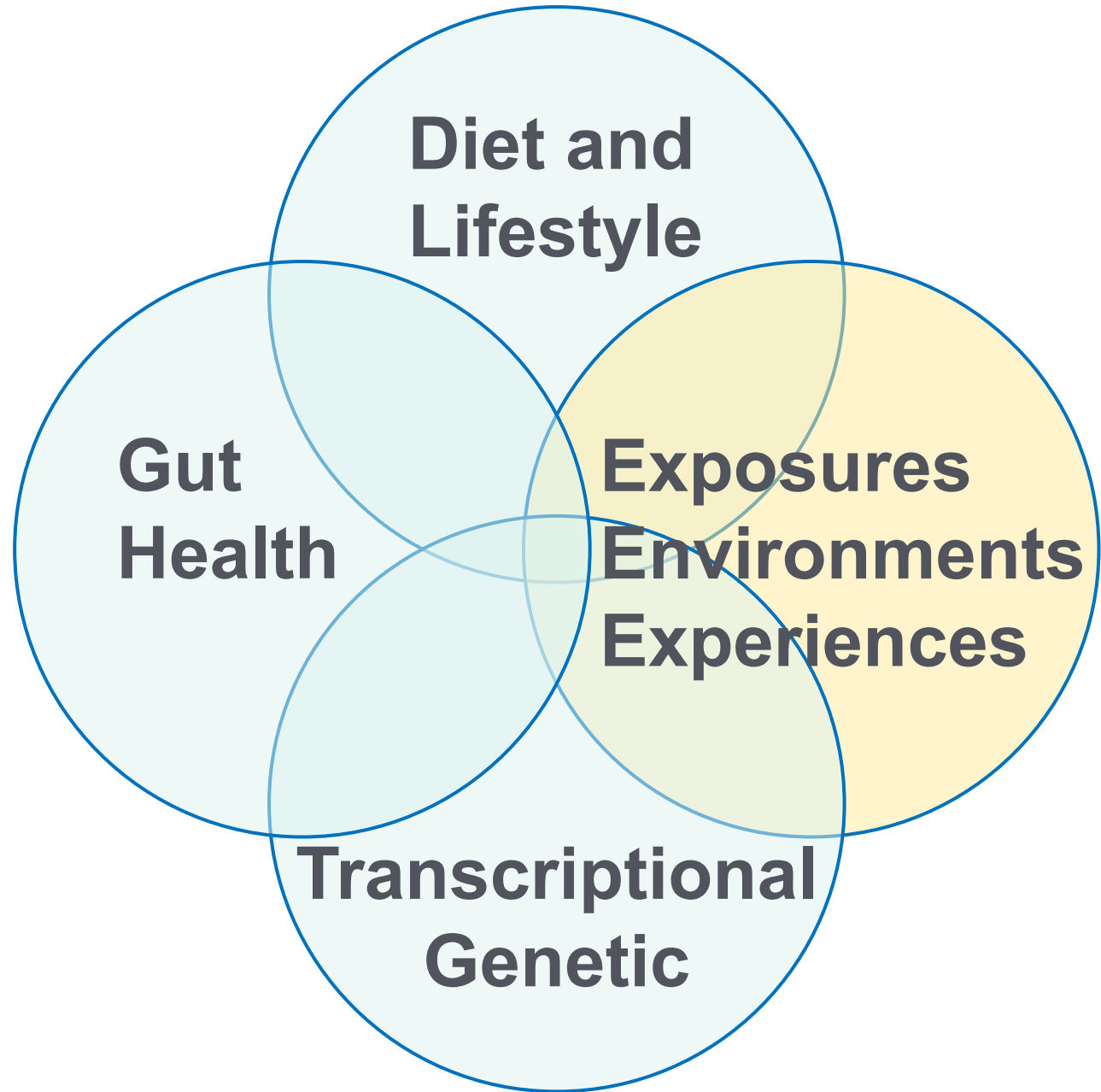
- Inflammatory Bowel Disease
- Autoimmune Thyroid Disease
- Type 1 Diabetes Mellitus
- Autoimmune Liver Diseases
- Primary Biliary Cirrhosis
- Primary Sclerosing Cholangitis
- Autoimmune Hepatitis
- Rheumatoid Arthritis
- Immune Thrombocytopenic
- Systemic Lupus Erythematosus
- Sjögren's Syndrome
- Autoimmune Dermatoses

Clinical Focus

- Pathophysiological mechanisms of chronic inflammation as mediated by experiences, exposures, and environments.



Most Common Antecedents, Triggers, and Mediators of Chronic Inflammation



Long Description: The Most Common A T Ms of Chronic Inflammation

- Overlapping circles representing the A T M's of chronic inflammation with a focus on exposures, environments, and experiences.

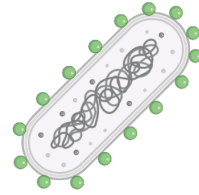
External and Internal Exposures

Mechanisms of Chronic Inflammation

EXTERNAL TRIGGERS OF INFLAMMATION



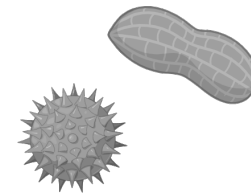
Trauma
(mechanical, chemical, thermal, electromagnetic)



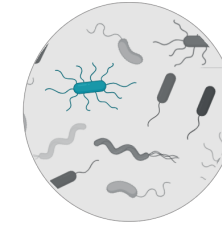
PAMPs
(Pathogen-Associated Molecular Patterns)



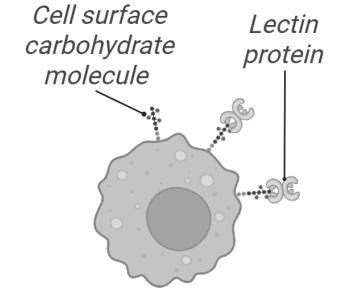
Toxins
(organic chemicals, heavy metals)



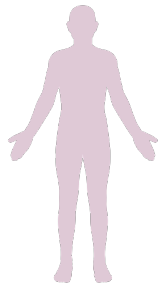
Allergens
(food, airborne, contact)



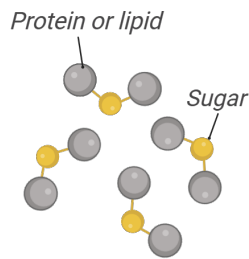
Pathogens
(microbial, multicellular)



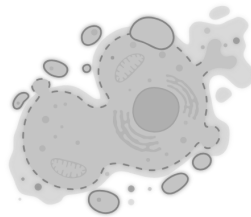
Lectins
(proteins, glycoproteins, glycolipids, synthetic chemicals)



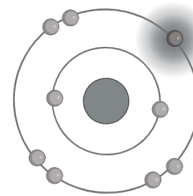
INTERNAL MEDIATORS OF INFLAMMATION



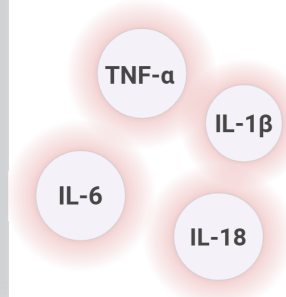
AGEs
(Advanced Glycation End products)



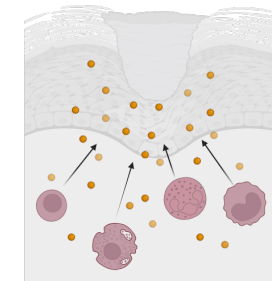
Cellular Debris
(DAMPs: nucleic acids, uric acid, etc.)



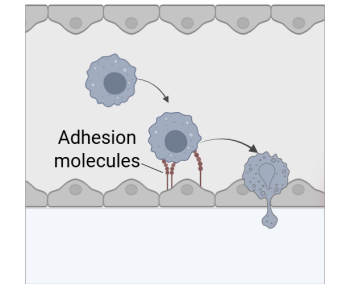
Free Radicals
(oxidized lipids, glucose, proteins)



Inflammatory Cytokines
(TNF-α, IL-1β, IL-6, IL-18)



Proinflammatory Chemokines
(IL-8, IL-7, CCL2-5, CX3CL1, etc.)



Adhesion Molecules
(Cadherins, selectins, integrins, immunoglobulin-like adhesion molecules)

See: "References: Triggers of Inflammation"

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Long Description: External and Internal Exposures: Mechanisms of Chronic Inflammation

- External triggers of inflammation
 - Trauma: mechanical, chemical, thermal, electromagnetic
 - PAMPs (pathogen-associated molecular patterns)
 - Toxins: organic chemicals, heavy metals
 - Allergens: food, airborne, contact
 - Pathogens: microbial, multicellular
 - Lectins: proteins, glycoproteins, glycolipids, synthetic chemicals
- Internal mediators of inflammation
 - AGE's (advanced glycation end products)
 - Cellular debris: DAMPs: nucleic acids, uric acid, etc.
 - Free radicals: oxidized lipids, glucose, proteins
 - Inflammatory cytokines: TNF alpha, IL 1 beta, IL 6, IL 18
 - Proinflammatory chemokines: IL 8, IL 7, CCL 2 5, CX3CL 1, etc.
 - Adhesion molecules: cadherins, selectins, integrins, immunoglobulin-like adhesion molecules

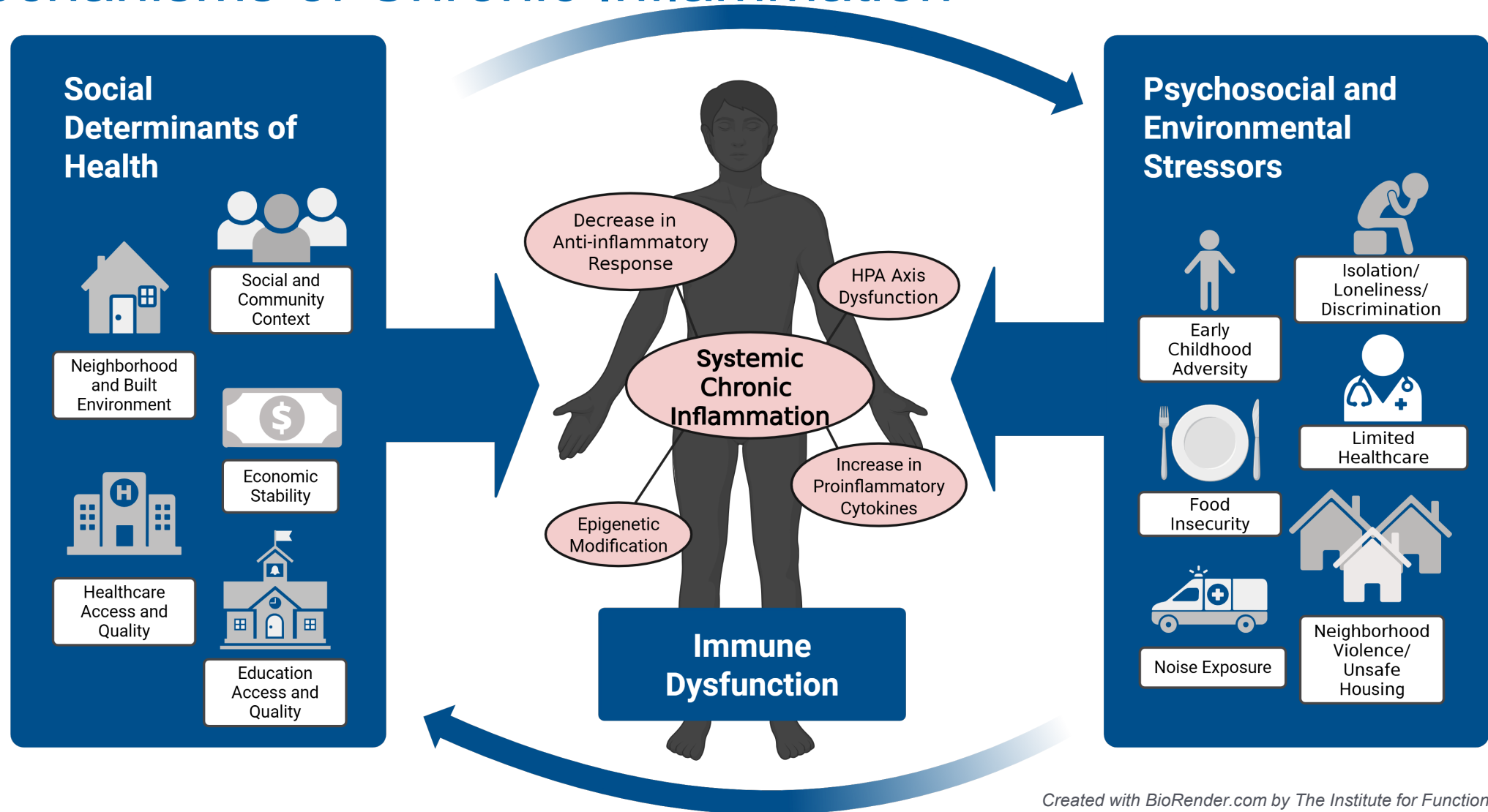
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Life Experiences and Living Environments

Mechanisms of Chronic Inflammation



Long Description: Life Experiences and Living Environments: Mechanisms of Chronic Inflammation

Contributors to immune dysfunction

- Social determinants of health
 - Neighborhood and built environment
 - Social and community context
 - Economic stability
 - Education access and quality
 - Healthcare access and quality
- Psychosocial and environmental stressors
 - Early childhood adversity
 - Isolation, loneliness, discrimination
 - Limited healthcare
 - Food insecurity
 - Noise exposure
 - Neighborhood violence, unsafe housing
- Systemic chronic inflammation
 - Decrease in anti-inflammatory response
 - H P A axis dysfunction
 - Increase in pro-inflammatory cytokines
 - Epigenetic modification

References

IMMUNE DYSFUNCTION AND SOCIAL DETERMINANTS OF HEALTH

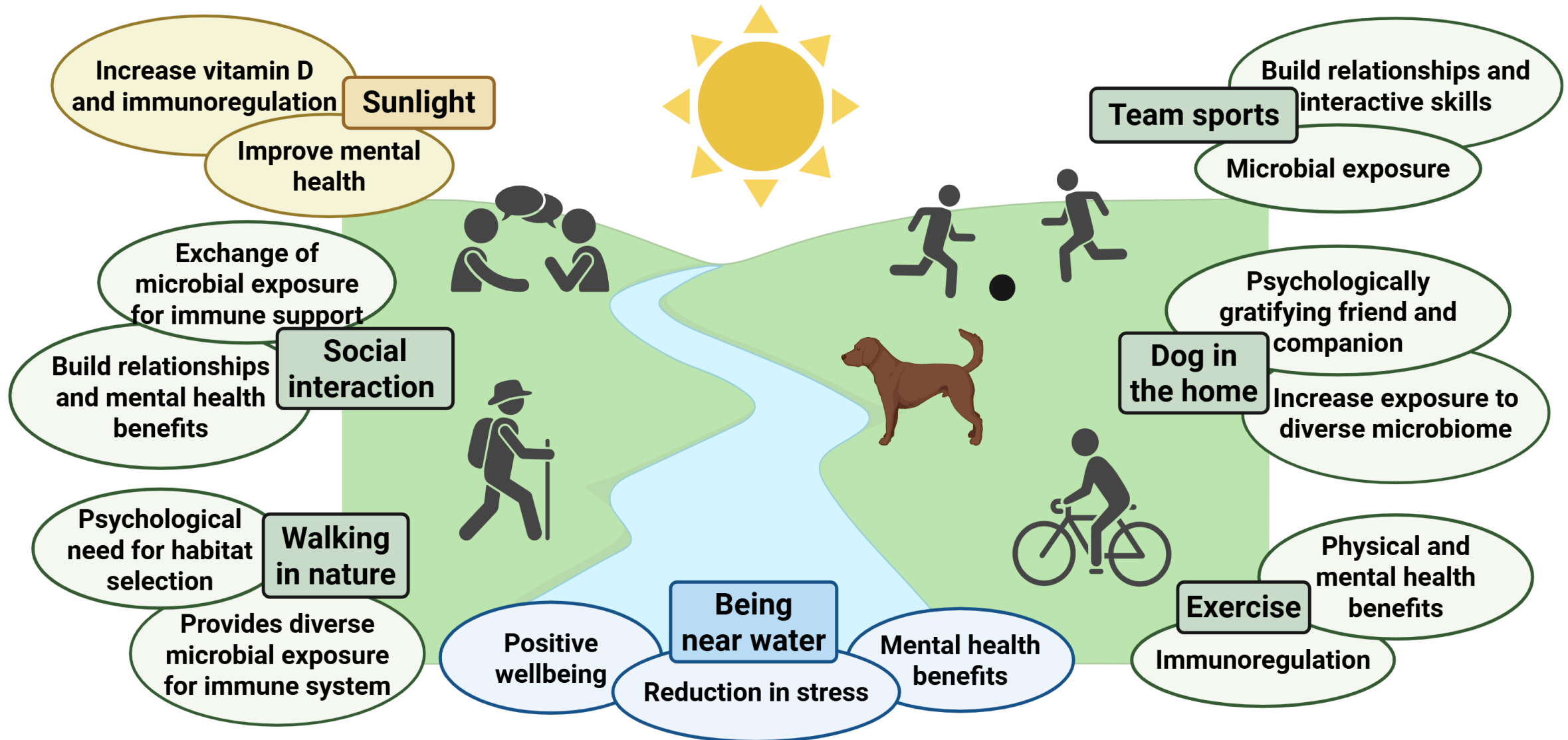
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Physiological and Psychological Effects of Green and Blue Spaces



Long Description:

Physiological and Psychological Effects of Green and Blue Spaces

- Physiological and psychological effects of green and blue spaces
- Sunlight
 - Increase vitamin D and immunoregulation
 - Improve mental health
- Social interaction
 - Exchange of microbial exposure for immune support
 - Build relationships and mental health benefits
- Walking in nature
 - Psychological need for habitat selection
 - Provides diverse microbial exposure for immune system
- Being near water
 - Positive wellbeing
 - Reduction in stress
 - Mental health benefits
- Exercise
 - Physical and mental health benefits
 - Immunoregulation
- Dog in the home
 - Psychologically gratifying friend and companion
 - Increase exposure to diverse microbiome
- Team sports
 - Build relationships and interactive skills
 - Microbial exposure

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Shinrin-Yoku (Forest Bathing)

Representative Example

- Reduces NK Cells
- Reduces IL-6
- Reduces IL-8
- Reduces Interferon- γ
- Reduces TNF- α
- Reduces CRP
- Reduces Salivary Cortisol

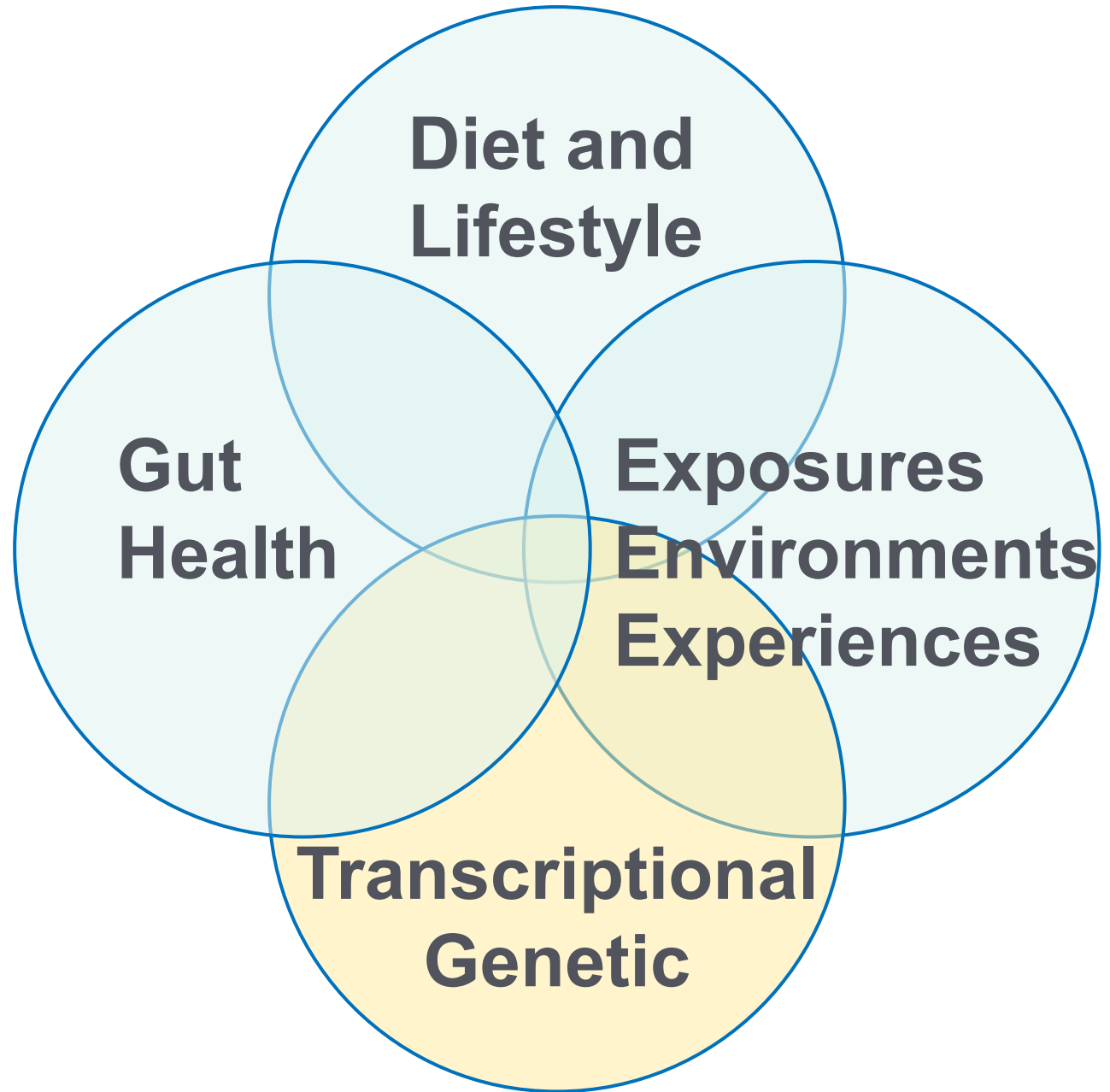


Clinical Focus

- Individual susceptibility to chronic inflammation based on genetic polymorphisms.



Most Common Antecedents, Triggers, and Mediators of Chronic Inflammation



Long Description: The Most Common A T Ms of Chronic Inflammation

- Overlapping circles representing the A T M's of chronic inflammation with a focus on transcriptional genetic factors.

Genes

Polymorphisms cause different individuals to have a wide spectrum of susceptibility to environmental exposures (toxins, stress, diet, etc.)



Function

Shifts physiologic state into “alarm” reaction characterized by inflammatory process



Symptoms of Inflammation

-osis becomes -itis with increasing severity



Genetic Polymorphisms and Increased Susceptibility IBD as Representative Example

A significant number of SNPs have been discovered that are associated with the development of IBD in humans.

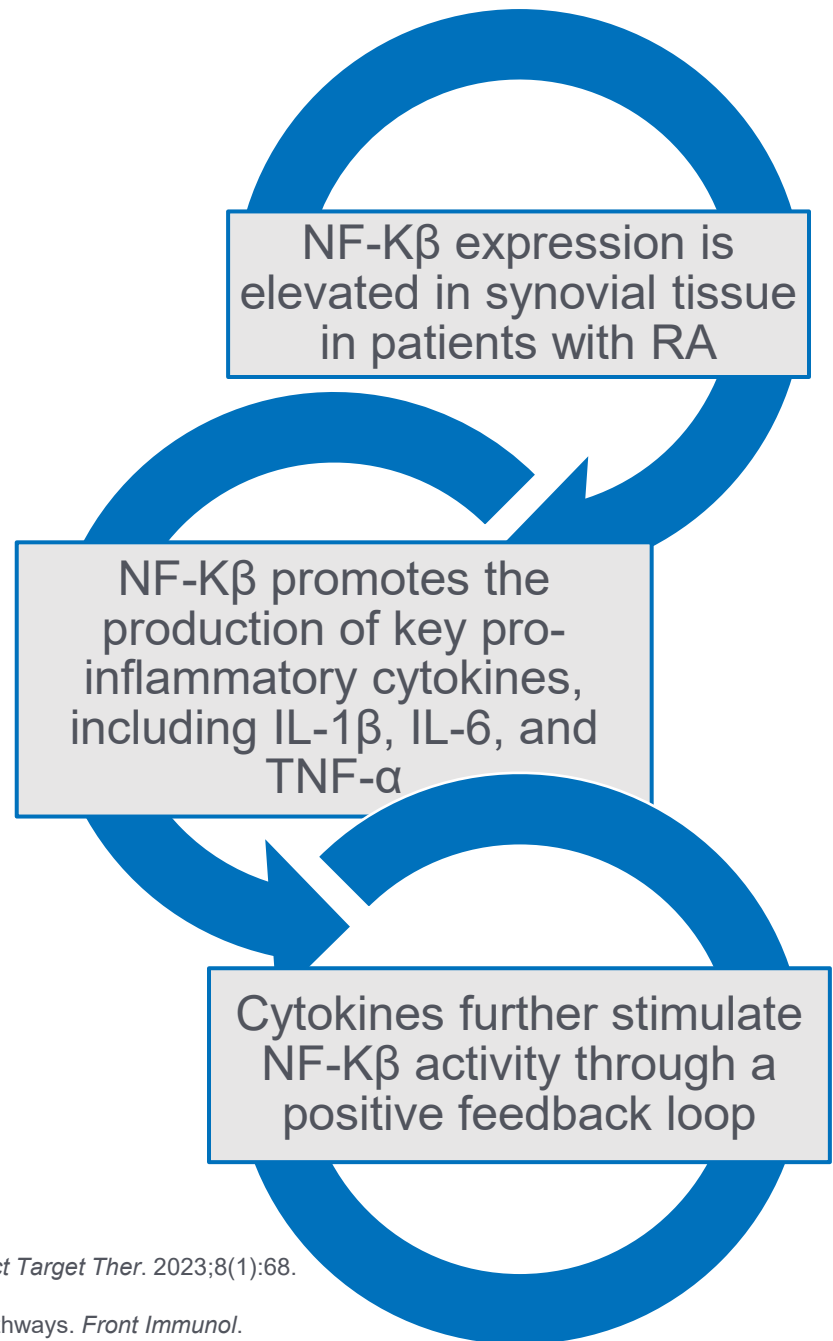
- Genetic SNPs Associated With IBD
- **Crohn's Disease**
 - NOD2, TLR4, IL23, ATG16L1/IRGM
- **Ulcerative Colitis**
 - FCGR2A, ORMDL3, ECM1, 12q15, 1p36, IL2/IL21



Transcription Factors in Rheumatoid Arthritis (RA)

REPRESENTATIVE EXAMPLE

The vicious cycle continues



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Defense and Repair: Part 3

Immune Regulation of Inflammation

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Physiological dysfunctions on other nodes of the matrix that are mediators of chronic inflammation.



Functional Medicine Matrix

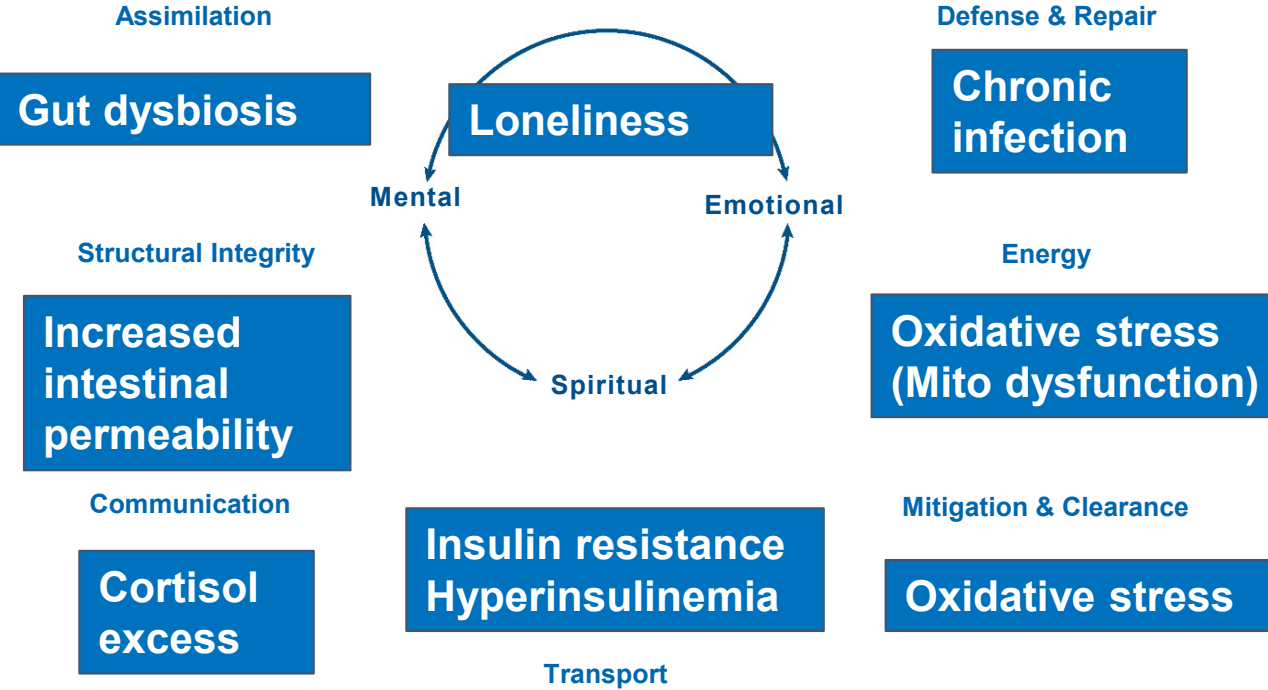
Clinical Imbalances and Physiological Dysfunctions

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators



Matrix
Dysfunctions as
Mediators of
Chronic
Inflammation:
Representative
Examples

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Long Description: Functional Medicine Matrix

- Defense and repair
 - Chronic infection
- Energy
 - Oxidative stress, mitochondrial dysfunction
- Mitigation and clearance
 - Oxidative stress
- Transport
 - Insulin resistance
 - Hyperinsulinemia
- Communication
 - Cortisol excess
- Structural integrity
 - Increased intestinal permeability
- Assimilation
 - Gut dysbiosis
- Mental, emotional, spiritual
 - Loneliness

Clinical Focus

- Therapeutic targets to mitigate chronic inflammation:
- NF- κ B
- Arachidonic acid



Inflammatory Insult

Example: Ultra-Processed Food Components

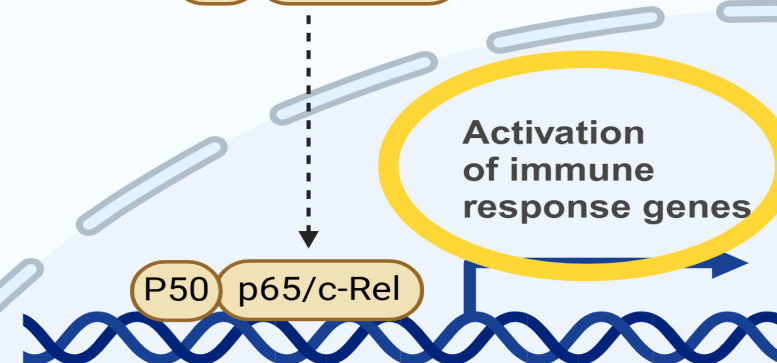
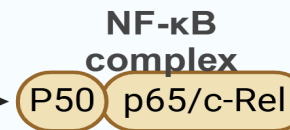
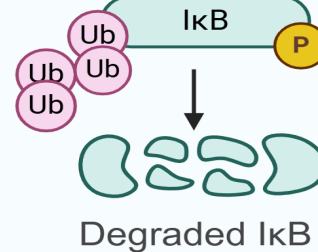
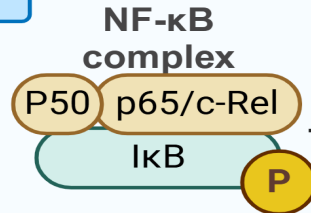


Trigger: Ultra-Processed Food

TLR

Vitamins C & E stabilize cell membranes

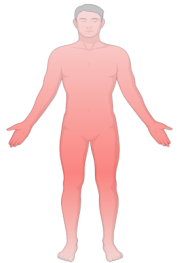
NF- κ B present in cell cytoplasm in inactive state: bound to inhibitory proteins



Activation of immune response genes

Interacts with >400 genes (genes turned on or off); Coordinates further inflammatory & Immune Responses

upregulation of COX & LOX enzymes → INFLAMMATION



Long Description: Inflammatory Insult Example

- The immune system recognizes components of ultra-processed foods as damage-associated molecular patterns, which bind to toll-like receptors. This initiates a signaling cascade that activates N F kappa B, upregulating pro-inflammatory genes and promoting cytokine release, driving further inflammatory and immune responses.

Foundational Support Spices that Target NFK-B

Available in
Your Toolkit

Accessible version of this resource in your toolkit.



Health Benefits of Spices

There are many benefits of eating a diet rich in spices. Spices provide a wide variety of flavor combinations to enhance the taste of healthy foods, and they contain phytonutrients that protect our bodies by reducing harmful amounts of inflammation and oxidative stress. As with all plant foods, eat a variety of spices regularly for the greatest health benefits.

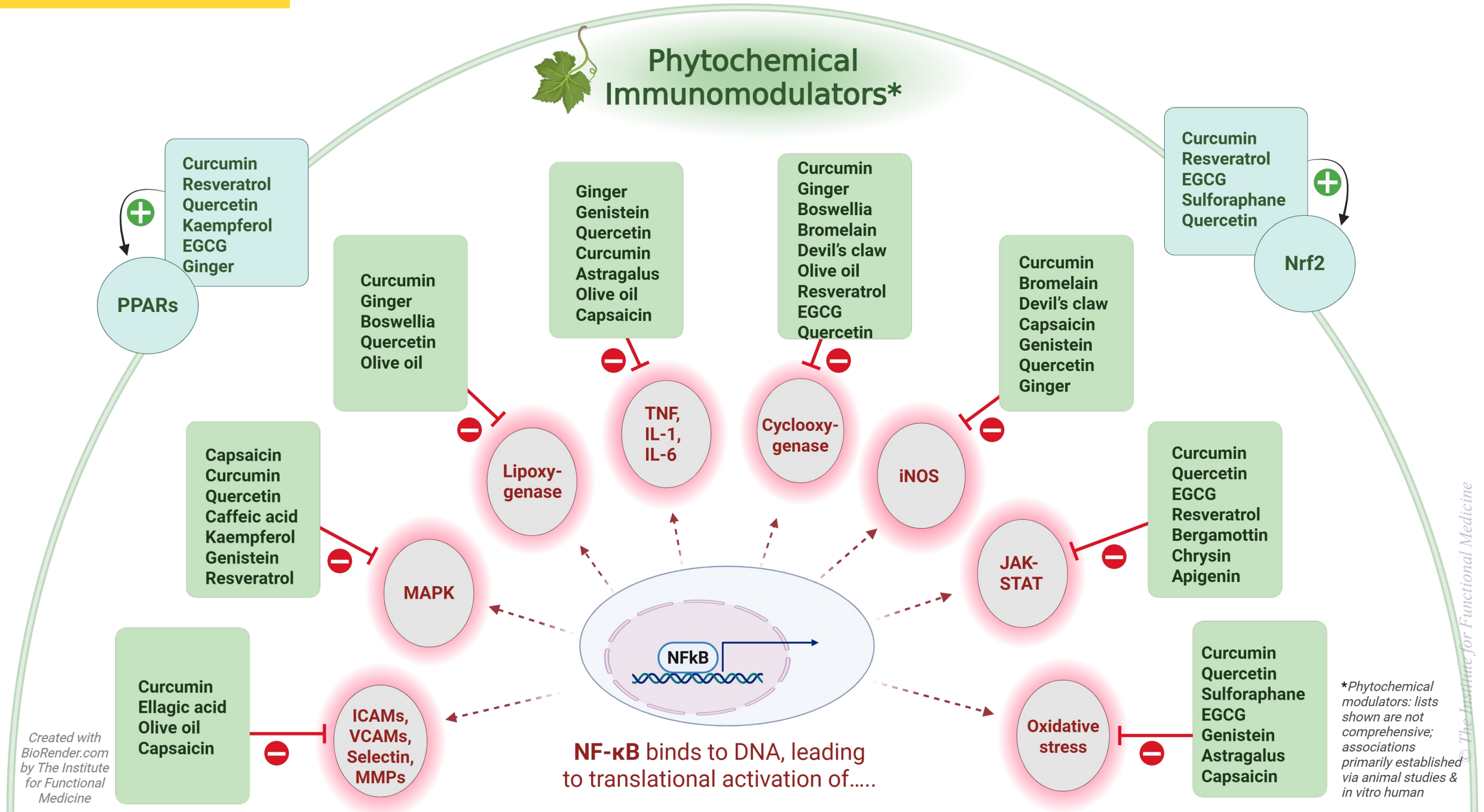
Health Benefits

Eating a diet rich in spices has been associated with improved blood sugar health, blood vessel health, digestive health, a greater sense of fullness from meals, and even a lower risk of death. Below are some of the health benefits shown in research studies.

- **Cardamom:** anti-inflammatory, blood sugar health, cell protection, heart health
- **Cinnamon:** anti-inflammatory, blood sugar health, heart health
- **Cumin:** blood sugar health, cell protection, heart health
- **Fennel:** digestive health, hormonal health
- **Garlic:** anti-inflammatory, blood sugar health, blood vessel health, cell protection, heart health, liver health
- **Ginger:** anti-inflammatory, blood sugar health, cell protection, digestive health, heart health, pain relief
- **Saffron:** blood sugar health, brain health, cell protection, heart health
- **Turmeric:** anti-inflammatory, blood sugar health, brain health, digestive health, heart health

While very high amounts of spices (or isolated chemicals within spices) are typically used in medical research, some studies have shown that even culinary amounts of spices have health benefits.





See: "References: Phytochemical Immunomodulators"

*Phytochemical
modulators: lists
shown are not
comprehensive;
associations
primarily established
via animal studies &
in vitro human

Long Description: Phytochemical Immunomodulators

- N F K B binds to DNA, leading to translational activation of the following molecules, which are then induced or inhibited by the corresponding phytochemical immunomodulators:
- P P A Rs Inducers: Curcumin, Resveratrol, Quercetin, Kaempferol, E G C G, Ginger
- N R F 2 Inducers: Curcumin, Resveratrol, E G C G, Sulforaphane, Quercetin
- Inhibitors of I C A Ms, V C A Ms, Selectin, M M Ps: Curcumin, Ellagic acid, Olive oil, Capsaicin
- Inhibitors of M A P K: Capsaicin, Curcumin, Quercetin, Caffeic acid, Kaempferol, Genistein, Resveratrol
- Inhibitors of Lipoxygenase: Curcumin, Ginger, Boswellia, Quercetin, Olive oil
- Inhibitors of T N F, I L-1, I L-6: Ginger, Genistein, Quercetin, Astragalus, Olive oil, Capsaicin
- Inhibitors of Cyclooxygenase: Curcumin, Ginger, Boswellia, Bromelain, Devil's Claw, Olive oil, Resveratrol, E G C G, Quercetin
- Inhibitors of I N O S: Curcumin, Bromelain, Devil's Claw, Capsaicin, Genistein, Quercetin, Ginger
- Inhibitors of J A K S T A T: Curcumin, Quercetin, E G C G, Resveratrol, Bergamottin, Chrysin, Apigenin
- Inhibitors of Oxidative Stress: Curcumin, Quercetin, Sulforaphane, E G C G, Genistein, Astragalus, Capsaicin

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Cold-water fish, flax oil

Cold-water fish, flax oil

Alpha-linolenic acid
(ALA)
18:3n-3



Stearidonic acid
18:4n-3

The diagram shows a yellow oval with a blue border. Inside the oval, the text "Stearidonic acid" is written in black, followed by "18:4n-3" in a slightly smaller black font. Two black arrows point towards the oval from the right side.

Eicosatetraenoic acid
20:4 n -3

Cold-water fish, wild game,
enriched eggs

Eicosapentaenoic acid
(EPA)
20:5n-3

Docosapentaenoic acid
22:5n-3

Cold-water fish, wild game,
enriched eggs, algae

Docosahexaenoic acid
(DHA)
 $22:6n-3$

Vegetable oils, grains

Vegetable oils, grains

Linoleic acid
(LA)
18:2n-6

Gamma-linolenic acid
(GLA)
18:3n-6

Dihomo-gamma-linolenic acid
20:3n-6

Arachidonic acid (AA) $20:4n-6$

Lipoxins (anti-inflammatory)

More inflammatory eicosanoids

Animal fat, dairy, shellfish

Arachidonic acid (AA)
20:4n-6



Adrenic acid
22:4n-6

Docosapentaenoic acid
22:5n-6

Delta-6-desaturase
(Inhibited by age, alcohol)

A diagram showing a central light blue diamond shape with the text "Delta-6-elongase" inside. It is connected to other nodes (partially visible) by curved arrows, indicating its role in a metabolic pathway.

Delta-5-desaturase
(upregulated by insulin)

Beta oxidation

 Inhibition

Retro conversion



Retro conversion

Adapted from: Created by David R. Throop, vectorized by Fvasconcellos

Long Description

- The fatty acid cascade with emphasis on arachidonic acid.

References

FATTY ACID PATHWAY CASCADE

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FATTY ACID PATHWAY CASCADE

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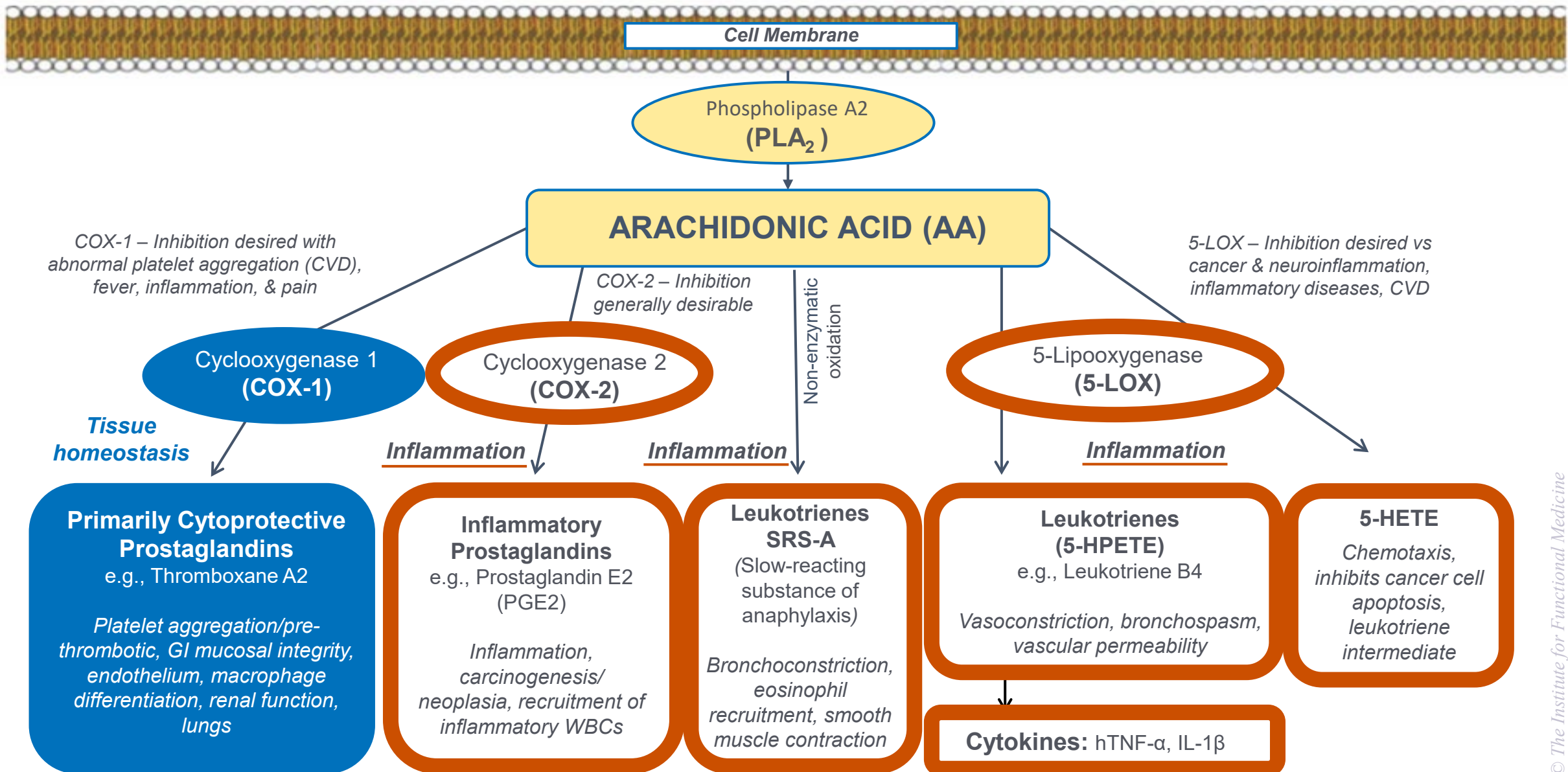
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FATTY ACID PATHWAY CASCADE

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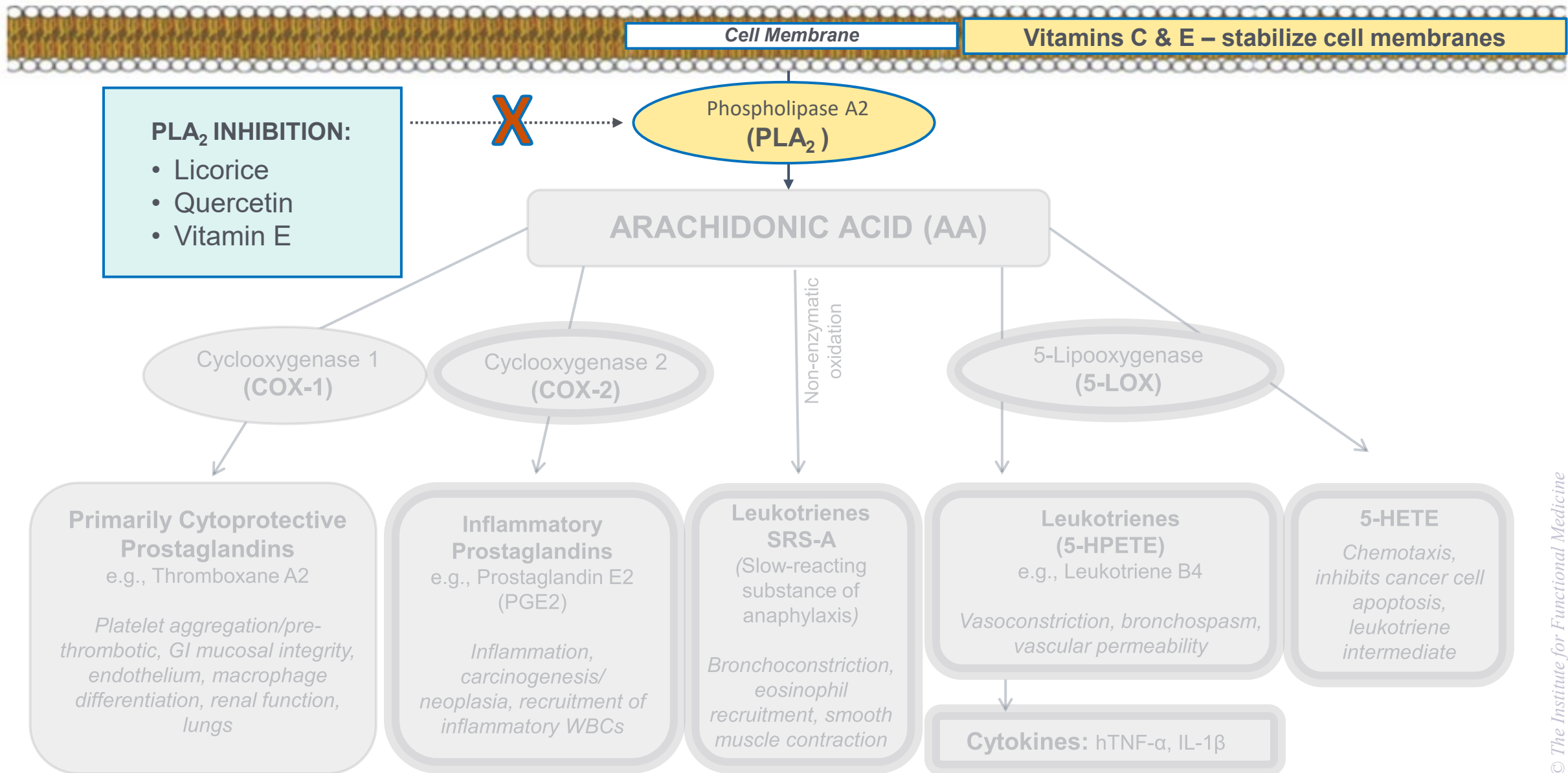
ARACHIDONIC ACID CASCADE



Long Description: Arachidonic Acid Cascade

- The arachidonic acid cascade starts when phospholipase A 2 cleaves arachidonic acid from the cell membrane in response to injury or inflammation. Arachidonic acid is then metabolized through various pathways. In the cyclooxygenase pathway, COX 1 produces primarily cytoprotective prostaglandins such as thromboxane A 2. Products of COX 2 include inflammatory prostaglandins such as P G E 2. The 5 lipoxygenase pathway produces leukotrienes and 5 H E T E. Nonenzymatic oxidation of arachidonic acid can occur, producing leukotrienes. COX 1 maintains tissue homeostasis, while COX 2, 5 LOX, and nonenzymatic oxidation drive inflammation.

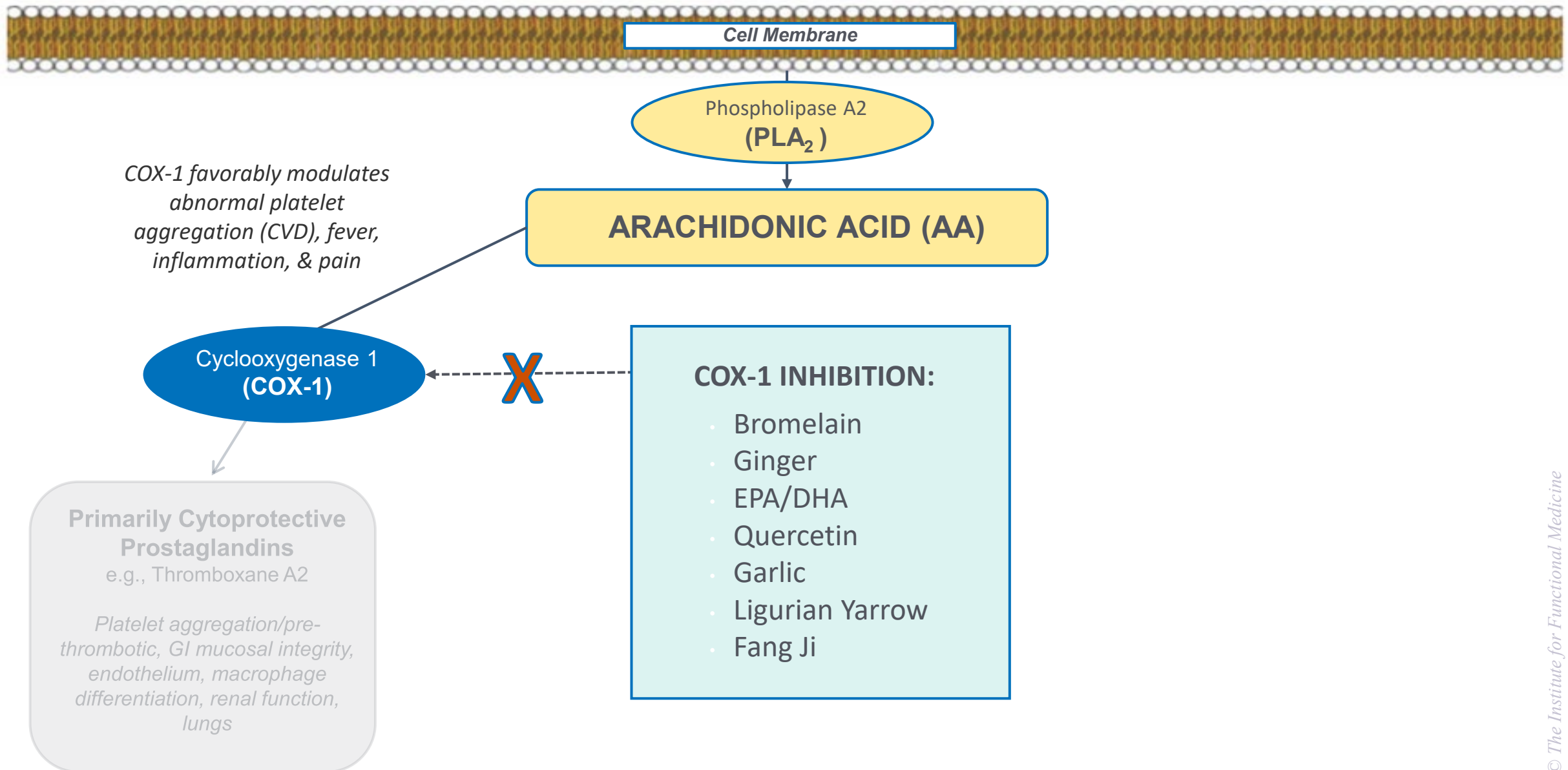
ARACHIDONIC ACID CASCADE



Long Description: Arachidonic Acid Cascade

- Phospholipase A 2 is inhibited by licorice, quercetin, and vitamin E.

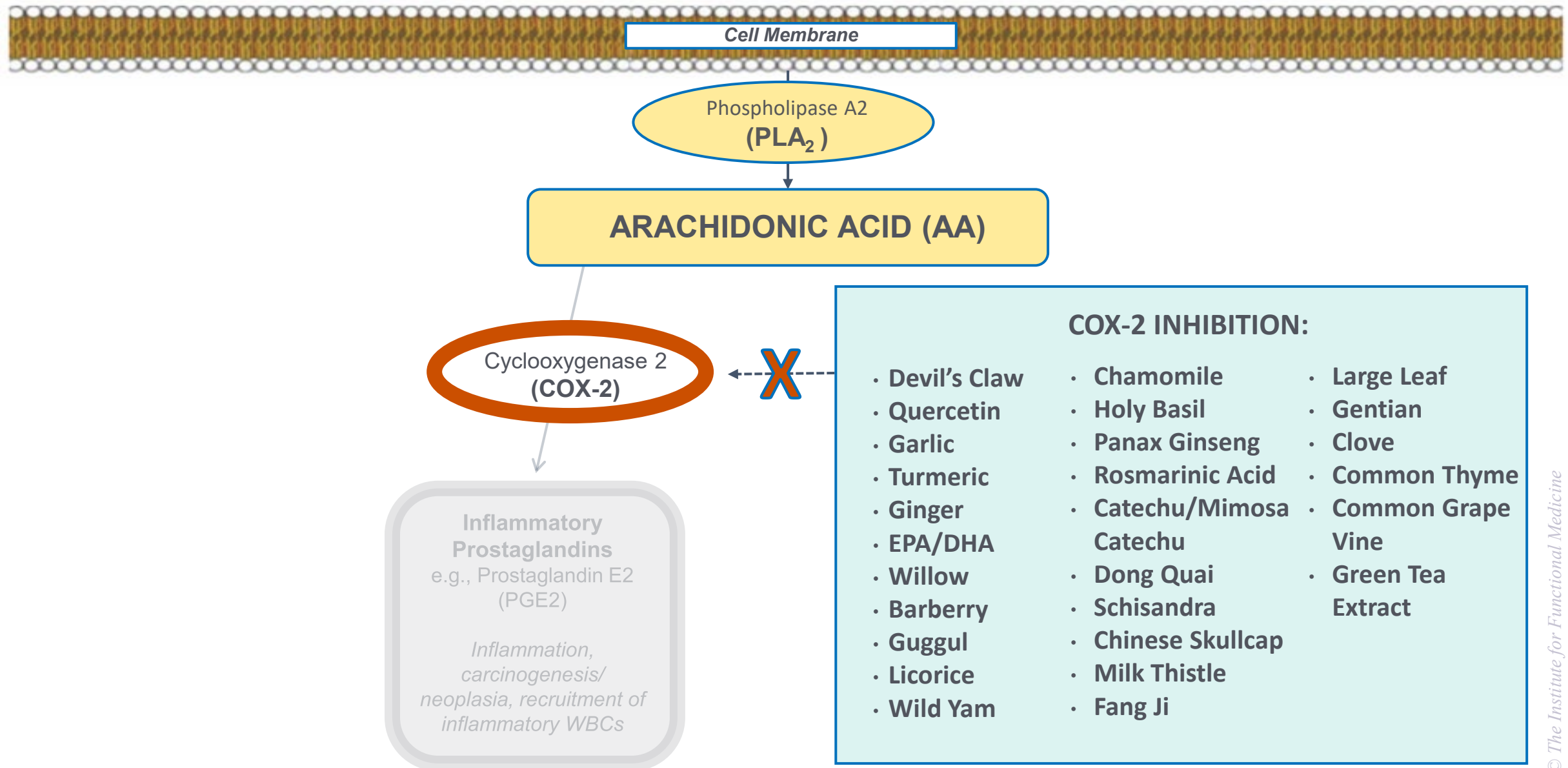
ARACHIDONIC ACID CASCADE



Long Description: Arachidonic Acid Cascade

- COX 1 is inhibited by bromelain, ginger, E P A, D H A, quercetin, garlic, ligurian yarrow, and fang ji.

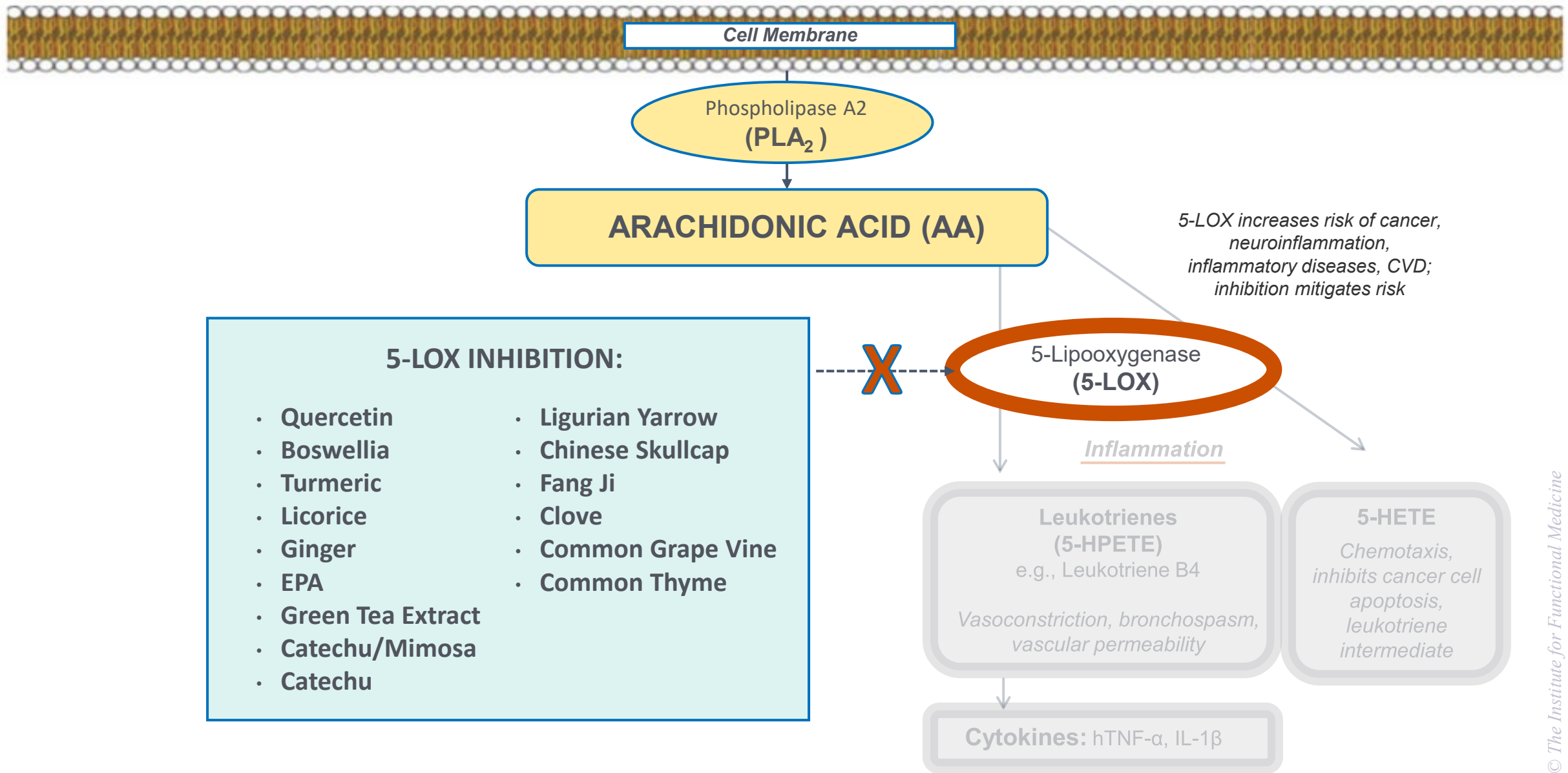
ARACHIDONIC ACID CASCADE



Long Description: Arachidonic Acid Cascade

- COX 2 is inhibited by Devil's Claw, quercetin, garlic, turmeric, ginger, E P A, D H A, willow, barberry, guggul, licorice, wild yam, chamomile, holy basil, panax ginseng, rosmarinic acid, catechu, dong quai, schisandra, Chinese skullcap, milk thistle, fang ji, large leaf, gentian, clove, common thyme, common grape vine, and green tea extract.

ARACHIDONIC ACID CASCADE



Long Description: Arachidonic Acid Cascade

- 5 LOX is inhibited by quercetin, boswellia, turmeric, licorice, ginger, E P A, green tea extract, catechu, ligurian yarrow, Chinese skullcap, fang ji, clove, common grape vine, common thyme.

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BOTANICAL & NUTRIENT MODULATION OF THE ARACHIDONIC CASCADE

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Clinical Focus

- Foundational therapeutic approach to chronic inflammation.
- Lifestyle, food, and gut health interventions.



Foundational Therapeutic Approach

Immune Function & Inflammation – Defense & Repair



LIFESTYLE FIRST



FOOD FIRST



GUT FIRST

Therapeutic Lifestyle Interventions for Chronic Inflammation

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

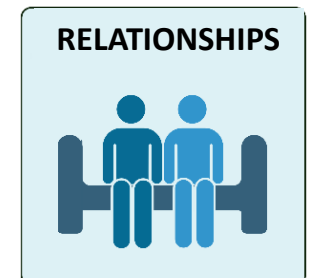
Inadequate Sleep
Sleep Disturbances

Sedentarism
Overexercise

Processed Foods
Inflammatory Foods

Stress
Gut-Brain Axis Issues

Loneliness
Lack of Support



References

THERAPEUTIC LIFESTYLE INTERVENTIONS FOR CHRONIC INFLAMMATION

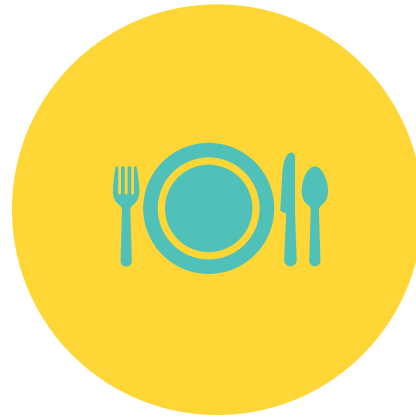
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Foundational Therapeutic Approach

Immune Function & Inflammation – Defense & Repair



LIFESTYLE FIRST

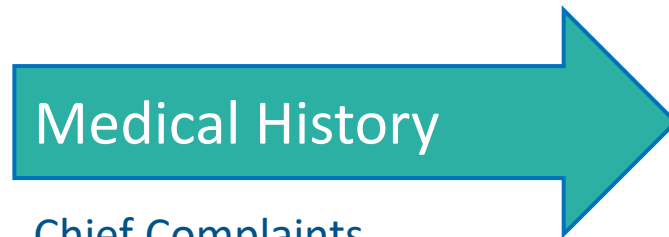


FOOD FIRST



GUT FIRST

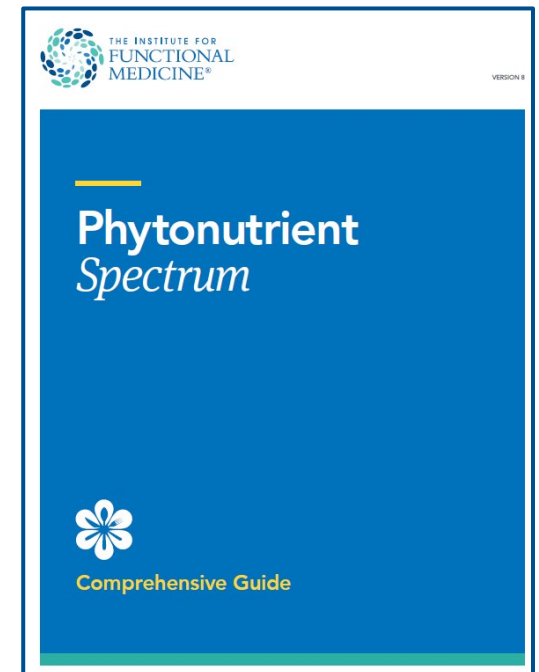
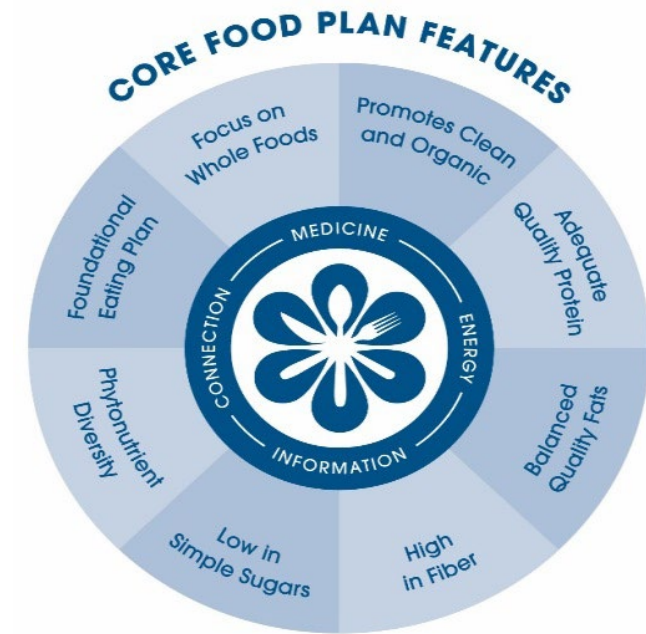
The Clinical Path to the Foundational Food Plans



Chief Complaints
Conditions
Timeline and ATMs
Medication Review

ABCDs of Nutrition Evaluation

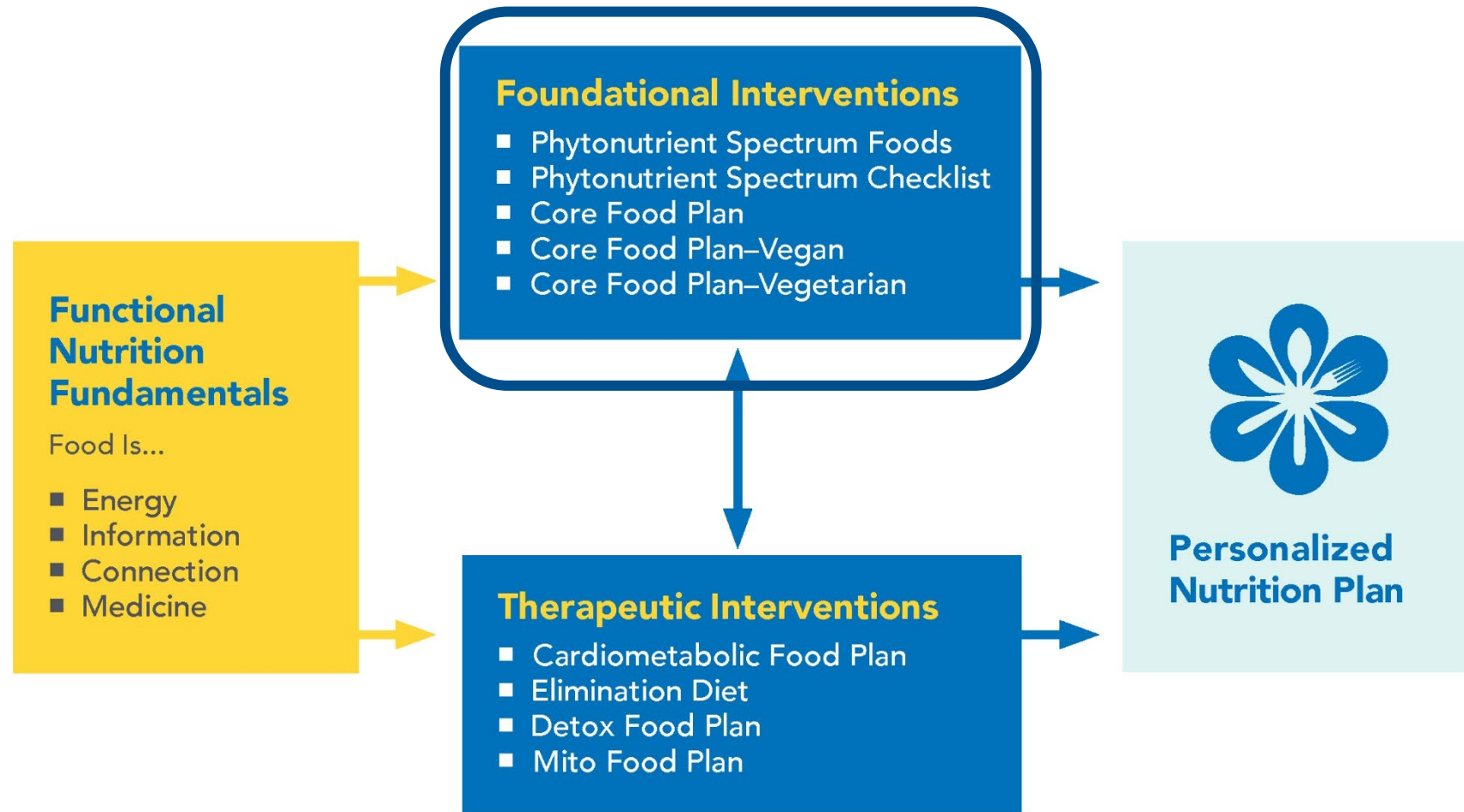
Anthropometrics
Biomarkers and Labs
Clinical Indications From NPE
Diet and Lifestyle Review
Matrix Review



Long Description: The Clinical Path to the Foundational Food Plans

- Medical history
 - Chief complaints
 - Conditions
 - Timeline and A T M's
 - Medication review
- A B C D's of Nutrition Evaluation
 - Anthropometrics
 - Biomarkers and labs
 - Clinical indications from N P E
 - Diet and lifestyle review
 - Matrix review
- Core Food Plan
- Phytonutrient spectrum

Nutrition Interventions

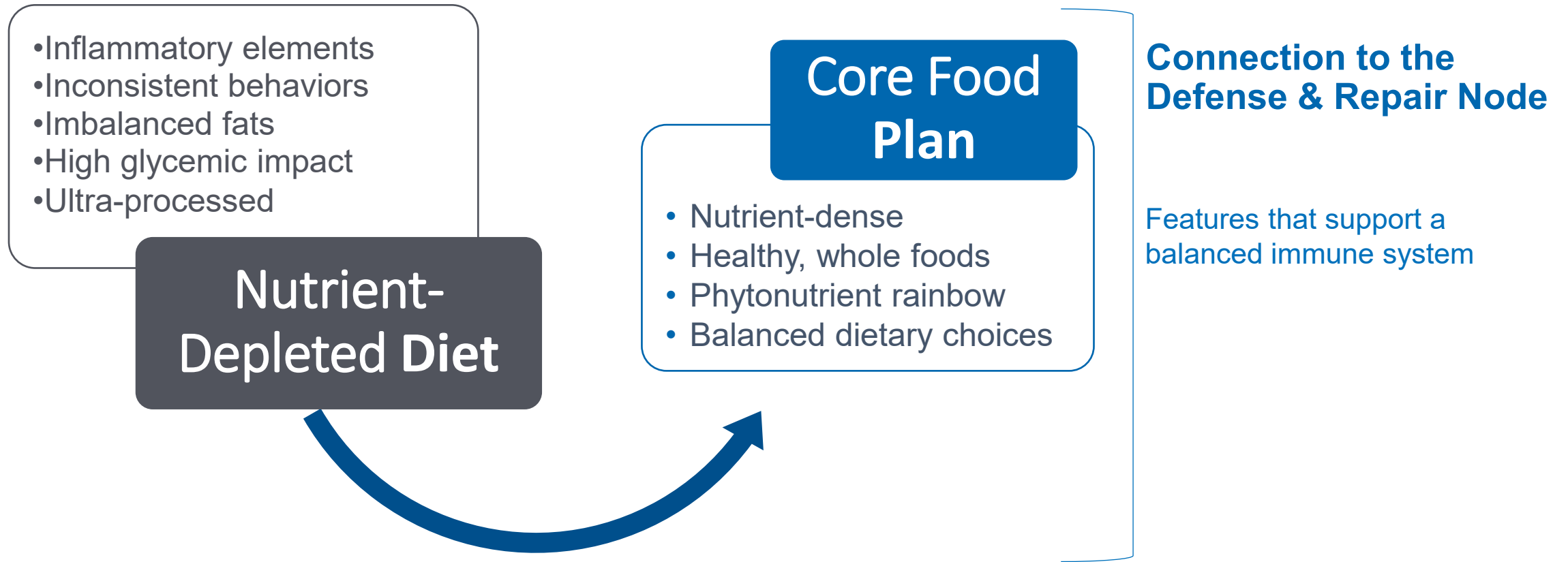


Long Description: Nutrition Interventions

- Core Food Plan Features
 - Focus on whole foods
 - Encourages organic
 - Adequate quality protein
 - Balanced quality fats
 - High in fiber
 - Low in simple sugars
 - Phytonutrient diversity
 - Foundational eating plan
- Clinical Toolkit Item: Phytonutrient Spectrum
 - Accessible version available in course materials

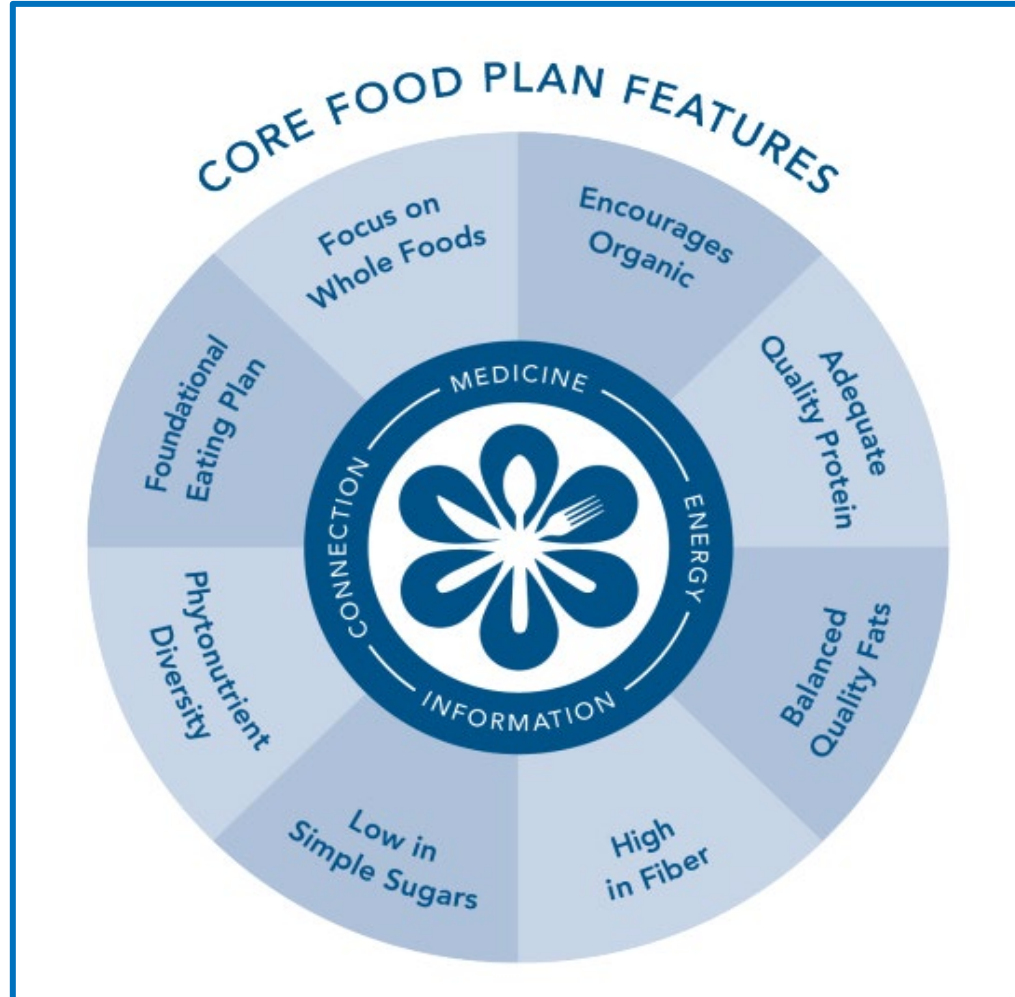
Recommended “Core” First Step for Most Patients

A More Nutrient-Dense Diet



Food First

Core Food Plan and Phytonutrient Spectrum



Accessible version of this resource in your toolkit.

Long Description: Food First

- Core Food Plan Features
 - Focus on whole foods
 - Encourages organic
 - Adequate quality protein
 - Balanced quality fats
 - High in fiber
 - Low in simple sugars
 - Phytonutrient diversity
 - Foundational eating plan
- Clinical Toolkit Item: Phytonutrient Spectrum
 - Accessible version available in course materials

Foundational Anti-Inflammatory Foods



**Phytonutrient
Spectrum**



Omega-3s



Ginger



Garlic



**Fermented
Foods**



Turmeric



Green Tea



Cayenne

Long Description: Foundational Anti-Inflammatory Foods

- Phytonutrient spectrum
- Omega 3's
- Ginger
- Garlic
- Fermented foods
- Turmeric
- Green tea
- Cayenne

Various Resources Available in Your Toolkit



Anti-Inflammatory Foods

Ongoing, low-grade inflammation underlies many common health problems. These include heart disease, type 2 diabetes, certain cancers, digestive issues, arthritis, and various brain disorders. Among the top causes of chronic inflammation are unhealthy eating, physical inactivity, and obesity (especially belly fat). Chronic infections, unbalanced gut bacteria, and toxin exposure also promote inflammation. A healthy lifestyle, including nutritious eating habits, helps tame inflammation.

How Foods Fight Inflammation

Whole foods like vegetables, fruits, whole grains, legumes, nuts, and seeds are rich in inflammation-fighting compounds. These include many vitamins, minerals, phytonutrients (plant compounds), and fiber. Omega-3 fats and probiotics (good bacteria), found in some animal and plant foods, can also calm inflammation.

The nutritional components of foods work in different ways to control inflammation. Some prevent the body from making as many inflammatory molecules. Others work as antioxidants to protect cells against damage from inflammation. Some promote a healthy community of gut bacteria, which can also help combat inflammation.

The following sections highlight foods with anti-inflammatory benefits, as shown in research studies. Similar foods in these categories could also have anti-inflammatory benefits but may be less well-studied.

Vegetables

Vegetables offer fiber and antioxidants, including beta-carotene, which gives an orange color. Vegetables are also super sources of green and red phytonutrients that fight inflammation. Some of the top anti-inflammatory vegetables are:

- **Dark green leafy vegetables:** lettuce (such as romaine and green leaf), spinach, Swiss chard
- **Cruciferous vegetables:** broccoli, broccoli sprouts, cabbage, cauliflower, collards, kale, mustard greens
- **Other vegetables:** beets, carrots, garlic, onions, tomatoes

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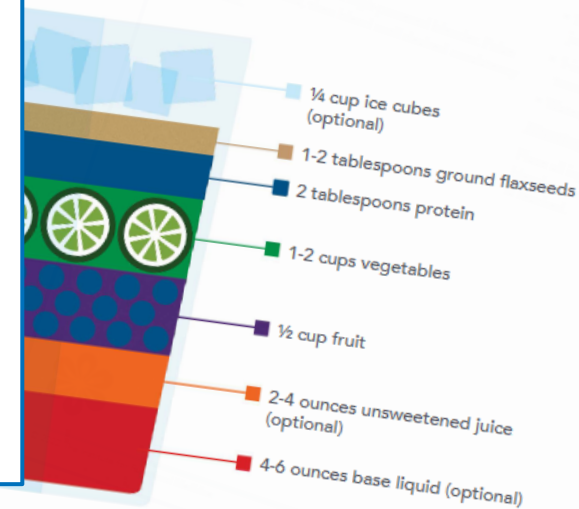
Version 1

THE INSTITUTE FOR
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MEDICINE

Building a Better Smoothie

Smoothies can be a great complement to your food plan and can help make "food as medicine" convenient and delicious. The illustration below outlines the basic components of a smoothie, and this handout provides ideas for each ingredient category. When possible, choose organic ingredients. Put the ingredients into a high-powered blender in the order described. Process the mixture until your smoothie is the desired consistency. Your healthcare provider may suggest specific ingredients or direct you to a recipe in this handout.

Healthy Smoothie Ingredients



Version 4 (references available in separate bibliography)

Recipes (continued)

Anti-Inflammatory Smoothie

Ingredients (about 12 ounces):

- 1/2 cup blackberries or black raspberries (or 1/2 cup black raspberry powder)
- 1/2 cup orange or tropical fruit (orange, tangerine, papaya, etc.)
- 1/2 cup green leafy vegetable
- 1/2 cup fresh ginger root (or 1/4 teaspoon ground)
- 1/2 cup flax seeds (or 2 ounces pomegranate)
- 1/2 cup turmeric root (or 1/2 teaspoon ground)
- 1/2 cup unsweetened juice
- 1/2 cup base liquid

Instructions: Put ingredients into a high-powered blender. Pulse, then blend until desired consistency.

Long Description

- Accessible versions of various clinical toolkit items available in course materials including Anti-Inflammatory Foods, Building a Better Smoothie, and Phytonutrient Smoothie Recipes

Foundational Therapeutic Approach

Immune Function & Inflammation – Defense & Repair



LIFESTYLE FIRST

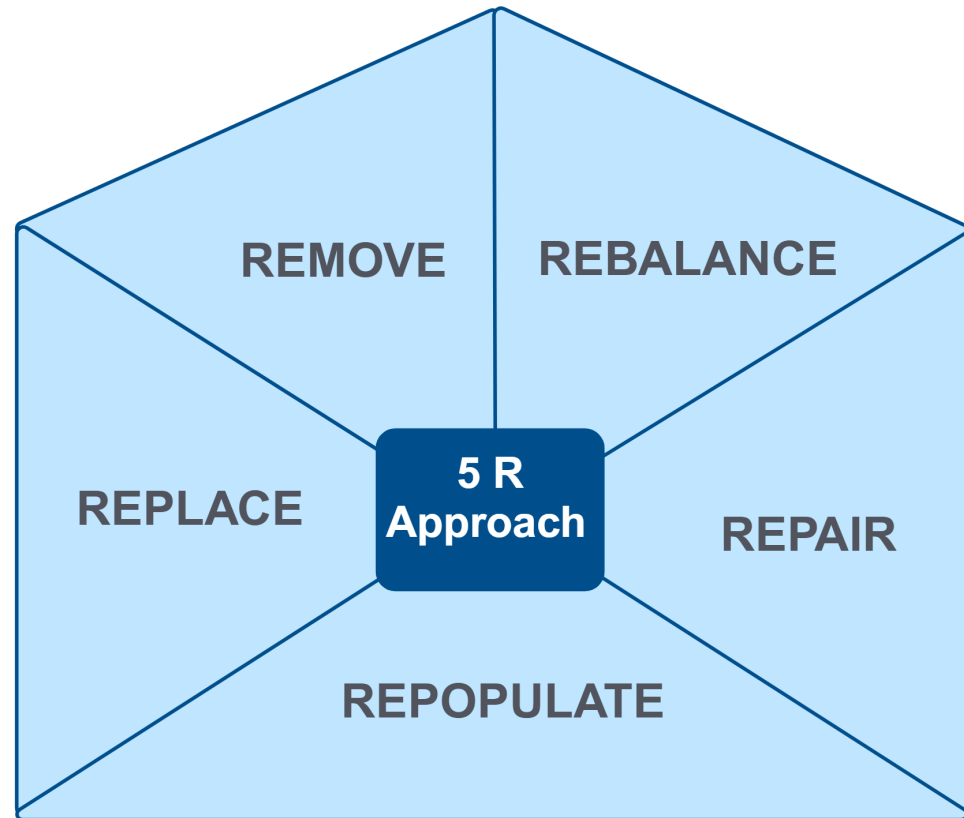


FOOD FIRST



GUT FIRST

The 5R Approach for Gut Restoration





The 5R Framework for Gut Restoration

A properly functioning digestive system is critical to good health. Problems with the gastrointestinal (GI) tract can cause more than just stomach upset, gas, bloating, and diarrhea. GI issues may underlie chronic health problems that seem unrelated to digestive health, including autoimmune diseases (such as rheumatoid arthritis), skin problems (such as eczema and acne rosacea), and heart disease, among others.

In the bigger picture, how can we deal with all that can go wrong in the gut? In functional medicine, we use a program that goes by the simple acronym of the **5Rs: remove, replace, reinoculate, repair, and rebalance**. When applied to various chronic health issues, the 5R program can lead to a dramatic improvement in symptoms and sometimes even complete resolution. The elements of the 5R program are described briefly here.

- 1. Remove**
Remove stressors. Get rid of things that negatively affect the environment of the GI tract, including foods you are sensitive to, parasites, and potential problematic bacteria or yeast. This might involve using an elimination diet to find out what foods are triggering GI symptoms. It may also involve taking medications or herbs to get rid of an infection.
- 2. Replace**
Replace digestive secretions. Add back things like digestive enzymes, bile acids, and hydrochloric acid that are required for proper digestion and that may be compromised by diet, medications, diseases, aging, or other factors.
- 3. Reinoculate**
Help beneficial bacteria flourish by eating probiotic-rich foods or taking supplements that contain "good" bacteria, such as *Bifidobacteria* and *Lactobacillus* species, and by eating high-fiber foods that supply prebiotics to nourish good microbes.
 - **Probiotics** are beneficial microorganisms found in the gut and are also called "friendly bacteria." The use of antibiotics kills both good and bad bacteria. Probiotics from supplements or food are often needed to help



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Version 4 (image courtesy of 123rf)

Long Description: the 5 R Approach for Gut Restoration

- 5 R Approach
 - Remove
 - Replace
 - Rebalance
 - Repair
 - Repopulate
- Toolkit item available in course materials: The 5 R framework for gut restoration

Second-Tier Anti-Inflammatory Support

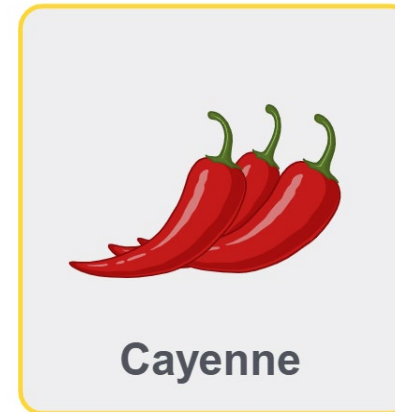
If the foundational diet, lifestyle, and 5R gut health interventions do not achieve desired results, consider second-tier botanicals, nutraceuticals, and supplements.



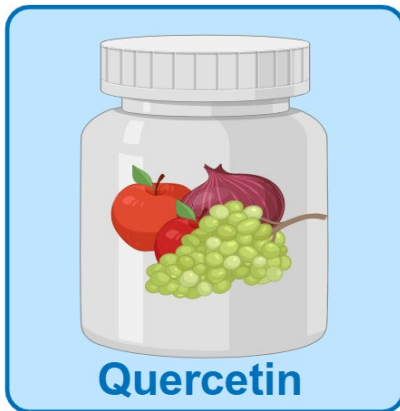
Second-Tier Botanicals, Nutraceuticals, Supplements



Omega-3 Dose:
EPA/DHA 2,000-
4,000 mg/day



Cayenne Dose (Topical):
At least one 8% capsaicin
patch used for 30-60 min
per day 0.075% capsaicin
cream 1-3 times/day

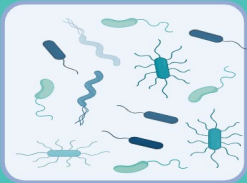


Quercetin Dose:
500-1,000 mg/day



Turmeric Dose:
750-1,500 mg/day

Second-Tier Botanicals, Nutraceuticals, Supplements



Probiotics

Probiotics
Dose: 10-100
Billion CFUs



Green Tea

Green Tea Dose:
1 gram of green
tea extract
(equivalent to ~5
cups of green tea)



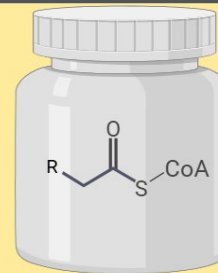
Garlic

Garlic Dose:
1,200-2,400
mg/day of garlic
extract



Ginger

Ginger Dose:
500-1,000 mg
TID (raw or
capsule form)



**N-Acetyl L-
Cysteine**

NAC Dose:
1,200-2,000mg/day

References

SELECTED ANTI-INFLAMMATORY BOTANICAL AGENTS

Omega-3 DHA/EPA reduces inflammation (CRP) (Moderate Evidence)

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Turmeric (Curcumin) reduces CRP (Strong Evidence)

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Ginger reduces CRP (Strong Evidence)

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SELECTED ANTI-INFLAMMATORY BOTANICAL AGENTS

Garlic reduces CRP (Moderate Evidence)

- Gadidala SK, Johny E, Thomas C, Nadella M, Undela K, Adela R. Effect of garlic extract on markers of lipid metabolism and inflammation in coronary artery disease (CAD) patients: A systematic review and meta-analysis. *Phytother Res.* 2023;37(6):2242-2254. doi:10.1002/ptr.7729
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Cayenne (topical capsaicin) reduces local pain (Strong Evidence)

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SELECTED ANTI-INFLAMMATORY BOTANICAL AGENTS

Green tea (EGCG) increased TAC and reduced fecal calprotectin (Moderate Evidence)

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SELECTED ANTI-INFLAMMATORY BOTANICAL AGENTS

Quercetin reduces CRP (Moderate, Conflicting Evidence)

- Mohammadi-Sartang M, Mazloom Z, Sherafatmanesh S, Ghorbani M, Firoozi D. Effects of supplementation with quercetin on plasma C-reactive protein concentrations: a systematic review and meta-analysis of randomized controlled trials. Eur J Clin Nutr. 2017;71(9):1033-1039. doi:10.1038/ejcn.2017.55
- Ou Q, Zheng Z, Zhao Y, Lin W. Impact of quercetin on systemic levels of inflammation: a meta-analysis of randomised controlled human trials. Int J Food Sci Nutr. 2020;71(2):152-163. doi:10.1080/09637486.2019.1627515

NAC reduces CRP (Strong Evidence)

- Askari M, Faryabi R, Mozaffari H, Darooghegi Mofrad M. The effects of N-Acetylcysteine on serum level of inflammatory biomarkers in adults. Findings from a systematic review and meta-analysis of randomized clinical trials. Cytokine. 2020;135:155239. doi:10.1016/j.cyto.2020.155239

References

SELECTED ANTI-INFLAMMATORY BOTANICAL AGENTS

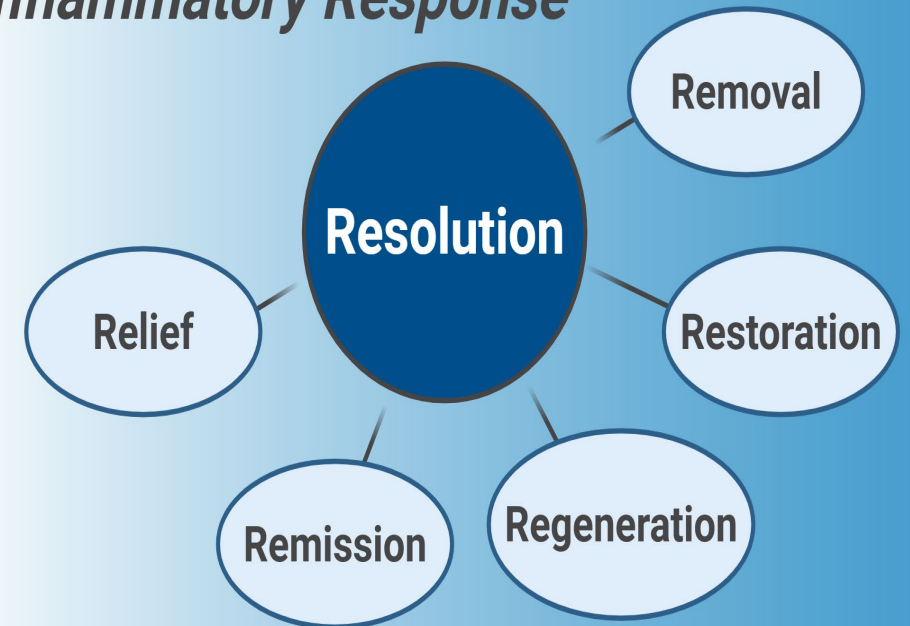
Probiotics reduce CRP (Strong Evidence)

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Chronic Inflammation: Therapeutic Principles

- Pharmaceutical anti-inflammatory agents most often suppress the downstream effects of inflammation.
- Functional medicine targets “upstream” factors that trigger or mediate inflammation.
- Chronic inflammation can most effectively be mitigated by avoiding further exposure to triggers and by modulating inflammatory mediators with lifestyle, diet, and gut interventions.

Resolution of Inflammatory Response





Thank you

AFMCP

Applying Functional Medicine in Clinical Practice

Clinical Focus

- Conclusion and Key Takeaways
- Clinical Flow Path for Chronic Inflammation

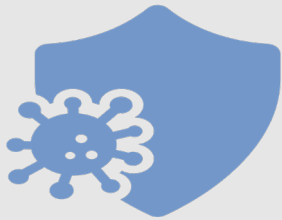


Defense and Repair

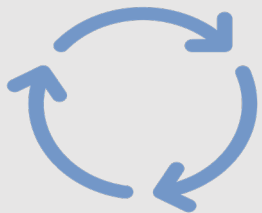
Immune Regulation of Inflammation: Key Clinical Takeaways



Acute inflammation is the body's normal physiological response to various insults and injury such as trauma, infection, toxins, and food antigens or poor diet.



Most chronic diseases have been linked to excessive or persistent chronic inflammation.



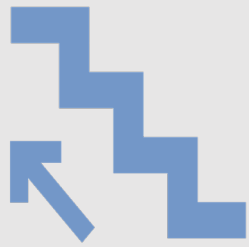
Chronic inflammation occurs when the insult, injury, trigger, or mediator is ongoing, or when there is a dysfunctional process of resolution or a dysregulated immune system.

Defense and Repair

Immune Regulation of Inflammation: Key Clinical Takeaways



Pharmaceutical interventions most often focus on the downstream effects, or final pathway, of inflammation.

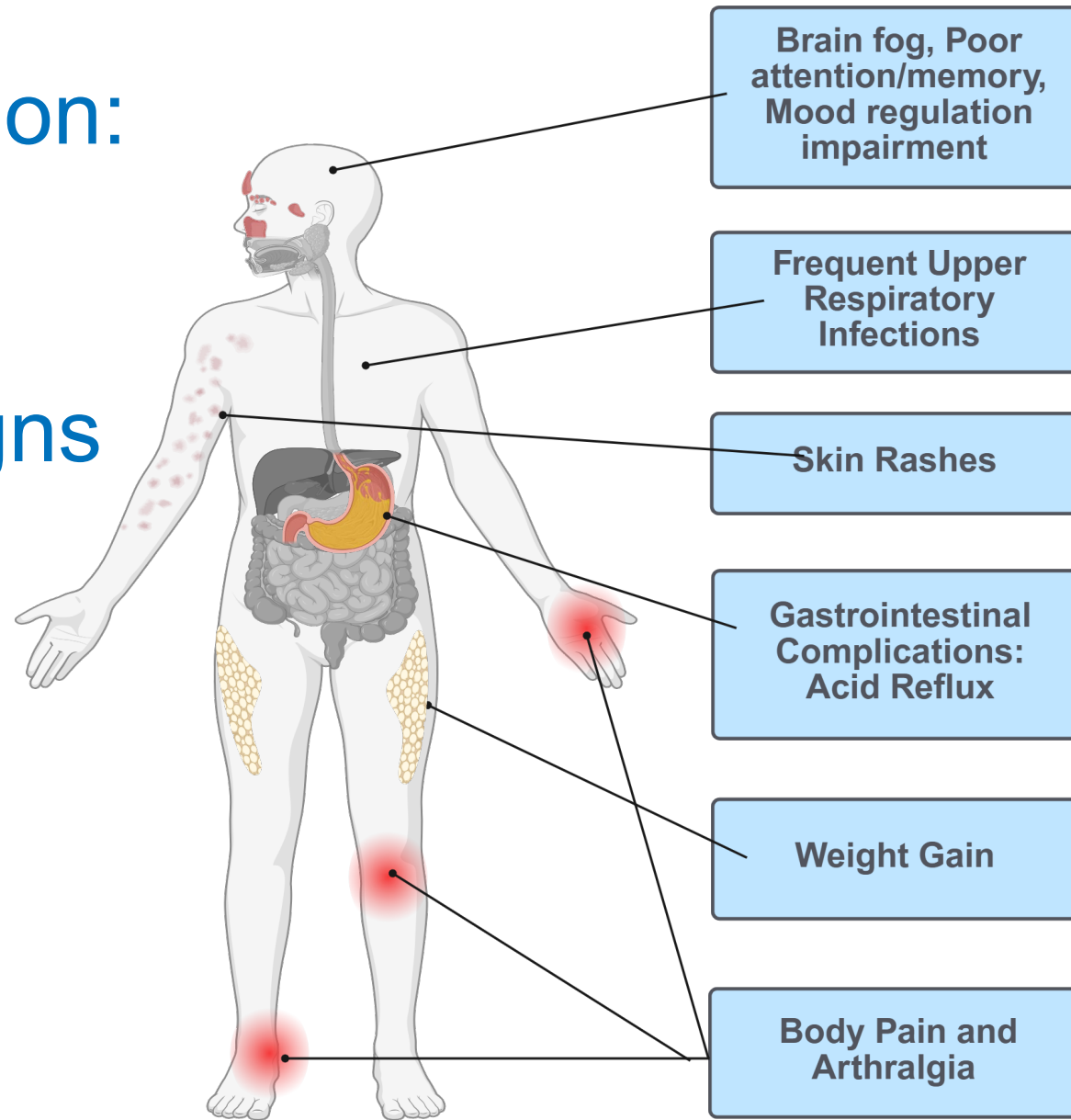


Functional medicine works upstream by addressing the underlying conditions and root causes that initiate or perpetuate inflammation (including ATMs, MLFs, & MES dysfunctions).



Inflammation can be best mitigated by avoiding exposure to triggers and by modulating inflammatory mediators with lifestyle, diet, and gut health interventions.

Chronic Inflammation: Clinical Findings Examples of Symptoms and Signs



Long Description: Signs and Symptoms of Chronic Inflammation

- Brain fog
- Poor attention, poor memory
- Mood regulation impairment
- Frequent upper respiratory infections
- Skin rashes
- Gastrointestinal complications, acid reflux
- Weight gain
- Body pain and arthralgia

Chronic Inflammation

Disease Associations

- Atopic syndrome/eczema
- Rheumatologic: autoimmune disorders
- Gastroenterological: IBD, gluten intolerance, IBS, SIBO, candida
- Cardiovascular disease: atherosclerosis
- Neurodegenerative disease: PD, AD, MS
- Cancer
- Endocrinologic: diabetes
- Psychiatric disorders: schizophrenia, depression, anxiety, mood disorders

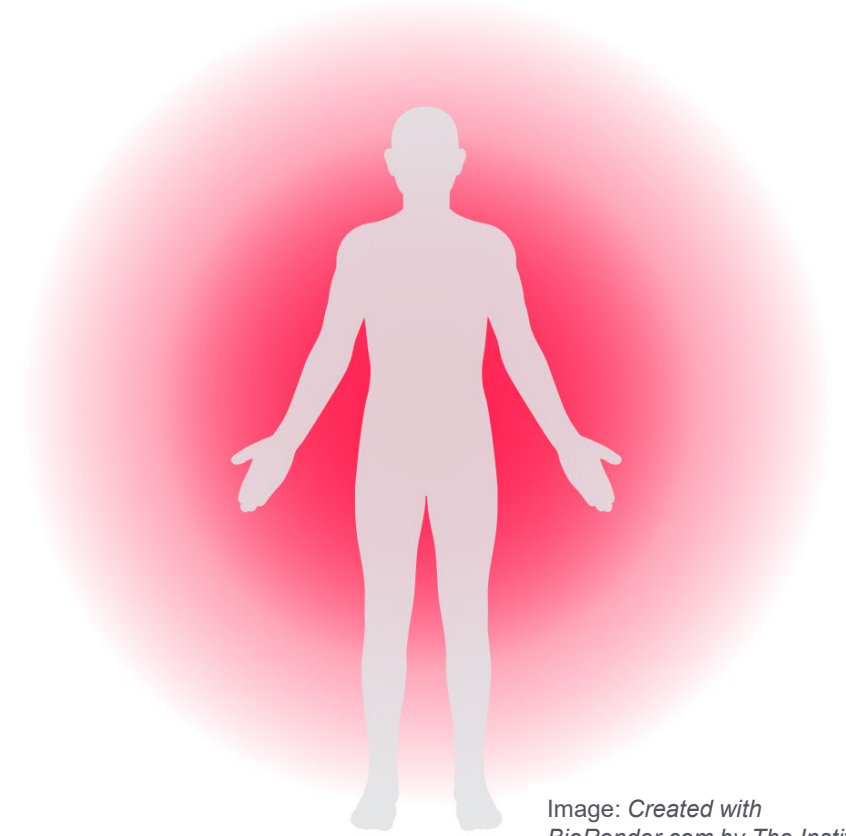
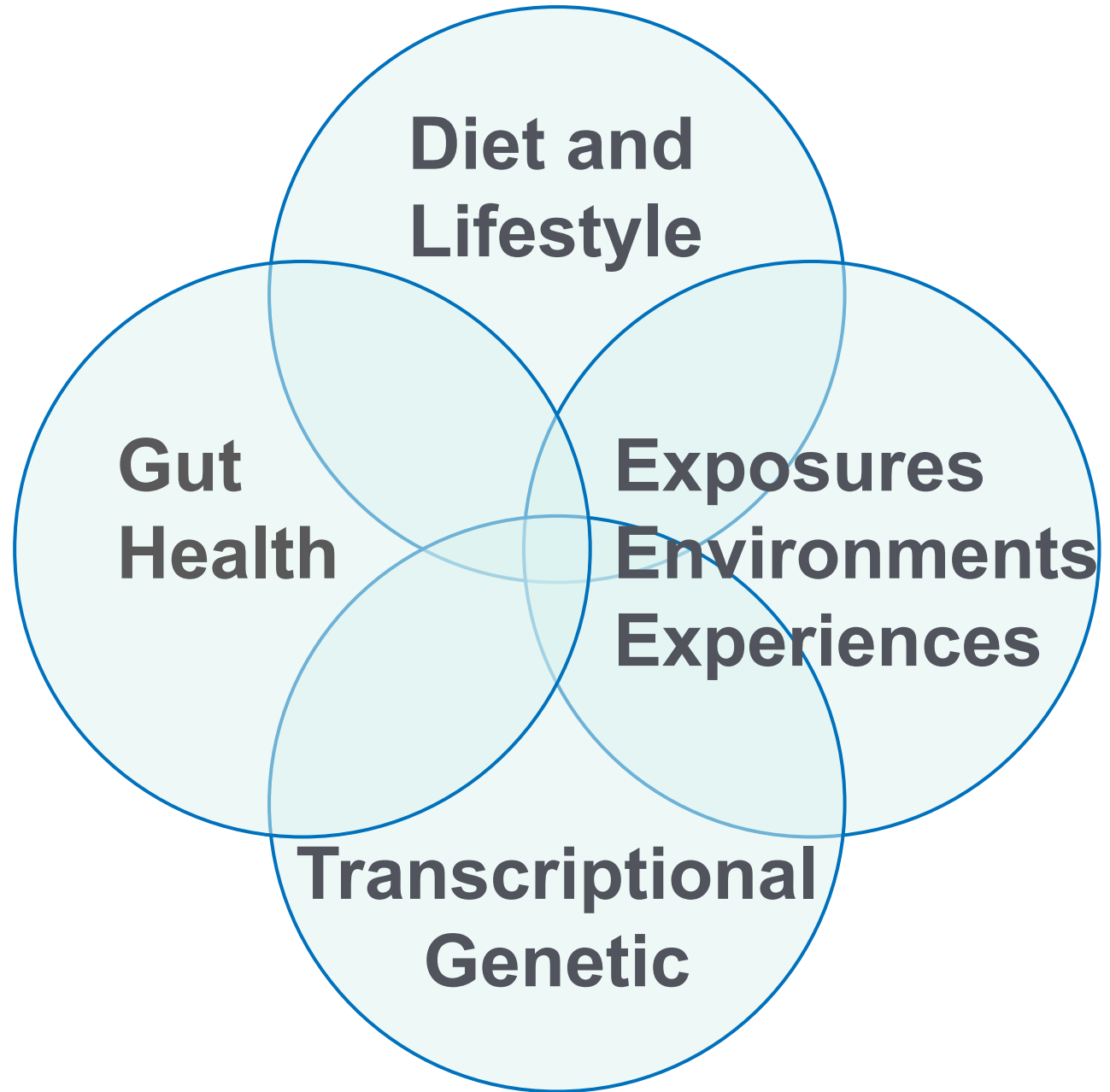


Image: Created with
BioRender.com by The Institute for
Functional Medicine

Most Common Antecedents, Triggers, and Mediators of Chronic Inflammation



Long Description: The Most Common A T Ms of Chronic Inflammation

- Overlapping circles representing the A T M's of chronic inflammation including diet and lifestyle, exposures, environments, and experiences, transcriptional, genetic, and gut health.

Other Common ATMs of Chronic Inflammation



Medications



Trauma



Toxins



Infections

Laboratory Clinical Findings of Chronic Inflammation

Lab	Associated Clinical Indicators/ATMs	Clinical Pearls
Leukopenia WBC <4,300 cells/mm ³	- Chronic viral infections are the most common cause; HIV, herpes, hepatitis, etc. - Folate, B-12, or copper deficiency	3,000-4,000 cells/mm³, often benign - Downward trend more significant than absolute number – rule out myelodysplastic syndromes
Leukocytosis WBC >11K	- Infection - Inflammation - Allergic reactions - Malignancy: leukemia, myeloproliferative disorders - Obesity - Stressors: trauma, overexertion, seizures, emotional stress Etiology by subclass (obtain differential) - Neutrophilia: acute bacterial infection - Lymphocytosis: acute viral infection - Monocytosis: chronic infection, inflammation - Eosinophilia: allergic, parasitic	Acute phase reactant

Laboratory Clinical Findings of Chronic Inflammation

Lab	Associated Clinical Indicators/ATMs	Clinical Pearls
Neutrophilia	Antecedent: Familial Triggers: Drugs, infection, pregnancy, post-splenectomy Mediators: • Smoking • Obesity • Smoking and obesity • Idiopathic, idiopathic-inflammatory • Myeloproliferative disorders • Lymphoproliferative disease • Multiple causes • Inflammatory conditions • Tumors	Drug-induced neutropenia (often administered for chronic inflammatory conditions) • Antibiotics • Anticonvulsants (carbamazepine, valproate) • Anti-thyroid (PTU, methimazole) • Anti-depressants & antipsychotics (clozapine) • Anti-inflammatories • Cardiovascular meds • Cancer chemotherapy: can be long term • Environmental toxins • Heavy metals: arsenic, mercury, gold
Elevated ESR	Diagnosis and/or monitoring • Chronic or persistent infections • Osteomyelitis • Pelvic inflammatory disease • Tuberculosis • Systemic lupus erythematosus • Polymyalgia rheumatica • Temporal arteritis • Rheumatoid arthritis • Inflammatory bowel disease • Multiple myeloma • Hodgkin's lymphoma • Chronic kidney disease	Used to help distinguish between inflammatory vs. allergic conditions

Laboratory Clinical Findings of Chronic Inflammation

Lab	Associated Clinical Indicators/ATMs	Clinical Pearls
Elevated CRP	• Inflammatory bowel disease • Coronary artery disease risk • Metabolic syndrome/T2DM • Myocardial damage after ischemia • Periodontal disease • Obesity • Arthritis (some forms) • Pelvic inflammatory disease • Systemic lupus erythematosus • Infection after surgery • Thyroid disease (hs-CRP)	Limitations • Does not increase with all types of inflammation • May have normal levels with active inflammatory disease, cancer, SLE • May be blunted with liver disease and protein deficiency Correlations With Loneliness and Immune Function • Higher CRP levels have shown to correlate with loneliness and immune function
Elevated hs-CRP	• Patients with stable coronary artery disease • Stable and unstable angina • Post-CABG surgery • AFib • Persons with T2DM	• Assessment of risk in cardiovascular or metabolic disease with other compounding risk factors present • Assess low-grade inflammation in person without overt symptoms such as inflammatory arthritis • If result comes back above >3-10 mg/L, consider doing regular CRP at that time (depending on type of analyzer used – indeterminate at higher levels) Helps Make Decisions With: • Statin therapy • Dietary recommendations • Hormone replacement therapy (E2)

Functional Medicine Matrix

Clinical Findings of Chronic Inflammation

Clinical Imbalances and Physiological Dysfunctions

The Patient's Story

Antecedents (Inherited & Acquired)

- Prenatal history/exposure
- Adverse childhood events (ACEs)
- Strong family history (autoimmune or CVD)

Triggers & Precipitating Events

- Infection
- Injury
- Toxin exposure
- Traumatic life event

Mediators/Perpetuators

- Chronic stress
- Gut dysfunction
- Environmental exposures
- Obesity

Assimilation

- Constipation, diarrhea, and acid reflux
- Gastroenterological: IBD, gluten intolerance
- Malabsorption

Structural Integrity

- Weight gain or loss (unintentional)
- Sarcopenia
- Obesity

Communication

- Insulin resistance
- Hormonal dysfunction
- Thyroid dysfunction

Defense & Repair

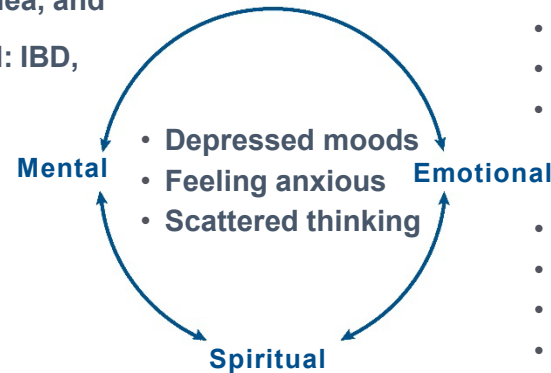
- Frequent infections
- Allergy, atopy, sensitivities
- Cancer
- Autoimmune

Energy

- Body pain, arthralgia, myalgia
- Chronic fatigue and insomnia
- Mental fatigue
- Chronic sleep dysfunction

Mitigation & Clearance

- Toxin exposure (mold, metal, chemical)
- Multiple chemical sensitivity history
- Constipation
- Lack of sweating



Transport

- Cardiovascular disease: atherosclerosis, hypertension
- Edema, lymphedema

Modifiable Lifestyle Factors

- Sleep**
- Sleep dysfunction/disorders
 - Chronic sleep deprivation
 - Sleep apnea

- Physical Activity**
- Sedentary lifestyle
 - Exercise intolerance
 - Excessive exercise

- Nutrition**
- Ultra-processed foods
 - Low protein
 - Nutrient devoid
 - Calorie excess

- Stressors**
- Chronic stress
 - Post-traumatic stress
 - Moral injury
 - Existential crisis

- Relationships**
- Social isolation
 - Low power/autonomy
 - Marital strain
 - Social rejection

Pahwa R, Goyal A, Jialal I. *Chronic Inflammation*. StatPearls Publishing; 2023. Updated August 7, 2023. Accessed May 29, 2025. <https://www.ncbi.nlm.nih.gov/books/NBK493173/>

Accessible version in your toolkit.

Long Description: Functional Medicine Matrix

- Clinical findings of chronic inflammation
- The patient's story
 - Antecedents
 - Prenatal history and exposure
 - Adverse childhood events
 - Strong family history
 - Triggers and precipitating factors
 - Infection
 - Injury
 - Toxin exposure
 - Traumatic life event
 - Mediators and perpetuators
 - Chronic stress
 - Gut dysfunction
 - Environmental exposures
 - Obesity
- Modifiable lifestyle factors
 - Sleep
 - Sleep dysfunction and disorders
 - Chronic sleep deprivation
 - Sleep apnea
 - Physical Activity
 - Sedentary lifestyle
 - Exercise intolerance
 - Excessive exercise
 - Nutrition
 - Ultra-processed foods
 - Low protein
 - Nutrient devoid
 - Calorie excess
 - Stressors
 - Chronic stress
 - P T S D
 - Moral injury
 - Existential crisis
- Relationships
 - Social isolation
 - Low power or autonomy
 - Marital strain
 - Social rejection
- Clinical imbalances and physiological dysfunction
 - Assimilation
 - Constipation, diarrhea, and acid
 - Gastroenterological: I B D, gluten
 - Malabsorption
 - Defense and repair
 - Frequent infections
 - Allergy, atopy, sensitivities
 - Cancer
 - Autoimmune
 - Energy
 - Body pain, arthralgia, myalgia
 - Chronic fatigue and insomnia
 - Mental fatigue
 - Chronic sleep dysfunction
 - Mitigation and clearance
 - Toxin exposure: mold, metal, ch

Functional Medicine Matrix

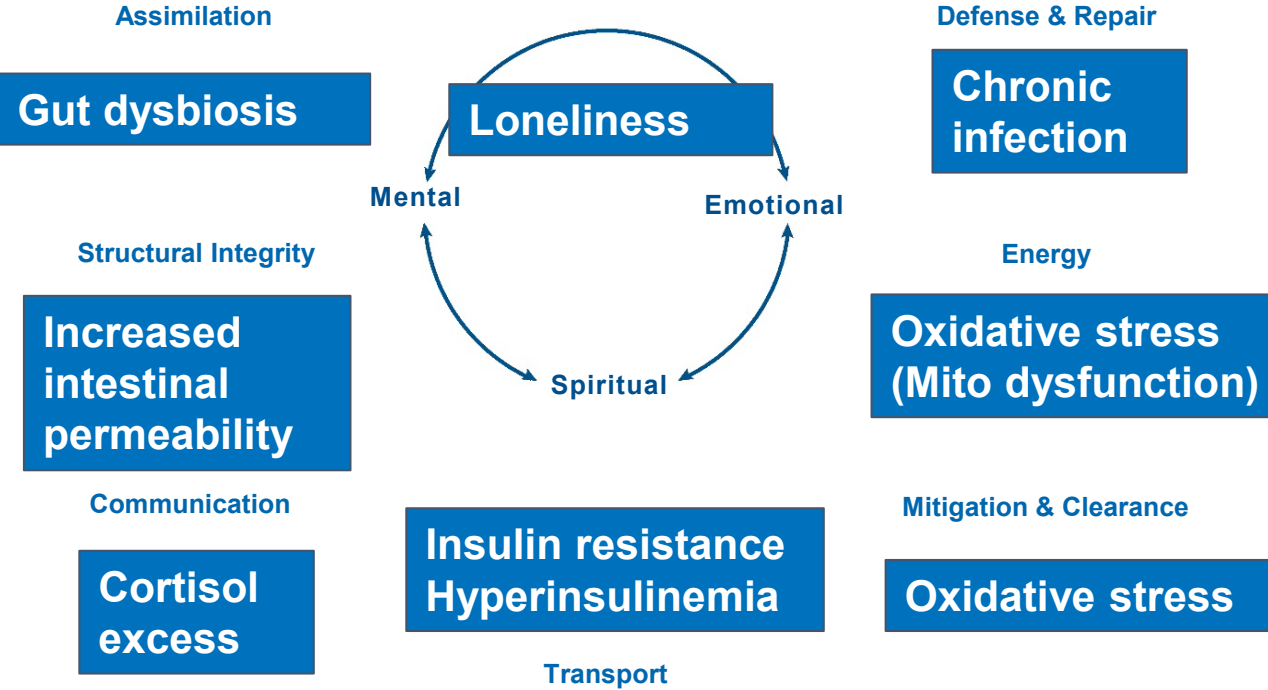
Clinical Imbalances and Physiological Dysfunctions

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators



Matrix
Dysfunctions as
Mediators of
Chronic
Inflammation:
Representative
Examples

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

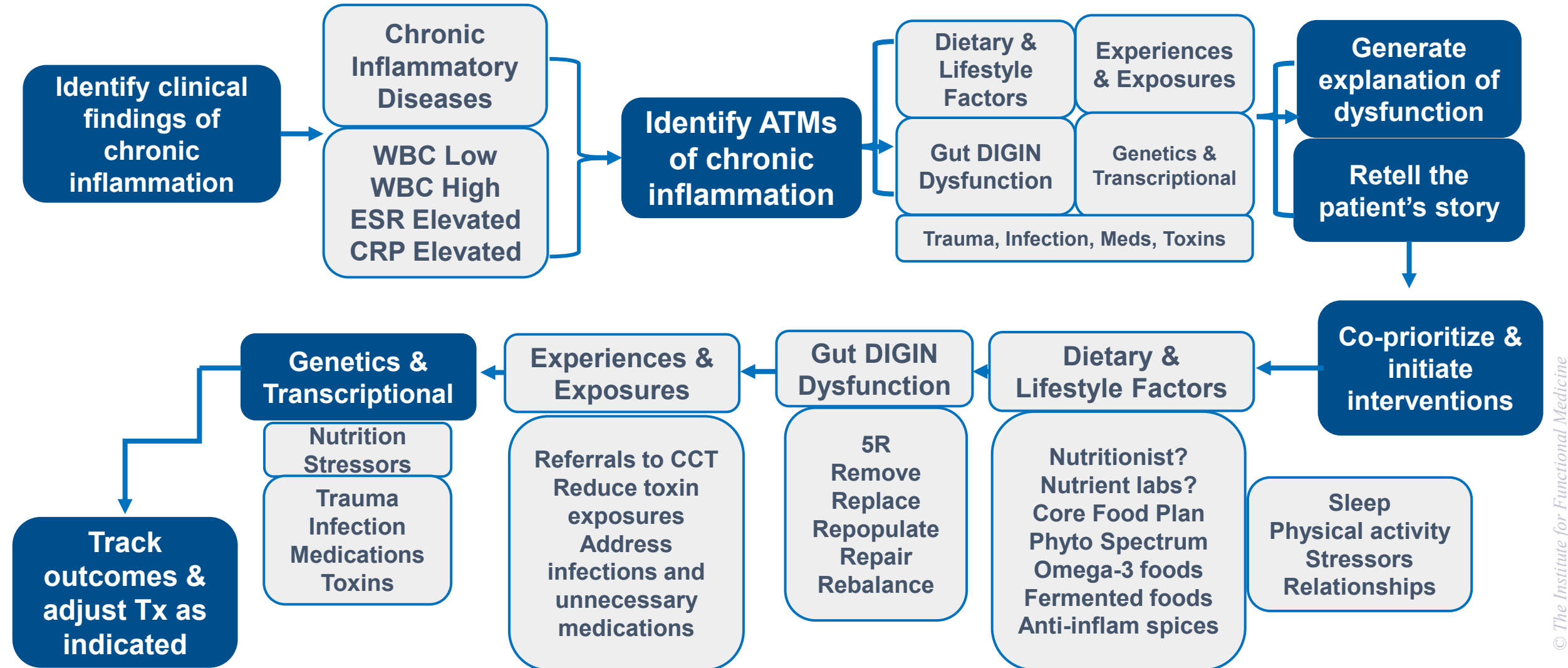
Stressors

Relationships

Long Description: Functional Medicine Matrix

- Defense and repair
 - Chronic infection
- Energy
 - Oxidative stress, mitochondrial dysfunction
- Mitigation and clearance
 - Oxidative stress
- Transport
 - Insulin resistance
 - Hyperinsulinemia
- Communication
 - Cortisol excess
- Structural integrity
 - Increased intestinal permeability
- Assimilation
 - Gut dysbiosis
- Mental, emotional, spiritual
 - Loneliness

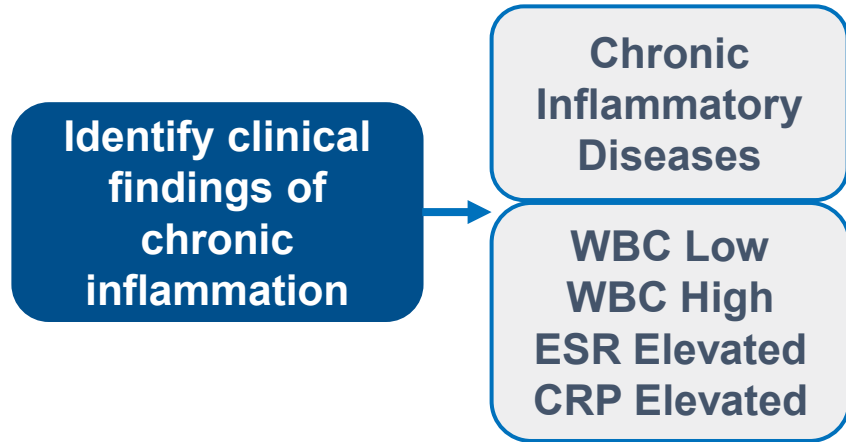
Clinical Flow Path for Chronic Inflammation



Long Description: Clinical Flow Path for Chronic Inflammation

- Identify clinical findings of chronic inflammation
- Identify A T M's of chronic inflammation
- Generate explanation of dysfunction
- Re tell the patient's story
- Co-prioritize and initiate interventions
- Track outcomes and adjust treatment as indicated

Clinical Flow Path for Chronic Inflammation



Long Description: Clinical Flow Path for Chronic Inflammation

- Here the emphasis is on identifying chronic inflammatory diseases, which may include low or elevated W B C, elevated E S R, and elevated C R P

Defense and Repair

Immune Regulation of Inflammation

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE